

A D10 Boundary-Current Route To The de Vries Electroweak Operator

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Abstract

We formulate a ten-dimensional boundary-current route to the branch operator

$$\mathcal{A}(J) = \begin{pmatrix} 0 & \sqrt{J} \\ \sqrt{J} & -J \end{pmatrix}, \quad J = j(j+1),$$

whose positive eigenvalues at $j = \frac{1}{2}, 1$ reproduce the on-shell weak mixing angle. The constructive layer is Type IIA on $M_4 \times K_6$, with $K_6 = \mathbb{CP}^2 \times \mathbb{CP}^1$ and a flag/current refinement. The active construction is entirely ten-dimensional. The D9 endpoint is the Q -boundary sector $(\mathrm{SU}(3)_c \times \mathrm{U}(1)_Q)/\mathbb{Z}_3$. Under the Stueckelberg Schur assumption with unit current metric, the projected D10 fluctuation operator gives the de Vries pole relation. An almost-commutative finite layer supplies the scalar dictionary: the Higgs expectation is the finite sheet length $\ell_H = \sqrt{2}/v_{\mathrm{EW}}$, the Q -neutral vacuum vector gives the $W/Z/\gamma$ Hessian, and modern scalar extensions add a neutral singlet Hessian block. The route emphasized here is the underused Q -boundary mechanism: the D8-like boundary $\Gamma_{\mathrm{EW}} = M_4 \times \mathbb{CP}^2 \times S_Q^1$, the RR/current contribution to $L_{2,\mathrm{vert}}^2$, and the boundary auxiliary field that generates the negative diagonal.

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1 D10 Boundary-Current Setup

The constructive object is the Type IIA compactification

$$M_4 \times K_6, \quad K_6 = \mathbb{CP}^2 \times \mathbb{CP}^1,$$

with retained Type IIA C_1, F_2 data and with a boundary/current sector carrying the electroweak operator. The electroweak endpoint is carried by a different circle, the diagonal weak-hypercharge frame circle

$$S_Q^1 \subset P_{T_3} \times P_Y, \quad Q = T_3 + Y.$$

This S_Q^1 cycle is the circle used by the localized Γ_{EW} boundary action.

The manuscript develops the following conditional implication. If the D10 boundary/current sector gives a projected operator

$$\Pi_j \mathcal{K}_{D10,*} \Pi_j = \mu_*^2 \begin{pmatrix} 0 & \sqrt{j(j+1)} \\ \sqrt{j(j+1)} & -j(j+1) \end{pmatrix}, \quad j = 0, \frac{1}{2}, 1,$$

then

$$M_\gamma^2 = 0, \quad M_W^2 = \mu_*^2 X_+(3/4), \quad M_Z^2 = \mu_*^2 X_+(2),$$

where

$$X_+(J) = \frac{-J + \sqrt{J^2 + 4J}}{2}.$$

The pole weak angle is then

$$\sin^2 \theta_{\text{pole}} = 1 - \frac{X_+(3/4)}{X_+(2)} = 0.223101 \dots$$

The localized boundary matrix defines L_Q , and L_Q fixes the absolute electromagnetic normalization of a concrete realization.

2 Weak Sectors And The Schur Operator

The weak $\mathbb{CP}^1$ factor carries the Borel–Weil sectors

$$H^0(\mathbb{CP}^1, \mathcal{O}(2j)) \simeq V_j, \quad j = 0, \frac{1}{2}, 1.$$

The affine-current filter supplies the same finite set at level $k = 2$, since integrable $\text{SU}(2)_k$ sectors obey

$$j = 0, \frac{1}{2}, \dots, \frac{k}{2}.$$

For this current-algebra filter the conformal weights are

$$\Delta_j = \frac{j(j+1)}{k+2}, \quad k = 2,$$

with conversion table

j	$J = j(j+1)$	$\Delta_j = J/4$	$4\Delta_j$
0	0	0	0
$\frac{1}{2}$	3/4	3/16	3/4
1	2	1/2	2

The string/interface Regge role is the mass template

$$M^2(n, j, \sigma) = \frac{n - a_0}{\alpha'} + \mu^2 \lambda_\sigma(j(j+1)) + \Delta M_{\text{compactification}}^2.$$

The string oscillator term carries the tower. The internal Schur term carries the finite electroweak branch. The target-space operator uses finite zero modes. The conformal-weight L_0 route supplies the control normalization. Let

$$A_j = \sum_{a=1}^3 T_a^{(j)} \otimes e_a,$$

with e_a an orthonormal current frame. Then

$$A_j^\dagger A_j = \sum_a T_a^{(j)} T_a^{(j)} = j(j+1) \mathbf{1}_{V_j} = J \mathbf{1}_{V_j}.$$

The same A_j admits an Atiyah–Manton-type holonomy reading [27]: a connection determines a Wilson line, and the first variation of the holonomy inserts the Lie-algebra generator in the same current frame. The candidate realization list retained for comparison is

Dolbeault/Dirac, boundary supercharge, current zero mode, brane localized auxiliary.

The boundary-current branch below supplies the coefficient calculation through the Payen endpoint Hilbert space and the localized Γ_{EW} action.

The algebraic boundary Hessian on $V_j \oplus \text{Im } A_j$ is

$$\mathcal{H}_j = \begin{pmatrix} 0 & A_j^\dagger \\ A_j & -A_j A_j^\dagger \end{pmatrix}.$$

Projection to a normalized image vector gives

$$\mathcal{H}_j \longrightarrow \begin{pmatrix} 0 & \sqrt{J} \\ \sqrt{J} & -J \end{pmatrix}.$$

Consequently the secular equation is

$$X^2 + JX - J = 0.$$

If the retained current metric is

$$\langle e_a, e_b \rangle_{\text{cur}} = \kappa_{\text{cur}} \delta_{ab}, \quad c = \sqrt{\kappa_{\text{cur}}},$$

the projected block becomes

$$\begin{pmatrix} 0 & c\sqrt{J} \\ c\sqrt{J} & -c^2 J \end{pmatrix}, \quad X^2 + c^2 JX - c^2 J = 0.$$

Thus the physical normalization is the statement

$$\kappa_{\text{cur}} = 1.$$

3 The Stueckelberg Normalization

The coefficient c can be isolated without committing to a complete compactification. On the unit S^3 , the scalar Casimir sector gives

$$\Delta_0 = 4J,$$

and the co-exact one-form sector has the Weitzenbock shift

$$\Delta_1|_{\text{co-exact}} = \Delta_0 + 1 = 4J + 1 = (2j + 1)^2.$$

The de Vries operator uses the algebraic Casimir $A_j^\dagger A_j = J$. The shifted geometric operator supplies a concrete failure channel whenever the diagonal and off-diagonal are built from independently shifted operators. Writing the local quadratic terms as

$$\alpha_{\text{mix}} \text{Re} \langle a, A_j \phi \rangle - \beta_{\text{diag}} \langle \phi, A_j^\dagger A_j \phi \rangle,$$

and normalizing the off-diagonal to $\sqrt{J}$, the coefficient in the projected block is

$$c = \frac{\beta_{\text{diag}}}{\alpha_{\text{mix}}^2}.$$

The single-coupling condition is the equality

$$\beta_{\text{diag}} = \alpha_{\text{mix}}^2.$$

Proposition 1 (Stueckelberg Schur normalization). *Suppose the order-parameter scalar ϕ in the spin- j sector has vanishing fluxless mass and couples to the background through a single Stueckelberg coupling*

$$a^\dagger A_j \phi + h.c.,$$

where the same A_j satisfies $A_j^\dagger A_j = J$. The reduced fluctuation block is the single-operator Schur completion

$$\begin{pmatrix} 0 & A_j^\dagger \\ A_j & -A_j A_j^\dagger \end{pmatrix},$$

and the coefficient in the deformed block is

$$c = 1.$$

Proof. The fluxless diagonal term of ϕ vanishes by hypothesis. The single coupling gives the mixing operator A_j . The rank-one self-energy generated by the same coupling is $-A_j A_j^\dagger$, so projection to $\text{Im } A_j$ gives the block

$$\begin{pmatrix} 0 & \sqrt{J} \\ \sqrt{J} & -J \end{pmatrix}.$$

The ratio of the diagonal coefficient to the square of the off-diagonal coefficient is therefore 1. $\square$

The same one-coupling pattern has a standard higher-dimensional home in gauge-Higgs unification [11, 12]. Splitting a higher-dimensional weak gauge field into A_μ and internal components A_a , the internal components are Wilson-line or Hosotani scalar variables. The cross term $F_{\mu a}$ supplies the Stueckelberg coupling, and on the $\mathbb{CP}^1$ line-bundle modes the weak current zero mode acts as

$$J_0^a|_{V_j} = T_a^{(j)}.$$

Thus the gauge–Higgs coupling uses the algebraic current operator

$$A_j = \sum_a T_a^{(j)} \otimes e_a, \quad A_j^\dagger A_j = J.$$

The shifted curl eigenvalue belongs to the cotangent-bundle control operator; the gauge–Higgs coupling supplies the bare-Casimir normalization needed by the Schur block. The Schur completion itself is then the D10 boundary/current statement.

The coefficient test is local. Let

$$U_H(\xi) = \mathcal{P} \exp \left(i\mu_\Gamma \int_{\gamma_H} \xi^a e_a T_a^{(j)} \right),$$

where e_a is the same current frame used in A_j . Then

$$\left. \frac{\partial U_H}{\partial \xi^a} \right|_0 = i\mu_\Gamma T_a^{(j)}, \quad \left. \frac{\partial^2 U_H}{\partial \xi^a \partial \xi^b} \right|_0 = -\mu_\Gamma^2 T_a^{(j)} T_b^{(j)}.$$

Thus the Wilson-line expansion gives

$$\alpha_{\text{mix}} = \mu_\Gamma, \quad \beta_{\text{diag}} = \mu_\Gamma^2, \quad \frac{\beta_{\text{diag}}}{\alpha_{\text{mix}}^2} = 1,$$

after the common physical factor has been extracted. For the stabilizer cycle,

$$U_Q(\theta) = \exp(i\theta Q), \quad Q\eta_0 = 0,$$

the first two insertions on η_0 vanish. The Hosotani cycle must therefore come from the retained transverse weak/current or Chan–Paton boundary-complex sector.

The proposition separates two issues. The ratio $c = 1$ follows from the single Stueckelberg Schur structure. The de Vries number additionally uses the bare-Casimir argument J in $X_+(J)$. The curvature-shifted alternatives provide numerical controls:

$$J = \frac{3}{4}, 2 \quad \rightsquigarrow \quad \sin^2 \theta_W = 0.2231 \dots,$$

The independent shifted assignment gives

$$c_W = \frac{4(3/4) + 1}{4(3/4)} = \frac{4}{3}, \quad c_Z = \frac{4(2) + 1}{4(2)} = \frac{9}{8},$$

and hence

$$\sin^2 \theta_W|_{\text{shifted}} = 0.267 \dots$$

4 The Q -Protected Boundary Cycle

The weak doublet section is

$$\eta_0 \in H^0(\mathbb{CP}^1, \mathcal{O}(1)), \quad T_3 \eta_0 = -\frac{1}{2} \eta_0, \quad Y \eta_0 = \frac{1}{2} \eta_0.$$

Therefore

$$Q\eta_0 = (T_3 + Y)\eta_0 = 0,$$

and the photon direction is protected by the same line sector that carries the $j = \frac{1}{2}$ slot.

Let P_{T_3} and P_Y be the unit-frame bundles for the weak weight line and the hypercharge line. The neutral character is

$$\chi_N(u_T, u_Y) = u_T^{-1} u_Y.$$

The Q -circle sector is

$$P_Q = \ker \chi_N \subset P_{T_3} \times P_Y.$$

For the selected line data,

$$c_1(L_Y) = 6h + 7\eta, \quad c_1(L_{T_3}) = \eta,$$

so the neutral line has

$$c_1(L_N) = c_1(L_Y) - c_1(L_{T_3}) = 6h + 6\eta = 6(h + \eta).$$

The diagonal six-unit line class also decomposes the selected class as

$$6h + 7\eta = 6(h + \eta) + \eta,$$

namely a diagonal six-unit line class plus one weak $\mathcal{O}(1)$ unit.

Using integer hypercharge $y = 6Y$, the ultraviolet diagonal element is

$$\zeta_6 = (\omega_3, -1, e^{i\pi/3}), \quad \omega_3 = e^{2\pi i/3}.$$

On a surviving state with color triality t , weak weight $T_3 = m$, and charge $Q = m + Y$, its action becomes

$$\omega_3^t \exp(i2\pi m) \exp(i\pi y/3) = \omega_3^t \exp(i2\pi Q).$$

Writing $q_3 = 3Q$, the residual generator is

$$\zeta_3 = (\omega_3, e^{2\pi i/3}),$$

and the endpoint group is

$$G_{D9} = \frac{\mathrm{SU}(3)_c \times \mathrm{U}(1)_Q}{\mathbb{Z}_3}.$$

5 Finite-Higgs Sheet And Scalar Vacuum

The same Q -neutral weak line admits an almost-commutative finite-geometric reading. In the Connes–Lott electroweak construction and in the later distance calculations for scalar fluctuations, the Higgs field is a finite-direction connection and its expectation value is a metric datum [13, 14, 20]. For a two-point finite direction,

$$D_F(\Phi) = \begin{pmatrix} 0 & \Phi \\ \Phi^\dagger & 0 \end{pmatrix},$$

and for $a = (\lambda_L, \lambda_R)$,

$$[D_F(\Phi), a] = \begin{pmatrix} 0 & (\lambda_R - \lambda_L)\Phi \\ (\lambda_L - \lambda_R)\Phi^\dagger & 0 \end{pmatrix}.$$

Consequently

$$\|[D_F(\Phi), a]\| = |\lambda_L - \lambda_R| \|\Phi\|, \quad d_{\text{sheet}} = \frac{1}{\|\Phi\|}.$$

For the project scalar,

$$\Phi_0 = v_F \eta_0, \quad Q\eta_0 = 0,$$

this gives

$$\ell_H = \frac{1}{|v_F|} = \frac{\sqrt{2}}{v_{\text{EW}}}.$$

Thus the electroweak vacuum amplitude is simultaneously a finite sheet length and a D10 scalar coordinate.

The finite spectral datum underlying this calculation is

$$\mathcal{A}_F^{\text{EW}} = \mathbb{C}_r \oplus \mathbb{H}_w, \quad \mathcal{H}_F^{\text{EW}} = H_r \oplus H_w, \quad \Phi : H_r \longrightarrow H_w.$$

The represented edge matched to the project is

$$H_r \xrightarrow{\Phi_0 = v_F \eta_0} H_w, \quad Y(H_r) = 0, \quad Y(H_w) = \frac{1}{2}, \quad Q\eta_0 = 0.$$

Inner fluctuation of D_F gives the finite connection Φ . The spectral distance gives ℓ_H . The electroweak covariant derivative gives the $W/Z/\gamma$ Hessian. The D10 boundary-current action compares this finite Hessian with the Schur image Hessian through the common vector η_0 .

The finite layer used here is the electroweak Glashow–Weinberg–Salam sector of Connes–Lott, together with the scalar-fluctuation distance formula. It supplies the local $U(1) \times SU(2)$ gauge data, a finite Higgs doublet, and the auxiliary-field vocabulary. The full Standard-Model finite triple supplies the KO6, Poincare, orientability, and unimodularity controls used below.

The local finite potential supplies the radial Higgs curvature. With

$$V_F(\Phi) = \lambda_F (\langle \Phi, \Phi \rangle - v_F^2)^2, \quad \Phi = v_F \eta_0 + \delta\Phi,$$

and

$$h = \sqrt{2} \operatorname{Re} \langle \eta_0, \delta\Phi \rangle,$$

the quadratic term is

$$V_F^{(2)} = 2\lambda_F v_F^2 h^2 = \frac{1}{2} m_H^2 h^2, \quad m_H^2 = 4\lambda_F v_F^2.$$

Equivalently,

$$m_H^2 = 2\lambda_F v_{\text{EW}}^2, \quad \lambda_F = \frac{m_H^2}{2v_{\text{EW}}^2}.$$

The value v_F fixes the finite sheet scale; the potential shape requires the finite triple, spectral-action coefficients, Yukawa data, cutoff normalization, and running. The KK side supplies a separate effective potential for compactification variables. The common object is therefore

$$V_{\text{eff}}(\rho, \lambda, v_F, \theta_H, \sigma, \dots),$$

with stationarity equations for the finite amplitude, the weak/base shape, the overall compactification scale, and possible Wilson-line or singlet directions. A D10 electroweak phase additionally requires the threshold window

$$\Lambda_{9 \rightarrow 10}(\rho_*, \lambda_*) < E_{\text{probe}} < \Lambda_{\text{UV}, 10}(\rho_*, \lambda_*).$$

Equivalently, writing the resolved weak-projective scale as

$$R_{\text{EW}}(\ell) = \ell \hat{R}_{\text{EW}},$$

the stationary vacuum must satisfy

$$R_{\text{EW}}(\ell_*) > \Lambda_{\text{UV}, 10}^{-1}.$$

This inequality gives the D10 probe regime. Collapse of the weak/projective direction gives the D9 endpoint. For the elementary quartic finite-sector model with $\lambda_F > 0$,

$$V_F = \lambda_F (\langle \Phi, \Phi \rangle - v_F^2)^2 \implies V_F \rightarrow +\infty \quad \text{as} \quad |\Phi| \rightarrow \infty.$$

In that phase diagram $v_F = 0$ can lose stability as the quadratic coefficient changes sign, a finite v_0 becomes the electroweak minimum, and the $v_F \rightarrow \infty$ boundary has infinite energy. A D9 endpoint in the present model therefore requires a coupled KK direction whose large-field asymptotic flattens or turns the effective potential along a compactification-collapse coordinate. A factorized potential,

$$V_{\text{eff}} = V_{\text{KK}}(\rho, \lambda, \tau) + V_F(v_F, \sigma, \dots),$$

leaves the finite electroweak scale and the dimension window as separate stationarity problems. The D10 electroweak phase therefore requires mixed Hessian entries such as

$$K_{\rho v} = \left. \frac{\partial^2 V_{\text{eff}}}{\partial \rho \partial v_F} \right|_*, \quad K_{\lambda v} = \left. \frac{\partial^2 V_{\text{eff}}}{\partial \lambda \partial v_F} \right|_*,$$

or a shared boundary/current scale

$$\mu_*^2 = \mu_*^2(\rho_*, \lambda_*, v_F, \theta_H, \dots).$$

The reduced radial coefficient skeleton retained for the one-scale pass is

$$V_{\text{rad}}(\ell) = -B_R \ell^{-2} + B_H \ell^{-6} + B_0 \ell^6 + B_2 \ell^2 + B_4 \ell^{-2} + B_6 \ell^{-6} + V_{\text{int}}.$$

The stationary calculation must pair this HKT scaling block with the finite quartic V_F and the localized boundary term V_Γ . The gauge Hessian uses the same vacuum vector:

$$K_{AB}^{\text{gauge}} = v_F^2 g_A g_B \langle T_A \eta_0, T_B \eta_0 \rangle.$$

Together with $T_3 \eta_0 = -\frac{1}{2} \eta_0$, $Y \eta_0 = \frac{1}{2} \eta_0$, this gives

$$M_W^2 = \frac{g_2^2 v_{\text{EW}}^2}{4}, \quad M_Z^2 = \frac{(g_2^2 + g_Y^2) v_{\text{EW}}^2}{4}, \quad M_\gamma^2 = 0.$$

Using Weinberg's current-length normalization, let $C = 2\pi\sqrt{16\pi G}$. The KK current-length dictionary is

$$g_2 = \frac{C}{L_2}, \quad g_Y = \frac{C}{L_Y}, \quad e = \frac{C}{L_Q}, \quad L_Q^2 = L_2^2 + L_Y^2.$$

Hence

$$\sin^2 \theta_W = \frac{L_2^2}{L_Q^2}, \quad \tan^2 \theta_W = \frac{L_2^2}{L_Y^2}.$$

The same dictionary rewrites the vector masses as

$$M_W = \frac{e v_{\text{EW}}}{2} \frac{L_Q}{L_2} = \frac{e v_{\text{EW}}}{2 \sin \theta_W}, \quad M_Z = \frac{e v_{\text{EW}}}{2} \frac{L_Q^2}{L_2 L_Y} = \frac{e v_{\text{EW}}}{2 \sin \theta_W \cos \theta_W}.$$

Equivalently, with

$$G_{\text{EW}}(\rho, \lambda, \dots) = \frac{1}{L_2(\rho, \lambda, \dots)^2} + \frac{1}{L_Y(\rho, \lambda, \dots)^2},$$

one has

$$M_Z^2 = \frac{C^2}{4} v_{\text{EW}}^2 G_{\text{EW}}, \quad \mu_*^2 = \frac{C^2}{4X_+(2)} v_{\text{EW}}^2 G_{\text{EW}}.$$

Thus the current-length dependence produces mixed scale sensitivity,

$$\frac{\partial^2 M_Z^2}{\partial \rho \partial v_{\text{EW}}} = \frac{C^2}{2} v_{\text{EW}} \partial_\rho G_{\text{EW}}, \quad \frac{\partial^2 M_Z^2}{\partial \lambda \partial v_{\text{EW}}} = \frac{C^2}{2} v_{\text{EW}} \partial_\lambda G_{\text{EW}}.$$

The localized source term with the physical electroweak amplitude uses the canonically normalized finite mode. Let

$$I_0 = \int_{\Gamma_{\text{EW},\text{int}}} \sqrt{\gamma_\Gamma} |\eta_0|^2, \quad I_{AB} = \int_{\Gamma_{\text{EW},\text{int}}} \sqrt{\gamma_\Gamma} \langle T_A \eta_0, T_B \eta_0 \rangle.$$

For an unnormalized boundary-profile amplitude v_b , the common DBI/open-boundary coefficient Z_0 gives

$$v_{\text{EW}}^2 = Z_0 I_0 v_b^2.$$

The physical source term is therefore

$$V_\Gamma = \frac{1}{2} v_{\text{EW}}^2 \Sigma_\Gamma^{AB} \mathcal{G}_{AB}(\rho, \lambda, \dots), \quad \Sigma_\Gamma^{AB} = \frac{I_{AB}}{I_0}, \quad Q\eta_0 = 0.$$

For the selected normalized $O(1)$ component,

$$T_3 \eta_0 = -\frac{1}{2} \eta_0, \quad Y \eta_0 = \frac{1}{2} \eta_0,$$

so

$$\Sigma_{33} = \frac{1}{4}, \quad \Sigma_{YY} = \frac{1}{4}, \quad \Sigma_{3Y} = -\frac{1}{4}.$$

Hence

$$\Sigma_\Gamma(Q, Q) = 0, \quad Q = T_3 + Y,$$

and the photon direction is protected at the source tensor level. The charged weak entries satisfy

$$\|T_1 \eta_0\|_\Gamma^2 = \|T_2 \eta_0\|_\Gamma^2 = \frac{1}{4}, \quad \text{Re} \langle T_1 \eta_0, T_2 \eta_0 \rangle_\Gamma = 0.$$

With $c_2 = C/L_2$ and $c_Y = C/L_Y$, the neutral source block is

$$K_{\text{neutral}} = \frac{v_{\text{EW}}^2}{4} \begin{pmatrix} c_2^2 & -c_2 c_Y \\ -c_2 c_Y & c_Y^2 \end{pmatrix} = \frac{C^2 v_{\text{EW}}^2}{4} \begin{pmatrix} L_2^{-2} & -(L_2 L_Y)^{-1} \\ -(L_2 L_Y)^{-1} & L_Y^{-2} \end{pmatrix}.$$

Its null eigenvector is proportional to (L_2, L_Y) , giving

$$A_\mu = \frac{L_2}{L_Q} W_\mu^3 + \frac{L_Y}{L_Q} B_\mu,$$

and its nonzero eigenvalue is $M_Z^2 = (C^2 v_{\text{EW}}^2 / 4) G_{\text{EW}}$. The localized boundary frame admits a general neutral length matrix

$$\mathcal{L}_\Gamma = \begin{pmatrix} L_2^2 & \epsilon_\Gamma \\ \epsilon_\Gamma & L_Y^2 \end{pmatrix}, \quad \det \mathcal{L}_\Gamma = L_2^2 L_Y^2 - \epsilon_\Gamma^2.$$

The Q -circle length is

$$L_Q^2 = (1, 1) \mathcal{L}_\Gamma \begin{pmatrix} 1 \\ 1 \end{pmatrix} = L_2^2 + L_Y^2 + 2\epsilon_\Gamma, \quad \alpha_\Gamma = \frac{C^2}{4\pi L_Q^2}.$$

The neutral source covector is $s = (-1, 1)$, hence the generalized neutral coefficient is

$$G_Z^\Gamma = s \mathcal{L}_\Gamma^{-1} s^T = \frac{L_2^2 + L_Y^2 + 2\epsilon_\Gamma}{L_2^2 L_Y^2 - \epsilon_\Gamma^2} = \frac{L_Q^2}{\det \mathcal{L}_\Gamma},$$

and

$$M_Z^2 = \frac{C^2 v_{\text{EW}}^2}{4} G_Z^\Gamma.$$

The diagonal source metric appears at $\epsilon_\Gamma = 0$, recovering

$$L_Q^2 = L_2^2 + L_Y^2, \quad G_Z^\Gamma = L_2^{-2} + L_Y^{-2}.$$

Let

$$R_{\text{dV}} = \frac{X_+(3/4)}{X_+(2)}.$$

Since

$$M_W^2 = \frac{C^2 v_{\text{EW}}^2}{4L_2^2},$$

the boundary length matrix gives

$$\frac{M_W^2}{M_Z^2} = \frac{\det \mathcal{L}_\Gamma}{L_2^2 L_Q^2} = \frac{L_2^2 L_Y^2 - \epsilon_\Gamma^2}{L_2^2 (L_2^2 + L_Y^2 + 2\epsilon_\Gamma)}.$$

The de Vries pass equation for the neutral boundary matrix is

$$\epsilon_\Gamma^2 + 2R_{\text{dV}} L_2^2 \epsilon_\Gamma + R_{\text{dV}} L_2^4 + (R_{\text{dV}} - 1) L_2^2 L_Y^2 = 0.$$

With

$$u = \frac{L_2^2}{L_Y^2}, \quad \delta = \frac{\epsilon_\Gamma}{L_Y^2},$$

this becomes

$$R_{\text{dV}} = \frac{u - \delta^2}{u(u + 1 + 2\delta)}, \quad \delta_\pm(u) = -R_{\text{dV}} u \pm \sqrt{u(1 - R_{\text{dV}})(1 - R_{\text{dV}} u)}.$$

The diagonal source metric sits at

$$\delta_+(u_{\text{dV}}) = 0, \quad u_{\text{dV}} = \frac{1 - R_{\text{dV}}}{R_{\text{dV}}} = \tan^2 \theta_{\text{pole}}.$$

Since $\mathcal{L}_\Gamma$ is a kinetic matrix, the allowed region is

$$u > 0, \quad |\delta| < \sqrt{u}, \quad u + 1 + 2\delta > 0.$$

The de Vries branches are real for

$$0 < u \leq R_{\text{dV}}^{-1} = 1.2871691363429836 \dots$$

At $u = u_{\text{dV}}$ the second branch gives

$$\delta_-(u_{\text{dV}}) = -2(1 - R_{\text{dV}}) = -0.4462026446017324 \dots$$

It also satisfies the kinetic inequalities. The localized Γ_{EW} action is therefore characterized by the neutral branch data, including L_Q , together with the charged current length. Equivalently, define the normalized neutral current correlation

$$\chi_\Gamma = \frac{\epsilon_\Gamma}{L_2 L_Y} = \frac{\delta}{\sqrt{u}}.$$

Then $-1 < \chi_\Gamma < 1$, and the pass equation reads

$$R_{\text{dV}} = \frac{1 - \chi_\Gamma^2}{u + 1 + 2\chi_\Gamma \sqrt{u}}.$$

Solving for the correlation gives

$$\chi_{\pm}(u) = -R_{\text{dV}}\sqrt{u} \pm \sqrt{(1-R_{\text{dV}})(1-R_{\text{dV}}u)}.$$

At the diagonal source point,

$$\chi_+(u_{\text{dV}}) = 0, \quad \chi_-(u_{\text{dV}}) = -2\sqrt{R_{\text{dV}}(1-R_{\text{dV}})} = -0.8326514812056032\dots$$

The neutral branch question is therefore a bounded current-correlation calculation in the localized boundary action. The same data factorize the matrix as

$$\mathcal{L}_{\Gamma} = L_Y^2 \begin{pmatrix} u & \chi_{\Gamma}\sqrt{u} \\ \chi_{\Gamma}\sqrt{u} & 1 \end{pmatrix}.$$

Consequently

$$\det \mathcal{L}_{\Gamma} = L_Y^4 u(1 - \chi_{\Gamma}^2), \quad L_Q^2 = L_Y^2(u + 1 + 2\chi_{\Gamma}\sqrt{u}),$$

and

$$G_Z^{\Gamma} = \frac{u + 1 + 2\chi_{\Gamma}\sqrt{u}}{L_Y^2 u(1 - \chi_{\Gamma}^2)}, \quad \alpha_{\Gamma} = \frac{C^2}{4\pi L_Y^2(u + 1 + 2\chi_{\Gamma}\sqrt{u})}.$$

The localized action target is the triple

$$S_{\Gamma} \rightsquigarrow (L_Y^2, u, \chi_{\Gamma}),$$

with

$$u = \frac{\int d\nu_{\Gamma} \|\theta_{\Gamma}^3\|^2}{\int d\nu_{\Gamma} \|\theta_{\Gamma}^Y\|^2}, \quad \chi_{\Gamma} = \frac{\int d\nu_{\Gamma} \gamma_{\Gamma}^{-1}(\theta_{\Gamma}^3, \theta_{\Gamma}^Y)}{L_2 L_Y}.$$

Writing $y = \sqrt{u}$, the pass equation is equivalently

$$R_{\text{dV}}y^2 + 2R_{\text{dV}}\chi_{\Gamma}y + R_{\text{dV}} + \chi_{\Gamma}^2 - 1 = 0,$$

so

$$y_{\pm}(\chi_{\Gamma}) = -\chi_{\Gamma} \pm \sqrt{\frac{(1-R_{\text{dV}})(1-\chi_{\Gamma}^2)}{R_{\text{dV}}}}, \quad u_{\pm}(\chi_{\Gamma}) = y_{\pm}(\chi_{\Gamma})^2.$$

At $\chi_{\Gamma} = 0$, the positive root is

$$u_+(0) = \frac{1 - R_{\text{dV}}}{R_{\text{dV}}}.$$

At $u = u_{\text{dV}}$, the correlated branch has

$$\chi_{\Gamma} = -2\sqrt{R_{\text{dV}}(1-R_{\text{dV}})}.$$

Define

$$D_{\Gamma} = u + 1 + 2\chi_{\Gamma}\sqrt{u}, \quad \Lambda_{\Gamma} = \frac{C^2}{L_Y^2}.$$

Then the factorized matrix gives

$$\alpha_{\Gamma} = \frac{\Lambda_{\Gamma}}{4\pi D_{\Gamma}}, \quad M_W^2 = \frac{\Lambda_{\Gamma} v_{\text{EW}}^2}{4u}, \quad M_Z^2 = \frac{\Lambda_{\Gamma} v_{\text{EW}}^2 D_{\Gamma}}{4u(1 - \chi_{\Gamma}^2)}.$$

Eliminating Λ_{Γ} gives

$$\frac{M_Z^2}{\pi \alpha_{\Gamma} v_{\text{EW}}^2} = \frac{D_{\Gamma}^2}{u(1 - \chi_{\Gamma}^2)}, \quad \frac{M_W^2}{\pi \alpha_{\Gamma} v_{\text{EW}}^2} = \frac{D_{\Gamma}}{u}.$$

The diagonal branch reduces this to

$$\frac{M_Z^2}{\pi\alpha_\Gamma v_{\text{EW}}^2} = \frac{1}{R_{\text{dV}}(1 - R_{\text{dV}})}, \quad \frac{M_W^2}{\pi\alpha_\Gamma v_{\text{EW}}^2} = \frac{1}{1 - R_{\text{dV}}}.$$

The electromagnetic normalization is the localized computation of $\Lambda_\Gamma = C^2/L_Y^2$. Given (u, χ_Γ) from the neutral current matrix, the three observable channels infer

$$\Lambda_\Gamma^{(\alpha)} = 4\pi\alpha_\Gamma D_\Gamma, \quad \Lambda_\Gamma^{(W)} = \frac{4uM_W^2}{v_{\text{EW}}^2}, \quad \Lambda_\Gamma^{(Z)} = \frac{4u(1 - \chi_\Gamma^2)M_Z^2}{v_{\text{EW}}^2 D_\Gamma}.$$

The D10 observable closure is

$$\Lambda_\Gamma^{(\alpha)} = \Lambda_\Gamma^{(W)} = \Lambda_\Gamma^{(Z)}.$$

Equivalently,

$$4\pi\alpha_\Gamma D_\Gamma = \frac{4uM_W^2}{v_{\text{EW}}^2} = \frac{4u(1 - \chi_\Gamma^2)M_Z^2}{v_{\text{EW}}^2 D_\Gamma}.$$

Define

$$S_W = \frac{M_W^2}{\pi\alpha_\Gamma v_{\text{EW}}^2}, \quad S_Z = \frac{M_Z^2}{\pi\alpha_\Gamma v_{\text{EW}}^2}, \quad R_{\text{obs}} = \frac{S_W}{S_Z}.$$

Then

$$D_\Gamma = S_W u, \quad 1 - \chi_\Gamma^2 = R_{\text{obs}} S_W u.$$

With $\varsigma_\Gamma = \text{sgn}(\chi_\Gamma)$,

$$\chi_\Gamma = \varsigma_\Gamma \sqrt{1 - R_{\text{obs}} S_W u}.$$

The inverse electroweak reconstruction equation is

$$S_W u = u + 1 + 2\varsigma_\Gamma \sqrt{u(1 - R_{\text{obs}} S_W u)}.$$

For the exact de Vries pass, $R_{\text{obs}} = R_{\text{dV}}$. The common scale is

$$\Lambda_\Gamma = 4\pi\alpha_\Gamma S_W u.$$

For canonical finite-mode normalization and frame-frozen Σ_Γ^{AB} ,

$$\partial_\rho \log G_{\text{EW}} = -\frac{1}{\rho}, \quad \partial_\lambda \log G_{\text{EW}} = -\frac{0.118746651783551}{\lambda}.$$

Therefore

$$\frac{\partial^2 M_Z^2}{\partial \rho \partial v_{\text{EW}}} = -\frac{2M_Z^2}{v_{\text{EW}} \rho}, \quad \frac{\partial^2 M_Z^2}{\partial \lambda \partial v_{\text{EW}}} = -0.237493303567102 \frac{M_Z^2}{v_{\text{EW}} \lambda}.$$

The same dimensionless coefficients apply to $\mu_*^2 = M_Z^2/X_+(2)$. For the charged weak block,

$$\frac{\partial^2 M_W^2}{\partial \rho \partial v_{\text{EW}}} = -\frac{2M_W^2}{v_{\text{EW}} \rho}, \quad \frac{\partial^2 M_W^2}{\partial \lambda \partial v_{\text{EW}}} = -0.838811351483078 \frac{M_W^2}{v_{\text{EW}} \lambda}.$$

The general mixed block is

$$K_{\rho v}^\Gamma = v_{\text{EW},*} [\Sigma_\Gamma^{AB} \partial_\rho \mathcal{G}_{AB} + (\partial_\rho \Sigma_\Gamma^{AB}) \mathcal{G}_{AB}]_*,$$

with the analogous λ -derivative. In the one-scale window, $L_A(\ell, \lambda) = \ell \widehat{L}_A(\lambda)$ and $\rho = c_\rho(\lambda)\ell^2$. At fixed λ ,

$$\mathcal{G}^{AB}(\rho) = \frac{c_\rho}{\rho} \widehat{\mathcal{G}}^{AB}, \quad \partial_\rho \mathcal{G}^{AB} = -\frac{1}{\rho} \mathcal{G}^{AB}.$$

For frame-frozen Σ_{Γ}^{AB} , a positive coefficient gives

$$\partial_{\rho} V_{\Gamma} = -\frac{1}{\rho} V_{\Gamma}, \quad K_{\rho v}^{\Gamma} = -\frac{1}{\rho_*} v_{\text{EW},*} \Sigma_{\Gamma}^{AB} \mathcal{G}_{AB}|_*$$

For an unnormalized boundary-profile amplitude, the DBI-scaled finite-section coefficient carries the D8 source scaling

$$Z_{\Gamma} \propto \rho \tau^{-3}.$$

At fixed τ , this cancels the radial scaling of $\mathcal{G}^{AB}$. With

$$\tau = \tau_* \left(\frac{\ell}{\ell_*} \right)^s, \quad \rho = c_{\rho} \ell^2,$$

one obtains

$$\partial_{\rho} \log(Z_{\Gamma} \mathcal{G}^{AB}) = -\frac{3s}{2\rho}.$$

In the canonical v_{EW} convention, the common DBI factor is absorbed into v_{EW} , leaving the normalized matrix Σ_{Γ}^{AB} . At fixed ρ , the selected source metric also gives local λ -derivatives. For

$$u(\beta) = \frac{L_2^2}{L_Y^2} = 3 \frac{n^2 q^2}{r^2} \frac{1 + \beta}{\beta}, \quad \lambda(\beta) = \frac{3 p^2}{4 q^2} \beta (3 - 4\beta)^2,$$

one obtains

$$\left. \frac{\partial \log u}{\partial \log \lambda} \right|_* = 1.347631602696264.$$

Combining this with the weak length coefficient gives

$$\partial_{\lambda} \log L_2^2 = \frac{0.419405675741539}{\lambda}, \quad \partial_{\lambda} \log L_Y^2 = -\frac{0.928225926954725}{\lambda}.$$

Consequently,

$$\partial_{\lambda} \log L_Q^2 = -\frac{0.627566902996737}{\lambda}, \quad \partial_{\lambda} \log G_{\text{EW}} = -\frac{0.118746651783551}{\lambda}.$$

The total λ -block for the canonical finite-section source adds $\partial_{\lambda} \Sigma_{\Gamma}^{AB}$. For the unnormalized boundary-profile control pass, at fixed ρ and fixed τ , write

$$Z_{\Gamma} = Z_{\Gamma,0} z_{\Gamma}(\lambda).$$

For a constant-profile boundary-cycle normalization on $\mathbb{CP}^2 \times S_Q^1$,

$$z_{\Gamma,\text{cp}}(\lambda) \propto c_{\rho}(\lambda)^{-5/2} \widehat{L}_Q(\lambda),$$

and the selected source metric gives

$$\partial_{\lambda} \log z_{\Gamma,\text{cp}} = -\frac{0.980450118165035}{\lambda}.$$

Consequently,

$$\partial_{\lambda} \log(z_{\Gamma,\text{cp}} G_{\text{EW}}) = -\frac{1.099196769948586}{\lambda}.$$

A canonically normalized four-dimensional finite mode can absorb the cycle-volume factor into v_{EW} . In that convention

$$z_{\Gamma,\text{unit}}(\lambda) = 1, \quad \partial_{\lambda} \log(z_{\Gamma,\text{unit}} G_{\text{EW}}) = -\frac{0.118746651783551}{\lambda}.$$

Standard KK scaling gives a second D9 boundary at finite electroweak vacuum value. Since $g_2 \propto L_2^{-1}$,

$$L_2 \rightarrow 0, \quad v_{\text{EW}} = v_0 < \infty \quad \implies \quad M_W, M_Z \rightarrow \infty.$$

With finite L_Y ,

$$e = \frac{g_2 g_Y}{\sqrt{g_2^2 + g_Y^2}} \rightarrow g_Y, \quad \sin^2 \theta_W \rightarrow 0,$$

so the surviving photon is the finite $U(1)_Q$ current at the D9 boundary. Indeed, with

$$A_\mu = \sin \theta_W W_\mu^3 + \cos \theta_W B_\mu, \quad Q = T_3 + Y,$$

the field direction tends to B_μ , and the charge generator remains Q because

$$g_2 \sin \theta_W = e, \quad g_Y \cos \theta_W = e.$$

The branch with $L_2, L_Y \rightarrow 0$ has $e \rightarrow \infty$ and gives a singular endpoint. This finite- v route is controlled by the current metric, and the large- v_F route is controlled by the scalar asymptotic. As a vacuum endpoint, finite v_{EW} with $M_W, M_Z \rightarrow \infty$ is a pathological decoupling limit; its value here is diagnostic for the Q -projection.

The de Vries operator enters through the boundary-current image channel. For

$$A_j = \sum_a T_a^{(j)} \otimes e_a, \quad A_j^\dagger A_j = J \mathbf{1},$$

the source/image Hessian is

$$\mathcal{H}_j = \begin{pmatrix} 0 & \sqrt{J} \\ \sqrt{J} & -J \end{pmatrix}, \quad \det \mathcal{H}_j = -J.$$

For $j = \frac{1}{2}$, the same neutral doublet gives $J = 3/4$. The finite potential, the gauge Hessian, and the Schur image Hessian therefore share the vacuum vector η_0 and occupy different quadratic sectors of the D10 interface. The corresponding Schur mass scale obeys

$$\mu_*^2 = \frac{M_Z^2}{X_+(2)} = \frac{g_2^2 + g_Y^2}{4X_+(2)} v_{\text{EW}}^2.$$

The canonical electroweak source tensor fixes the physical $j = 1$ neutral eigenvalue, so the dimensionful Schur scale is anchored by

$$\mu_*^2 X_+(2) = M_Z^2 = \frac{C^2 v_{\text{EW}}^2}{4} G_{\text{EW}}.$$

The coefficient split is

$$\text{scale: } \mu_*^2 = \frac{C^2 v_{\text{EW}}^2}{4X_+(2)} G_{\text{EW}}, \quad \text{ratio: } a = 1, \quad \kappa_{\text{cur}} = 1.$$

Anchoring the $j = 1$ channel converts the charged channel into the weak-angle length equation. The source block gives

$$\frac{M_W^2}{M_Z^2} = \frac{L_Y^2}{L_2^2 + L_Y^2} = \frac{1}{1 + u}, \quad u = \frac{L_2^2}{L_Y^2}.$$

The Schur block gives

$$\frac{M_W^2}{M_Z^2} = \frac{X_+(3/4)}{X_+(2)}.$$

Source/Schur agreement is therefore equivalent to

$$u = \frac{X_+(2)}{X_+(3/4)} - 1 = 0.2871691363429836.$$

Thus the source tensor supplies the common mass scale, and the Payen Wilson/current-frame calculation supplies the Schur ratio. Modern spectral-action variants can add a real singlet scalar σ coupled to the finite Higgs sector [15, 16, 17, 18, 19]. A local model is

$$V_{h\sigma} = \lambda_F(\langle\Phi, \Phi\rangle - v_F^2)^2 + \frac{1}{2}M_{\sigma,0}^2\sigma^2 + a_\sigma\sigma(\langle\Phi, \Phi\rangle - v_F^2).$$

In the canonical basis (h, σ) , the Hessian is

$$K_{h\sigma} = \begin{pmatrix} 2\lambda_F v_{\text{EW}}^2 & a_\sigma v_{\text{EW}} \\ a_\sigma v_{\text{EW}} & M_{\sigma,0}^2 \end{pmatrix}.$$

Writing $M_{h,0}^2 = 2\lambda_F v_{\text{EW}}^2$, the scalar eigenvalues are

$$m_\pm^2 = \frac{1}{2} \left(M_{h,0}^2 + M_{\sigma,0}^2 \pm \sqrt{(M_{h,0}^2 - M_{\sigma,0}^2)^2 + 4a_\sigma^2 v_{\text{EW}}^2} \right),$$

with mixing angle

$$\tan 2\alpha = \frac{2a_\sigma v_{\text{EW}}}{M_{\sigma,0}^2 - M_{h,0}^2}.$$

Since $Q\sigma = 0$, this mixing leaves the photon-protection equation unchanged and enlarges the scalar vacuum system.

The finite-triple control layer uses the Standard-Model KO6 structure [21, 22, 23]. A local four-summand lift is

$$A_{\text{lift}} = M_3(\mathbb{C})_c \oplus \mathbb{H}_w \oplus \mathbb{C}_r \oplus \mathbb{C}_q,$$

with project edge

$$H_{rq}^- \longrightarrow H_{wq}^+.$$

The first-order factors are

$$(a_w - a_r)(b_q - b_q) = 0, \quad (a_q - a_q)(b_w - b_r) = 0,$$

and the local charge equations are

$$Y(H_{rq}) = 0, \quad Y(H_{wq}) = \frac{1}{2}.$$

With $y_w = 0$, this fixes

$$y_r = y_q = -\frac{1}{2}.$$

The project edge contribution to the four-summand intersection form has

$$\det(\cap\Phi) = 0.$$

The full represented Krajewski matrix supplies the Standard-Model Poincare form, determinant-one hypercharge lift, and fermion Hilbert-space data. Modern finite-geometry checks add Hodge-duality and spectral-torsion constraints involving Yukawa and Majorana data [24, 25].

6 The D8-Like Boundary And Schur Auxiliary

The common boundary interface for Q protection and the Schur auxiliary channel is

$$\Gamma_{\text{EW}} = M_4 \times \mathbb{CP}^2 \times S_Q^1.$$

The localized action on this interface is split as

$$S_\Gamma = S_{\text{DBI}}^\Gamma + S_{\text{CS}}^\Gamma + S_{\text{Payen}}^\Gamma + S_{\text{aux}}^\Gamma.$$

The D8-like DBI term is

$$S_{\text{DBI}}^\Gamma = -T_8 \int_{\Gamma_{\text{EW}}} d^9 \xi e^{-\phi} \sqrt{-\det(P[g + B] + 2\pi\alpha' F_Q)}.$$

Its quadratic finite/current density is

$$d\nu_\Gamma = T_8 (2\pi\alpha')^2 e^{-\phi} \sqrt{\gamma_\Gamma} \mathcal{W}_\Gamma d^5 \sigma,$$

where $\mathcal{W}_\Gamma$ collects the orientifold-image, frame, and reduction factors. It is D8-like in Type IIA. The universal localized-source scaling gives

$$V_{Dp} \propto \rho^{(p-6)/2} \tau^{-3},$$

so for $p = 8$,

$$V_{D8} \propto \rho \tau^{-3}.$$

With one-scale variables

$$\rho = c_\rho \ell^2, \quad \tau = \tau_* \left(\frac{\ell}{\ell_*} \right)^s,$$

this contribution scales as

$$V_{D8} \propto x^{2-3s}, \quad x = \frac{\ell}{\ell_*}.$$

The Romans-mass jump takes the distributional form

$$dF_0 = N_8 \delta_{\Gamma_{\text{EW}}}.$$

The Wess–Zumino skeleton is

$$S_{\text{WZ}} = \mu_8 \int_{\Gamma_{\text{EW}}} C \wedge \text{ch}(E_Q) \sqrt{\frac{\widehat{A}(T\Gamma_{\text{EW}})}{\widehat{A}(N\Gamma_{\text{EW}})}}.$$

On the $\mathbb{CP}^2$ boundary the spin^c shift gives

$$\frac{F_Q}{2\pi} = 6h_Q + \frac{x}{2}, \quad x = (2m+1)h_Q.$$

For the minimal lift,

$$\frac{F_Q}{2\pi} = \frac{13}{2}h_Q, \quad \frac{2F_Q}{2\pi} = 13h_Q.$$

The boundary therefore carries the determinant signal $P + Q = 13$. In a realization with weak period $Q = 7$, the internal consistency check reads

$$r = Q(P + Q) = 7 \cdot 13 = 91.$$

This relation is used inside a specified realization as an internal consistency condition.

The source-backed boundary placement is Payen's Chan–Paton action for a group-valued endpoint field $g(\tau)$ [9]:

$$S_{\text{CP}} = \int_{\partial\Sigma} d\tau \kappa(-ig^{-1}(\partial_\tau + A_\tau)g).$$

Canonical quantization gives a finite-dimensional representation R of the endpoint group, and the boundary path integral gives the Wilson loop

$$\text{Tr } \mathcal{T}_{\partial\Sigma} \exp \left(i \int_{\partial\Sigma} d\tau R(A_\tau) \right).$$

For $G = \text{SU}(2)$, $R = V_j$, and

$$R_j(A_\tau) = A_\tau^a T_a^{(j)},$$

the Wilson expansion supplies the linear charge insertion $T_a^{(j)}$ and the bilinear insertion $T_a^{(j)} T_b^{(j)}$. With a boundary fluctuation

$$R_j(A_\tau) = \mu \xi^a T_a^{(j)},$$

the expansion has one tensor normalization:

$$\delta W_j|_{\xi=0} \sim i\mu T_a^{(j)} \delta \xi^a, \quad \delta^2 W_j|_{\xi=0} \sim -\mu^2 T_a^{(j)} T_b^{(j)} \delta \xi^a \delta \xi^b.$$

The representation product is therefore fixed once the boundary fluctuation scale μ is fixed. Equivalently, a rescaling $A_j \mapsto a A_j$ sends the image bilinear to $a^2 A_j A_j^\dagger$. The boundary exponential therefore gives the tensor relation $b = a^2$. After the common physical scale μ_*^2 is factored out, the orthonormal current frame gives the canonical dimensionless coefficient $a = 1$. The coefficient μ belongs to the boundary fluctuation and the retained current frame. Representation dependence enters through $T_a^{(j)}$ and its Casimir $j(j+1)$. Thus a single Γ_{EW} boundary sector uses one worldvolume scale μ_1^2 for $j = \frac{1}{2}$ and $j = 1$.

The same boundary can carry the auxiliary field $y \in V_j$. Before the Q projection, the weak Chan–Paton line sectors are $\mathcal{O}(0)$ and $\mathcal{O}(2j)$. Their mixed boundary-changing zero-mode space is

$$H^0(\mathbb{CP}^1, \text{Hom}(\mathcal{O}(0), \mathcal{O}(2j))) = H^0(\mathbb{CP}^1, \mathcal{O}(2j)) = V_j.$$

Thus the representation space for the auxiliary is the same Borel–Weil sector used by the current map. The D-brane charge formula uses a Chan–Paton bundle with connection on the wrapped worldvolume [6], and Payen's boundary action quantizes the endpoint degree into the representation V_j [9]. For the two-line weak interface set

$$E_0 = \mathcal{O}(0), \quad E_j = \mathcal{O}(2j), \quad \mathcal{E}_j = \text{Hom}(E_0, E_j) \simeq \mathcal{O}(2j).$$

The Hermitian metrics on E_0 and E_j induce one Hermitian metric on $\mathcal{E}_j$, hence on

$$V_j = H^0(\mathbb{CP}^1, \mathcal{E}_j).$$

Thus

$$y, \quad t, \quad A_j^\dagger n \in V_j$$

are measured by the same boundary-complex norm. The physical coefficient is the common worldvolume normalization multiplying this norm. For the neutral anchor $j = 1$, this boundary-complex metric is tied to the finite weak doublet metric by the line-square map. Let $L = \mathcal{O}(1)$. Then

$$V_{\frac{1}{2}} = H^0(\mathbb{CP}^1, L), \quad V_1 = H^0(\mathbb{CP}^1, L^{\otimes 2}) \simeq \text{Sym}^2 V_{\frac{1}{2}}.$$

If the Q -neutral section $\eta_0 \in V_{\frac{1}{2}}$ has unit norm, the induced tensor-square metric gives

$$\|\eta_0 \odot \eta_0\|_{V_1} = \|\eta_0\|_{V_{\frac{1}{2}}}^2 = 1.$$

Thus the $j = 1$ Schur scale anchor can use the same finite-section normalization that defines $v_{\text{EW}}^2 = Z_0 I_0 v_b^2$. Writing $Z_{\mathcal{E},1}$ for the localized line-square boundary-complex norm, the source-matched coefficient is

$$\mu_\Gamma^2 = \frac{C^2}{4X_+(2)} Z_{\mathcal{E},1} v_b^2 G_{\text{EW}}, \quad Z_{\mathcal{E},1} = Z_0 I_0.$$

In a standard spin basis, the Q -neutral choice

$$\eta_0 = |-\tfrac{1}{2}\rangle \in V_{\frac{1}{2}}$$

has tensor square

$$\eta_0 \odot \eta_0 = |-1\rangle \in V_1.$$

Its charges are

$$T_3(\eta_0 \odot \eta_0) = -(\eta_0 \odot \eta_0), \quad Y(\eta_0 \odot \eta_0) = \eta_0 \odot \eta_0,$$

so

$$Q(\eta_0 \odot \eta_0) = 0.$$

Consequently $\eta_0 \odot \eta_0$ is basic for the S_Q^1 boundary action and descends to the Γ_{EW} boundary. The spin-one Casimir then gives

$$\sum_a T_a^{(1)} T_a^{(1)} |-1\rangle = 2 |-1\rangle,$$

so the neutral Schur anchor carries $J = 2$ from the symmetric square of the same finite section that supplies the electroweak source tensor.

The scale equality is now a zero-mode normalization test. In the Hilbert normalization of the boundary zero modes,

$$\|\widehat{\eta}_0\|_{V_{\frac{1}{2}}} = 1, \quad \|\widehat{\eta}_0 \odot \widehat{\eta}_0\|_{\text{Sym}^2 V_{\frac{1}{2}}} = 1,$$

so the induced neutral-anchor coefficient is

$$Z_{\mathcal{E},1}^{\text{zm}} = Z_0 I_0, \quad \chi_\Gamma^{\text{zm}} = 1.$$

A pointwise tensor-profile reading gives a distinct diagnostic. With

$$I_2 = \int_{\Gamma_{\text{EW},\text{int}}} \sqrt{\gamma_\Gamma} |\eta_0|^4,$$

the same local density gives

$$Z_{\mathcal{E},1}^{\text{pt}} = Z_0 I_2, \quad \chi_\Gamma^{\text{pt}} = \frac{I_2}{I_0}.$$

Thus source matching selects the zero-mode Hilbert tensor metric, or a localized profile identity $I_2 = I_0$. The Payen boundary construction supports the zero-mode branch: the endpoint variable quantizes to a finite-dimensional Hilbert space carrying R , and the boundary path integral is the Wilson trace [9]. With

$$R = V_j = H^0(\mathbb{CP}^1, \mathcal{O}(2j)),$$

the line-square anchor is a tensor inside the endpoint Hilbert representation. The pointwise I_2/I_0 branch is a separate DBI profile assumption for a composite line-square field. Payen's non-oriented extension adds an orientifold compatibility check: the endpoint representation must be equivalent to its dual [9]. For $\text{SU}(2)$,

$$V_j^* \simeq V_j,$$

with integer j real and half-integer j pseudoreal. Hence the retained $j = \frac{1}{2}$ doublet and $j = 1$ triplet both pass the D8/O8 endpoint reality test. Endpoint reality is a survival test. The finite retained set

$$j = 0, \frac{1}{2}, 1$$

still comes from the line-bundle or level-two selection.

Proposition 2 (Zero-mode scale pass). *Assume that the localized Γ_{EW} boundary action uses the Payen endpoint Hilbert representation $R = V_j$, tensors the Q -neutral line inside $V_{\frac{1}{2}}$, and multiplies the finite source and neutral Schur anchor by the same D8/open-boundary density. Then*

$$Z_{\mathcal{E},1} = Z_0 I_0, \quad \chi_\Gamma = 1,$$

and

$$\mu_\Gamma^2 = \frac{C^2}{4X_+(2)} Z_0 I_0 v_b^2 G_{\text{EW}} = \mu_*^2.$$

Proof. The zero-mode tensor metric gives $\|\widehat{\eta}_0 \odot \widehat{\eta}_0\| = 1$. The shared D8/open-boundary density gives $Z_{\mathcal{E},1} = Z_0 I_0$. Substitution in $\chi_\Gamma = Z_{\mathcal{E},1}/(Z_0 I_0)$ gives $\chi_\Gamma = 1$, and the definition of μ_Γ^2 gives the stated scale. $\square$

The shared-density form can be written explicitly as

$$d\nu_\Gamma = T_8(2\pi\alpha')^2 e^{-\phi} \sqrt{\gamma_\Gamma} \mathcal{W}_\Gamma d^5\sigma,$$

where $\mathcal{W}_\Gamma$ collects the frame, orientifold-image, and reduction factors of the localized action. The finite source coefficient is

$$Z_0 I_0 = \int_{\Gamma_{\text{EW},\text{int}}} d\nu_\Gamma |\eta_0|^2.$$

For the endpoint Hilbert embedding,

$$Z_{\mathcal{E},1}^{\text{zm}} = \int_{\Gamma_{\text{EW},\text{int}}} d\nu_\Gamma |\eta_0|^2 \|\widehat{\eta}_0 \odot \widehat{\eta}_0\|^2 = Z_0 I_0.$$

Thus the coefficient datum is the same $d\nu_\Gamma$ in the finite source term and in the Payen endpoint Hilbert action. Consequently, for any modulus m whose dependence in both coefficients is carried by the same $d\nu_\Gamma$,

$$\partial_m \log Z_{\mathcal{E},1}^{\text{zm}} = \partial_m \log(Z_0 I_0), \quad \partial_m \log \chi_\Gamma^{\text{zm}} = 0.$$

Frame, orientifold-image, and reduction factors affect the common scale $Z_0 I_0$ and cancel from the scale-pass ratio. At fixed shape and finite profile, the D8-like density carries the source scaling

$$Z_0 I_0 \propto \rho \tau^{-3}, \quad G_{\text{EW}} \propto \rho^{-1}.$$

Hence the Schur scale has

$$\mu_\Gamma^2 \propto Z_0 I_0 G_{\text{EW}} \propto \tau^{-3}, \quad \partial_\rho \log(Z_0 I_0 G_{\text{EW}})|_{\tau,\lambda} = 0.$$

Along $\rho = c_\rho \ell^2$ and $\tau = \tau_*(\ell/\ell_*)^s$,

$$\partial_{\log \ell} \log \mu_\Gamma^2 = -3s.$$

For $J = j(j+1) > 0$, the image projector is fixed by the Casimir identity:

$$A_j^\dagger A_j = J \mathbf{1}_{V_j}, \quad P_{\text{im}}^{(j)} = \frac{1}{J} A_j A_j^\dagger.$$

The current codomain decomposes under the diagonal $SU(2)$ action as

$$V_j \otimes V_1 = V_{j+1} \oplus V_j \oplus V_{j-1},$$

with the absent negative-spin summand omitted. The map A_j is an equivariant embedding of V_j into the middle summand. Thus

$$n_{\text{im}} = P_{\text{im}}^{(j)} n, \quad n_{\perp} = (1 - P_{\text{im}}^{(j)}) n$$

gives a canonical image/orthogonal split. For $j = 0$, $A_0 = 0$ gives the photon slot. Equivalently, with

$$\mathcal{B}_j = \sum_a T_a^{(j)} \otimes S_a^{(1)}$$

on $V_j \otimes V_1$, the diagonal Casimir gives eigenvalues

$$j, \quad -1, \quad -(j+1)$$

on V_{j+1} , V_j , and V_{j-1} , respectively. Hence

$$P_{\text{im}}^{(j)} = -\frac{(\mathcal{B}_j - j)(\mathcal{B}_j + j + 1)}{j(j+1)}.$$

The complementary projector is

$$P_{\perp}^{(j)} = 1 - P_{\text{im}}^{(j)} = \frac{\mathcal{B}_j(\mathcal{B}_j + 1)}{j(j+1)}.$$

It has eigenvalue zero on the image summand and eigenvalue one on the orthogonal summands. Thus $M_{\perp}^2 \langle n, P_{\perp}^{(j)} n \rangle$ is a canonical local gap term for $M_{\perp}^2 > 0$. Equivalently, let $\Delta_{\Gamma,j}$ be the self-adjoint Laplacian of the localized boundary complex whose zero modes are the image summand. Compactness of the internal boundary image gives

$$\lambda_{\perp,j} = \inf_{\substack{n \in P_{\perp}^{(j)}(V_j \otimes V_1) \\ \|n\|=1}} \langle n, \Delta_{\Gamma,j} n \rangle > 0.$$

The complement gap can then be written as $M_{\perp}^2 = \lambda_{\perp,j}$ up to the common boundary kinetic normalization. With the shared density, the finite projector acts fibrewise:

$$\mathcal{H}_{\Gamma,j} = L^2(\Gamma_{\text{EW,int}}, d\nu_{\Gamma}) \otimes (V_j \otimes V_1),$$

$$\langle n, P_{\perp}^{(j)} n \rangle_{\Gamma} = \int_{\Gamma_{\text{EW,int}}} d\nu_{\Gamma} \langle n(\sigma), P_{\perp}^{(j)} n(\sigma) \rangle_{V_j \otimes V_1}.$$

All frame, orientifold-image, and reduction factors stay in $d\nu_{\Gamma}$. If

$$\ker \Delta_{\Gamma,j} = \text{Im } A_j,$$

then

$$M_{\perp,j}^2 = \inf_{\substack{n \in P_{\perp}^{(j)} \mathcal{H}_{\Gamma,j} \\ \|n\|_{\Gamma}=1}} \langle n, \Delta_{\Gamma,j} n \rangle_{\Gamma} > 0.$$

The residual dimensionless number is

$$r_{\perp,j} = \frac{M_{\perp,j}^2}{\mu_{\Gamma}^2}.$$

It leaves the image-block de Vries eigenvalue unchanged and assigns the complement datum to the localized boundary complex.

Let

$$t \in V_j, \quad n = n_{\text{im}} + n_{\perp}, \quad n_{\text{im}} \in \text{Im } A_j, \quad y \in V_j.$$

The dimensionful local auxiliary form is

$$S_{\Gamma,j}^{(2)} = \mu_{\Gamma}^2 \left[\|t\|^2 + \|y\|^2 + \langle y, t - A_j^{\dagger} n \rangle + \langle t - A_j^{\dagger} n, y \rangle \right] + M_{\perp}^2 \langle n, P_{\perp}^{(j)} n \rangle$$

eliminates y and leaves

$$\mu_{\Gamma}^2 \left[\langle t, A_j^{\dagger} n \rangle + \langle n, A_j t \rangle - \langle n, A_j A_j^{\dagger} n \rangle \right] + M_{\perp}^2 \langle n, P_{\perp}^{(j)} n \rangle.$$

Equivalently, setting $s = t - A_j^{\dagger} n$, the y -dependent part is

$$\mu_{\Gamma}^2 \left[\|t\|^2 + \|y\|^2 + \langle y, s \rangle + \langle s, y \rangle \right] = \mu_{\Gamma}^2 \left[\|y + s\|^2 + \|t\|^2 - \|s\|^2 \right].$$

Thus the equal-coefficient test is local: the D8 boundary complex provides the same Hermitian metric for y , for $t - A_j^{\dagger} n$, and for the endpoint field t . At the selected stationary point, the source-to-Schur scale equation is

$$\mu_{\Gamma}^2 = \mu_*^2 = \frac{C^2 v_{\text{EW}}^2}{4X_+(2)} G_{\text{EW}}.$$

Since $X_+(2) = \sqrt{3} - 1$, equivalently

$$\mu_{\Gamma}^2 = \frac{\sqrt{3} + 1}{2} M_Z^2.$$

In boundary-profile variables, $v_{\text{EW}}^2 = Z_0 I_0 v_b^2$ gives

$$\mu_{\Gamma}^2 = \frac{C^2}{4X_+(2)} Z_0 I_0 v_b^2 G_{\text{EW}}.$$

This boundary Schur term supplies the negative diagonal in the projected block on $V_j \oplus \text{Im } A_j$. The orthogonal image complement is gapped by $M_{\perp}^2$.

7 RR/Current Boundary Length

The weak coupling normalization follows Weinberg's rms-circumference rule,

$$g^2 = \frac{4\pi^2 \kappa^2}{L^2}, \quad \kappa^2 = 16\pi G.$$

The D10 current-frame normalization is the retained weak length from the localized current frame on Γ_{EW} . Write

$$\Pi_{\text{EW}} \iota_{\Gamma}^* C_1 = q_{\Gamma} \theta_{\Gamma}^3, \quad \Pi_{\text{EW}} \iota_{\Gamma}^* F_2 = q_{\Gamma} d_{\Gamma} \theta_{\Gamma}^3.$$

The $\text{SU}(2)$ -equivariant completion gives the triplet θ_{Γ}^a . The boundary-current length matrix is

$$K_{ab}^{\Gamma} = \int_{\Gamma_{\text{EW},\text{int}}} d\nu_{\Gamma} \gamma_{\Gamma}^{-1}(\theta_{\Gamma}^a, \theta_{\Gamma}^b).$$

The unit current metric condition is the D10 equation

$$K_{ab}^{\Gamma} = L_{2,\Gamma}^2 \delta_{ab}.$$

A useful split is

$$L_{2,\Gamma}^2 = L_{2,\text{horiz}}^2 + L_{2,\text{vert}}^2,$$

with the horizontal term supplied by the $\mathbb{CP}^1$ weak orbit and the vertical term supplied by the localized C_1, F_2 current channel. On a round $\mathbb{CP}^1 \simeq S^2$ weak factor of radius R_w ,

$$K_a = x \times e_a, \quad |K_a|^2 = R_w^2 - x_a^2,$$

and the sphere average gives

$$\frac{1}{\text{Vol}(\mathbb{CP}^1)} \int_{\mathbb{CP}^1} \sqrt{g} g(K_a, K_b) = \frac{2}{3} R_w^2 \delta_{ab}.$$

Thus

$$L_{2,\text{horiz}}^2 = \frac{2}{3} \ell^2 \lambda R_0^2.$$

The vertical term is the localized RR/current integral

$$L_{2,\text{vert}}^2 \delta_{ab} = \int_{\Gamma_{\text{EW},\text{int}}} d\nu_\Gamma \gamma_\Gamma^{-1}(\theta_{\Gamma,\text{vert}}^a, \theta_{\Gamma,\text{vert}}^b).$$

Equivalently, let $\tilde{K}_a$ be the retained boundary-current lift whose horizontal part is the $\mathbb{CP}^1$ rotation field K_a , with vertical part supplied by the localized C_1, F_2 current channel. The averaged metric condition is

$$\text{Av}_\Gamma g(\tilde{K}_a, \tilde{K}_b) = (L_{2,\text{horiz}}^2 + L_{2,\text{vert}}^2) \delta_{ab} = L_{2,\Gamma}^2 \delta_{ab}.$$

Thus the current frame used in the Schur operator is

$$e_a = \frac{\tilde{K}_a}{L_{2,\Gamma}}, \quad A_j = \sum_a T_a^{(j)} \otimes e_a, \quad A_j^\dagger A_j = j(j+1) \mathbf{1}.$$

The same density $d\nu_\Gamma$ appears in the finite source coefficient and in the Payen endpoint Hilbert coefficient, so the vertical length and the Schur scale are computed inside one boundary action.

7.1 Neutral Matrix Extraction

Set

$$\Theta_\Gamma = \begin{pmatrix} \theta_\Gamma^3 \\ \theta_\Gamma^Y \end{pmatrix}, \quad I_{AB}^\Gamma = \int_{\Gamma_{\text{EW},\text{int}}} d\nu_\Gamma \gamma_\Gamma^{-1}(\theta_\Gamma^A, \theta_\Gamma^B), \quad A, B \in \{3, Y\}.$$

The localized action extracts the neutral length matrix as

$$\mathcal{L}_\Gamma = \begin{pmatrix} I_{33}^\Gamma & I_{3Y}^\Gamma \\ I_{3Y}^\Gamma & I_{YY}^\Gamma \end{pmatrix},$$

so

$$L_2^2 = I_{33}^\Gamma, \quad L_Y^2 = I_{YY}^\Gamma, \quad \epsilon_\Gamma = I_{3Y}^\Gamma.$$

Equivalently,

$$u = \frac{I_{33}^\Gamma}{I_{YY}^\Gamma}, \quad \chi_\Gamma = \frac{I_{3Y}^\Gamma}{\sqrt{I_{33}^\Gamma I_{YY}^\Gamma}}, \quad D_\Gamma = u + 1 + 2\chi_\Gamma \sqrt{u}.$$

The common current scale is

$$\Lambda_\Gamma = \frac{C^2}{I_{YY}^\Gamma}.$$

The local boundary calculation then gives

$$\alpha_\Gamma = \frac{\Lambda_\Gamma}{4\pi D_\Gamma}, \quad M_W^2 = \frac{\Lambda_\Gamma v_{\text{EW}}^2}{4u}, \quad M_Z^2 = \frac{\Lambda_\Gamma v_{\text{EW}}^2 D_\Gamma}{4u(1 - \chi_\Gamma^2)}.$$

The de Vries pass condition becomes the integral constraint

$$R_{\text{dV}} = \frac{I_{33}^\Gamma I_{YY}^\Gamma - (I_{3Y}^\Gamma)^2}{I_{33}^\Gamma (I_{33}^\Gamma + I_{YY}^\Gamma + 2I_{3Y}^\Gamma)}.$$

The D10 task has therefore been reduced to constructing $\theta_\Gamma^3, \theta_\Gamma^Y, d\nu_\Gamma$ from the localized DBI/CS boundary system and evaluating these three current integrals.

A rank-one specialization gives a viability test. If

$$\theta_\Gamma^Y = c \theta_\Gamma^3$$

in the retained current Hilbert space, then

$$I_{YY}^\Gamma = c^2 I_{33}^\Gamma, \quad I_{3Y}^\Gamma = c I_{33}^\Gamma, \quad \det \mathcal{L}_\Gamma = 0.$$

The neutral coefficient $G_Z^\Gamma = L_Q^2 / \det \mathcal{L}_\Gamma$ lacks a finite value. Thus the S_Q^1 factor supplies photon protection, and a viable neutral matrix requires a second current direction from the $\mathbb{CP}^2$, Chan–Paton, or boundary-complex sector.

A rank-two normal form makes this requirement explicit. Let e_Q, e_N be orthonormal retained current directions, with e_Q the Q -protected photon direction, and write

$$\theta_\Gamma^3 = a_\Gamma e_Q + b_\Gamma e_N, \quad \theta_\Gamma^Y = c_\Gamma e_Q - b_\Gamma e_N.$$

Then

$$\theta_\Gamma^Q = \theta_\Gamma^3 + \theta_\Gamma^Y = (a_\Gamma + c_\Gamma) e_Q,$$

so the massive-neutral direction cancels from the photon current. The integrals are

$$I_{33}^\Gamma = a_\Gamma^2 + b_\Gamma^2, \quad I_{YY}^\Gamma = c_\Gamma^2 + b_\Gamma^2, \quad I_{3Y}^\Gamma = a_\Gamma c_\Gamma - b_\Gamma^2.$$

Consequently

$$L_Q^2 = (a_\Gamma + c_\Gamma)^2, \quad \det \mathcal{L}_\Gamma = b_\Gamma^2 (a_\Gamma + c_\Gamma)^2,$$

and

$$G_Z^\Gamma = \frac{1}{b_\Gamma^2}, \quad G_W^\Gamma = \frac{1}{a_\Gamma^2 + b_\Gamma^2}.$$

The de Vries pass condition is therefore the current-splitting condition

$$R_{\text{dV}} = \frac{M_W^2}{M_Z^2} = \frac{b_\Gamma^2}{a_\Gamma^2 + b_\Gamma^2},$$

or

$$b_\Gamma^2 = R_{\text{dV}} L_2^2, \quad a_\Gamma^2 = (1 - R_{\text{dV}}) L_2^2, \quad \frac{|b_\Gamma|}{|a_\Gamma|} = \sqrt{\frac{R_{\text{dV}}}{1 - R_{\text{dV}}}} = 1.866083698 \dots$$

The neutral datum is $L_Q = |a_\Gamma + c_\Gamma|$, equivalently

$$c_\Gamma = \sigma_Q L_Q - a_\Gamma, \quad \sigma_Q = \pm 1.$$

The second direction has a canonical boundary-line source. The combined weak-hypercharge frame bundle carries the neutral character

$$\chi_N(u_T, u_Y) = u_T^{-1} u_Y, \quad L_N = L_{T_3}^{-1} \otimes L_Y,$$

whose connection is represented by

$$\theta_\Gamma^N = \theta_\Gamma^Y - \theta_\Gamma^3.$$

For the neutral finite section,

$$T_3\eta_0 = -\frac{1}{2}\eta_0, \quad Y\eta_0 = \frac{1}{2}\eta_0,$$

and therefore

$$(T_3\theta_\Gamma^3 + Y\theta_\Gamma^Y)\eta_0 = \frac{1}{2}\theta_\Gamma^N\eta_0.$$

The massive-neutral current is the current-metric projection of this neutral line away from the photon direction:

$$P_{Q^\perp}\omega = \omega - \frac{\langle \omega, \theta_\Gamma^Q \rangle_\Gamma}{\langle \theta_\Gamma^Q, \theta_\Gamma^Q \rangle_\Gamma} \theta_\Gamma^Q, \quad e_N = -\frac{P_{Q^\perp}\theta_\Gamma^N}{\|P_{Q^\perp}\theta_\Gamma^N\|_\Gamma}.$$

The projection has a matrix expression:

$$\|P_{Q^\perp}\theta_\Gamma^N\|_\Gamma^2 = \|\theta_\Gamma^N\|_\Gamma^2 - \frac{\langle \theta_\Gamma^N, \theta_\Gamma^Q \rangle_\Gamma^2}{\|\theta_\Gamma^Q\|_\Gamma^2} = \frac{4 \det \mathcal{L}_\Gamma}{L_Q^2}.$$

The rank-two normal form gives

$$P_{Q^\perp}\theta_\Gamma^N = -2b_\Gamma e_N, \quad \|P_{Q^\perp}\theta_\Gamma^N\|_\Gamma^2 = 4b_\Gamma^2.$$

Thus the de Vries pass becomes the projected neutral-line norm condition

$$\|P_{Q^\perp}\theta_\Gamma^N\|_\Gamma^2 = 4R_{\text{dV}}L_2^2.$$

Equivalently,

$$\det \mathcal{L}_\Gamma = R_{\text{dV}}L_2^2L_Q^2.$$

This replaces the abstract second-current condition by concrete D10 data: L_N , its projected boundary norm using $d\nu_\Gamma$, and the comparison with the weak-current norm.

The associated line-lattice data are

$$c_1(L_Y) = 6h + 7\eta, \quad c_1(L_{T_3}) = \eta,$$

so

$$c_1(L_N) = c_1(L_Y) - c_1(L_{T_3}) = 6(h + \eta).$$

The topological class fixes the neutral line; the physical condition is the projected norm

$$\mathcal{N}_N^\Gamma = \int_{\Gamma_{\text{EW,int}}} d\nu_\Gamma \gamma_\Gamma^{-1} (P_{Q^\perp}\theta_\Gamma^N, P_{Q^\perp}\theta_\Gamma^N),$$

with

$$\frac{\mathcal{N}_N^\Gamma}{4L_2^2} = R_{\text{dV}} = 0.776898678 \dots$$

Thus the diagonal six-unit line class supplies the neutral line class, and the localized DBI/CS action supplies the current metric whose projected norm gives the de Vries ratio.

Equivalently, let ω_h, ω_η be the localized boundary representatives of h, η and define

$$H_{ij}^\Gamma = \int_{\Gamma_{\text{EW,int}}} d\nu_\Gamma \gamma_\Gamma^{-1}(\omega_i, \omega_j), \quad i, j \in \{h, \eta\}.$$

For

$$\omega_T = \omega_\eta, \quad \omega_Y = 6\omega_h + 7\omega_\eta,$$

the neutral and photon representatives are

$$\omega_N = 6(\omega_h + \omega_\eta), \quad \omega_Q = 6\omega_h + 8\omega_\eta.$$

Hence

$$\mathcal{N}_N^\Gamma = \langle \omega_N, \omega_N \rangle_\Gamma - \frac{\langle \omega_N, \omega_Q \rangle_\Gamma^2}{\langle \omega_Q, \omega_Q \rangle_\Gamma},$$

with

$$\langle \omega_N, \omega_N \rangle_\Gamma = 36 (H_{hh}^\Gamma + 2H_{h\eta}^\Gamma + H_{\eta\eta}^\Gamma),$$

$$\langle \omega_N, \omega_Q \rangle_\Gamma = 36H_{hh}^\Gamma + 84H_{h\eta}^\Gamma + 48H_{\eta\eta}^\Gamma,$$

and

$$\langle \omega_Q, \omega_Q \rangle_\Gamma = 36H_{hh}^\Gamma + 96H_{h\eta}^\Gamma + 64H_{\eta\eta}^\Gamma.$$

The neutral-line pass becomes a boundary Hodge-matrix equation:

$$\mathcal{N}_N^\Gamma = 4R_{\text{dV}}L_2^2.$$

Writing

$$A = H_{hh}^\Gamma, \quad B = H_{h\eta}^\Gamma, \quad C = H_{\eta\eta}^\Gamma,$$

the projection simplifies to

$$\mathcal{N}_N^\Gamma = \frac{36(AC - B^2)}{9A + 24B + 16C}.$$

Thus the localized pass equation is

$$\frac{9(AC - B^2)}{9A + 24B + 16C} = R_{\text{dV}}L_2^2.$$

The weak-current representative $\omega_T = \omega_\eta$ gives a useful specialization:

$$L_2^2 = C = H_{\eta\eta}^\Gamma.$$

With

$$r_H = \frac{A}{C}, \quad s_H = \frac{B}{C},$$

the pass condition is

$$\frac{9(r_H - s_H^2)}{9r_H + 24s_H + 16} = R_{\text{dV}},$$

so

$$r_H(s_H) = \frac{9s_H^2 + 24R_{\text{dV}}s_H + 16R_{\text{dV}}}{9(1 - R_{\text{dV}})}.$$

Along this curve,

$$r_H - s_H^2 = \frac{R_{\text{dV}}(3s_H + 4)^2}{9(1 - R_{\text{dV}})}, \quad 9r_H + 24s_H + 16 = \frac{(3s_H + 4)^2}{1 - R_{\text{dV}}}.$$

The diagonal Hodge case gives the concrete target

$$s_H = 0, \quad \frac{H_{hh}^\Gamma}{H_{\eta\eta}^\Gamma} = \frac{16R_{\text{dV}}}{9(1 - R_{\text{dV}})} = 6.190\dots$$

For comparison, the bulk KLT harmonic representatives

$$h = x + y, \quad \eta_{\text{harm}} = -\frac{1}{2}x + \frac{1}{2}y + z$$

with $x = e^{12}$, $y = e^{34}$, $z = e^{56}$, have weights

$$\|x\|^2 = \|y\|^2 \propto \lambda^{1/2}, \quad \|z\|^2 \propto \lambda^{-3/2}.$$

They give

$$A_{\text{bulk}} = 2\lambda^{1/2}, \quad B_{\text{bulk}} = 0, \quad C_{\text{bulk}} = \frac{1}{2}\lambda^{1/2} + \lambda^{-3/2},$$

so

$$r_H^{\text{bulk}}(\lambda) = \frac{4\lambda^2}{\lambda^2 + 2}.$$

On $2/5 \leq \lambda \leq 2$, this ranges from $0.296\dots$ to $2.666\dots$. The boundary-current pass therefore requires either a localized boundary Hodge measure or a weak-current frame factor beyond the unlocalized KLT harmonic metric. If

$$L_2^2 = \kappa_H C_{\text{bulk}},$$

then

$$\kappa_H(\lambda) = \frac{9\lambda^2}{R_{\text{dV}}(13\lambda^2 + 8)}.$$

At $\lambda = 0.514554893751227$, one obtains $\kappa_H = 0.268\dots$

A localized representative can also shift the horizontal component:

$$\eta(t) = t(x + y) + y + z, \quad q = \lambda^2.$$

For the same KLT weights,

$$A = 2q, \quad B = (2t + 1)q, \quad C = (2t^2 + 2t + 1)q + 1,$$

so

$$AC - B^2 = q(q + 2),$$

and the weak-current identification gives

$$9q(q + 2) = 2R_{\text{dV}}(16qt^2 + 40qt + 29q + 8)(q(2t^2 + 2t + 1) + 1).$$

For example, $\lambda = 1.915$ has roots

$$t = -0.811\dots, \quad (r_H, s_H) = (2.070\dots, -0.644\dots),$$

and

$$t = -0.939\dots, \quad (r_H, s_H) = (1.727\dots, -0.758\dots).$$

The quartic can be centered by setting $s = t + 7/8$. It becomes

$$F(q, s) = F_0(q) - 2R_{\text{dV}}q(7q + 32)s^2 - 64R_{\text{dV}}q^2s^4,$$

where

$$F_0(q) = \frac{576q^2 + 1152q - R_{\text{dV}}(625q^2 + 1600q + 1024)}{64}.$$

Consequently real representatives occur for $F_0(q) \geq 0$. The positive threshold is

$$q_c = \frac{32(25R_{\text{dV}} + 18\sqrt{1 - R_{\text{dV}}} - 18)}{576 - 625R_{\text{dV}}} = 3.511614004347\dots, \quad \lambda_c = 1.873930095907\dots$$

For $q \geq q_c$, the roots are

$$t_{\pm} = -\frac{7}{8} \pm \sqrt{\frac{-(7q + 32) + \sqrt{(7q + 32)^2 + 64F_0(q)/R_{\text{dV}}}}{64q}}.$$

At $\lambda = 2$, this gives

$$t_- = -0.983371\dots, \quad t_+ = -0.766629\dots$$

The localized representative t is therefore a computable boundary datum near $t = -7/8$. With a weak-current frame factor $L_2^2 = \kappa_H C$, the same calculation gives

$$\kappa_H(q, t) = \frac{9q(q+2)}{2R_{\text{dV}}(16qt^2 + 40qt + 29q + 8)(q(2t^2 + 2t + 1) + 1)}.$$

The maximum over this representative family occurs at $t = -7/8$:

$$\kappa_H^{\text{max}}(q) = \frac{576q(q+2)}{R_{\text{dV}}(25q + 32)^2}.$$

Thus $\kappa_H^{\text{max}} = 1$ gives the same threshold q_c . Below λ_c , the neutral boundary condition requires a reduced weak-current frame factor. The center also has a boundary-orthogonality interpretation. For horizontal weights

$$\langle x, x \rangle_{\partial} = w_x, \quad \langle y, y \rangle_{\partial} = w_y, \quad \langle x, y \rangle_{\partial} = 0,$$

the condition $\langle \eta(t), h \rangle_{\partial} = 0$ gives

$$t = -\frac{w_y}{w_x + w_y}.$$

Hence $t = -7/8$ corresponds to a localized horizontal split

$$w_y : w_x = 7 : 1.$$

For the weak-period value $Q = 7$, this split is the special case of

$$w_y : w_x = Q : 1, \quad t_Q = -\frac{Q}{Q+1}.$$

Together with the primitive line-lattice relation $P = Q - 1$, the retained class

$$\zeta = Ph + Q\eta(t_Q)$$

has the form

$$\zeta = Q(y + z) - \frac{h}{Q+1}.$$

Thus for $Q = 7$,

$$\zeta = 7(y + z) - \frac{1}{8}h, \quad 8\zeta = 56(y + z) - h.$$

The centered representative therefore converts the weak period and the single weak $\mathcal{O}(1)$ unit into an eightfold boundary lift. The D8/O8 bookkeeping uses

$$N_8 = 2n, \quad q_8^{\text{net}} = N_8 - 16 = 2(n - 8), \quad k = n - 8.$$

Identifying the lift denominator with the Chan–Paton half-rank,

$$n = Q + 1,$$

gives, for $Q = 7$,

$$n = 8, \quad N_8 = 16, \quad q_8^{\text{net}} = 0, \quad k = 0.$$

The same integer that centers the representative therefore selects the locally neutral D8/O8 endpoint. Borel–Weil supplies a concrete source for the half-rank relation. The weak-period line has

$$L_Q|_{\mathbb{CP}^1} \simeq \mathcal{O}(Q),$$

and hence

$$H^0(\mathbb{CP}^1, \mathcal{O}(Q)) \simeq \text{Sym}^Q(\mathbb{C}^2)^*, \quad \dim H^0(\mathbb{CP}^1, \mathcal{O}(Q)) = Q + 1.$$

Taking the Chan–Paton half-stack to be this zero-mode space gives

$$n = \dim H^0(\mathbb{CP}^1, \mathcal{O}(Q)) = Q + 1.$$

For $Q = 7$,

$$N_8 = 2 \dim H^0(\mathbb{CP}^1, \mathcal{O}(7)) = 16.$$

The same line gives the horizontal weight split through point evaluation. Fix $p \in \mathbb{CP}^1$. Since the ideal sheaf of p is $\mathcal{O}(-1)$, tensoring by $\mathcal{O}(Q)$ gives

$$0 \longrightarrow \mathcal{O}(Q-1) \longrightarrow \mathcal{O}(Q) \xrightarrow{\text{ev}_p} \mathcal{O}(Q)|_p \longrightarrow 0.$$

For $Q \geq 1$, global sections form the exact sequence

$$0 \longrightarrow H^0(\mathbb{CP}^1, \mathcal{O}(Q-1)) \longrightarrow H^0(\mathbb{CP}^1, \mathcal{O}(Q)) \xrightarrow{\text{ev}_p} \mathcal{O}(Q)|_p \longrightarrow 0,$$

with

$$\dim \ker \text{ev}_p = Q, \quad \dim \text{im } \text{ev}_p = 1.$$

In coordinates $[u : v]$ with $p = [1 : 0]$,

$$H^0(\mathbb{CP}^1, \mathcal{O}(Q)) = \text{span}\{u^Q, u^{Q-1}v, \dots, v^Q\},$$

and ev_p selects the coefficient of u^Q . If P_p projects onto u^Q and $K_p = 1 - P_p$, the trace metric gives

$$\text{Tr } P_p = 1, \quad \text{Tr } K_p = Q.$$

The same projector gives the local current metric on the transverse weak plane. Let $J_Q = Q/2$, let $|0\rangle = |J_Q, J_Q\rangle$, and define

$$G_{ab}^{(p)} = \text{Re } \langle 0 | T_a K_p T_b | 0 \rangle.$$

With

$$T_1 = \frac{1}{2}(T_+ + T_-), \quad T_2 = \frac{1}{2i}(T_+ - T_-), \quad T_3|0\rangle = J_Q|0\rangle,$$

and $T_-|0\rangle = \sqrt{Q}|J_Q, J_Q - 1\rangle$, one obtains

$$G_{ab}^{(p)} = \frac{Q}{4} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

With common localized density factor f_Γ^2 , the transverse Hosotani normalization is

$$f_{H,\perp}^2 = f_\Gamma^2 \frac{Q}{4}.$$

Thus a holonomy scale μ_Γ gives

$$\alpha_{\text{mix}} = \frac{2\mu_\Gamma}{\sqrt{Q} f_\Gamma}, \quad \beta_{\text{diag}} = \frac{4\mu_\Gamma^2}{Q f_\Gamma^2}, \quad \frac{\beta_{\text{diag}}}{\alpha_{\text{mix}}^2} = 1.$$

The endpoint line also carries the neutral weak-hypercharge character. Write Q_w for the weak-period integer here and set

$$J_w = \frac{Q_w}{2}, \quad |0\rangle = |J_w, J_w\rangle, \quad Y|0\rangle = -J_w|0\rangle.$$

Then $Q_{\text{em}} = T_3 + Y$ satisfies $Q_{\text{em}}|0\rangle = 0$. For (T_3, Y) , the endpoint charge vector is $q = (J_w, -J_w)$, hence

$$N_{AB}^{(p)} = \langle 0|q_A q_B|0\rangle = \frac{Q_w^2}{4} \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix}.$$

The null vector is $(1, 1)$, the Q_{em} direction. The vector $(1, -1)$ has eigenvalue $Q_w^2/2$. This endpoint-line block is the rank-one neutral mass block; the gauge-kinetic neutral lengths are supplied by the localized current matrix. With gauge fields (W_μ^3, B_μ) , this block becomes

$$M_N^2 = v_\Gamma^2 J_w^2 \begin{pmatrix} g_2^2 & -g_2 g_Y \\ -g_2 g_Y & g_Y^2 \end{pmatrix}.$$

Thus

$$A_\mu = \frac{g_Y W_\mu^3 + g_2 B_\mu}{\sqrt{g_2^2 + g_Y^2}}, \quad Z_\mu = \frac{g_2 W_\mu^3 - g_Y B_\mu}{\sqrt{g_2^2 + g_Y^2}},$$

and

$$M_\gamma^2 = 0, \quad M_Z^2 = v_\Gamma^2 J_w^2 (g_2^2 + g_Y^2).$$

The transverse K_p metric gives

$$M_W^2 = v_\Gamma^2 \frac{Q_w}{4} g_2^2.$$

For $Q_w = 1$, the $\mathcal{O}(1)$ doublet gives

$$M_W^2 = \frac{g_2^2 v_\Gamma^2}{4}, \quad M_Z^2 = \frac{(g_2^2 + g_Y^2) v_\Gamma^2}{4}.$$

The Connes finite-line amplitude satisfies

$$|v_F| = \frac{v_{\text{EW}}}{\sqrt{2}},$$

so the endpoint gauge-mass convention is

$$v_\Gamma = v_{\text{EW}} = \sqrt{2} |v_F|.$$

Equivalently,

$$M_W^2 = \frac{g_2^2 |v_F|^2}{2}, \quad M_Z^2 = \frac{(g_2^2 + g_Y^2) |v_F|^2}{2}.$$

The $Q_w = 7$ period stack supplies the boundary filtration, half-rank, and endpoint metric factors; electroweak mass normalization uses the $\mathcal{O}(1)$ doublet together with the localized current lengths. The endpoint convention then gives the localized Schur scale target

$$\mu_{\Gamma, \text{Schur}}^2 = \mu_*^2 = \frac{M_Z^2}{X_+(2)} = \frac{g_2^2 + g_Y^2}{4X_+(2)} v_\Gamma^2 = \frac{g_2^2 + g_Y^2}{2X_+(2)} |v_F|^2.$$

For a stationary Hosotani scalar with

$$f_H \theta_{H,*} = |v_F| = \frac{v_{\text{EW}}}{\sqrt{2}}, \quad v_\Gamma = \sqrt{2} f_H \theta_{H,*},$$

this becomes

$$\mu_{\Gamma, \text{Schur}}^2 = \frac{g_2^2 + g_Y^2}{2X_+(2)} f_H^2 \theta_{H,*}^2.$$

Thus the D10 matching datum is the dimensionless scale-ratio coordinate

$$\chi_H = \frac{2X_+(2) \mu_{\Gamma, \text{Schur}}^2}{(g_2^2 + g_Y^2) f_H^2 \theta_{H,*}^2} = 1.$$

With the point-projector normalization $f_H^2 = Q f_\Gamma^2/4$, the same target is

$$\mu_{\Gamma, \text{Schur}}^2 = \frac{Q(g_2^2 + g_Y^2)}{8X_+(2)} f_\Gamma^2 \theta_{H,*}^2.$$

Writing $x = w_H \theta_H$, the minimal localized Wilson potential with an interior stationary value is

$$V_H(\theta_H) = c_1 \cos x + c_2 \cos(2x).$$

Its stationary equation is

$$\partial_{\theta_H} V_H = -w_H \sin x [c_1 + 4c_2 \cos x],$$

so the interior branch satisfies

$$\cos x_* = -\frac{c_1}{4c_2}, \quad c_2 > 0, \quad |c_1| < 4c_2.$$

Matching the finite-sheet value

$$x_* = \frac{w_H v_{\text{EW}}}{\sqrt{2} f_H}$$

fixes the Fourier ratio

$$\frac{c_1}{c_2} = -4 \cos\left(\frac{w_H v_{\text{EW}}}{\sqrt{2} f_H}\right).$$

The canonical scalar curvature from this Wilson sector is

$$m_\theta^2 = \frac{1}{f_H^2} \partial_{\theta_H}^2 V_H|_{\theta_{H,*}} = \frac{4w_H^2 c_2}{f_H^2} \sin^2 x_*.$$

The full-curvature specialization $m_\theta^2 = M_H^2$ gives

$$c_2 = \frac{M_H^2 f_H^2}{4w_H^2 \sin^2 x_*}, \quad c_1 = -\frac{M_H^2 f_H^2 \cos x_*}{w_H^2 \sin^2 x_*}.$$

With $M_H^2 = 2\lambda_F v_{\text{EW}}^2$, the same coefficients are

$$c_2 = \frac{\lambda_F v_{\text{EW}}^2 f_H^2}{2w_H^2 \sin^2 x_*}, \quad c_1 = -\frac{2\lambda_F v_{\text{EW}}^2 f_H^2 \cos x_*}{w_H^2 \sin^2 x_*}.$$

Eliminating f_H by

$$f_H = \frac{w_H v_{\text{EW}}}{\sqrt{2} x_*}$$

gives the dimensionless coefficient target

$$\frac{c_2}{v_{\text{EW}}^4} = \frac{\lambda_F}{4x_*^2 \sin^2 x_*}, \quad \frac{c_1}{v_{\text{EW}}^4} = -\frac{\lambda_F \cos x_*}{x_*^2 \sin^2 x_*}.$$

A localized spectrum calculation of c_1/c_2 fixes

$$x_* = \arccos\left(-\frac{c_1}{4c_2}\right), \quad f_H = \frac{w_H v_{\text{EW}}}{\sqrt{2} x_*},$$

on the chosen $0 < x_* < \pi$ minimum branch. For the standard four-dimensional Coleman–Weinberg integral over one circle, a massless charge-one tower contributes

$$c_m^{(1)} = \frac{A_1}{m^5}.$$

Such a tower gives $c_1/c_2 = 32$. The interior minimum condition

$$\left| \frac{c_1}{c_2} \right| < 4$$

therefore demands additional charged sectors, massive thresholds, or localized boundary terms. With charge-one amplitude A_1 and charge-two amplitude A_2 at the fundamental period,

$$c_1 = A_1, \quad c_2 = \frac{A_1}{32} + A_2,$$

and the stationary electroweak angle fixes

$$\frac{A_2}{A_1} = -\frac{1}{4 \cos x_*} - \frac{1}{32}.$$

The local minimum condition $c_2 > 0$ gives two sign branches:

$$A_1 > 0 : \quad \frac{A_2}{A_1} > \frac{7}{32}, \quad \frac{\pi}{2} < x_* < \pi,$$

and

$$A_1 < 0 : \quad \frac{A_2}{A_1} < -\frac{9}{32}, \quad 0 < x_* < \frac{\pi}{2}.$$

More generally, let A_r be the total signed amplitude of massless towers with charge r under $x = w_H \theta_H$. Then

$$V_H(x) = \sum_{r \geq 1} A_r \sum_{m \geq 1} \frac{\cos(mrx)}{m^5} = \sum_{n \geq 1} c_n \cos(nx),$$

with divisor-sum coefficients

$$c_n = \frac{1}{n^5} \sum_{r|n} r^5 A_r.$$

Thus

$$c_1 = A_1, \quad c_2 = \frac{A_1}{32} + A_2, \quad c_3 = \frac{A_1}{243} + A_3, \quad c_4 = \frac{A_1}{1024} + \frac{A_2}{32} + A_4.$$

The two-harmonic approximation requires the higher harmonics to be small against $|c_1| + |c_2|$. Massive thresholds and localized boundary terms enter through the same harmonic bookkeeping with threshold-adjusted coefficients. In the Borel–Weil convention $H^0(\mathbb{CP}^1, \mathcal{O}(n)) \simeq V_{n/2}$, use integer current charge $q = 2m$. The charge set is

$$q = -n, -n+2, \dots, n-2, n.$$

Thus $\mathcal{O}(1)$ supplies $|q| = 1$, $\mathcal{O}(2)$ supplies $|q| = 2$, and $\mathcal{O}(7)$ supplies $|q| = 1, 3, 5, 7$. The physical doublet gives the charge-one tower. The active $\mathcal{O}(1) \oplus \mathcal{O}(2)$ charged set has $\gcd(1, 2) = 1$, and the $\mathcal{O}(7)$ stack has $\gcd(1, 3, 5, 7) = 1$. Thus

$$w_H = 1$$

in the integer current-charge convention. A global-form lift rescales the endpoint charge lattice by an explicit factor. The weak-period stack gives zero A_2 and higher odd harmonics. The minimal Borel–Weil second-harmonic source is

$$\mathcal{O}(2) \simeq \text{Sym}^2 \mathcal{O}(1),$$

with

$$A_1 > 0 : \quad A_2 > \frac{7}{32} A_1, \quad A_1 < 0 : \quad A_2 < -\frac{9}{32} A_1.$$

The line-square sector used for the $j = 1$ Schur anchor therefore supplies the Hosotani second harmonic in the minimal localized spectrum. Let the localized towers coupled to γ_H have integral charges q_α , one-loop signs σ_α , degeneracies d_α , and positive threshold weights τ_α . With common loop factor $\mathcal{C}_H > 0$,

$$A_r = \mathcal{C}_H \sum_{\alpha: |q_\alpha|=r} \sigma_\alpha d_\alpha \tau_\alpha.$$

For $\mathcal{O}(1) \oplus \mathcal{O}(2)$, write the signed charged weights as

$$S_1 = \sum_{\alpha \in \mathcal{O}(1), |q_\alpha|=1} \sigma_\alpha d_\alpha \tau_\alpha, \quad S_2 = \sum_{\alpha \in \mathcal{O}(2), |q_\alpha|=2} \sigma_\alpha d_\alpha \tau_\alpha.$$

Then $A_2/A_1 = S_2/S_1$, and the finite Hosotani minimum requires

$$S_1 > 0 : \quad \frac{S_2}{S_1} > \frac{7}{32}, \quad S_1 < 0 : \quad \frac{S_2}{S_1} < -\frac{9}{32}.$$

A same-sign charged-weight pass has $S_2 = S_1$. It gives

$$\frac{c_1}{c_2} = \frac{32}{33}, \quad \cos x_* = -\frac{8}{33}, \quad x_* = 1.815660181000518\dots,$$

and hence

$$\frac{f_H}{v_{\text{EW}}} = \frac{w_H}{\sqrt{2}x_*} = 0.389448856446748\dots w_H.$$

For $m_\theta^2 = M_H^2 = 2\lambda_F v_{\text{EW}}^2$,

$$\frac{c_2}{v_{\text{EW}}^4} = 0.080570282164285\dots \lambda_F, \quad \frac{c_1}{v_{\text{EW}}^4} = 0.078128758462337\dots \lambda_F.$$

The Borel–Weil weights occur in charge-conjugate pairs. If the localized real structure pairs $+r$ with $-r$ and preserves the threshold weight, then $S_r = 2\sigma_r d_r \tau_r$. The $\mathcal{O}(1)$ sector has $q = \pm 1$; the $\mathcal{O}(2)$ sector has $q = \pm 2$, with a θ_H -independent middle state at $q = 0$. Hence

$$\frac{S_2}{S_1} = \frac{\sigma_2 d_2 \tau_2}{\sigma_1 d_1 \tau_1} =: \chi_{12}.$$

The equal-weight pass is $\chi_{12} = 1$. More generally,

$$\sigma_1 d_1 \tau_1 > 0 : \quad \chi_{12} > \frac{7}{32}, \quad \sigma_1 d_1 \tau_1 < 0 : \quad \chi_{12} < -\frac{9}{32}.$$

For harmonic computations, split the threshold weight by wrapping number. For a flat local circle pass with charged mass M_r , set

$$z_r = M_r L_H, \quad F(z) = e^{-z} \left(1 + z + \frac{z^2}{3} \right).$$

The m -th wrapping of the charge- r pair carries $\tau_{r,m} = F(mz_r)$. With signed base pair weights

$$D_r = \mathcal{C}_H 2\sigma_r d_r, \quad \chi_{12}^{(0)} = \frac{D_2}{D_1},$$

the first two Fourier coefficients are

$$c_1 = D_1 F(z_1), \quad c_2 = \frac{D_1}{32} F(2z_1) + D_2 F(z_2).$$

Hence

$$\frac{c_2}{c_1} = \frac{F(2z_1)}{32F(z_1)} + \chi_{12}^{(0)} \frac{F(z_2)}{F(z_1)}.$$

For a chosen stationary angle,

$$\chi_{12}^{(0)} = \frac{F(z_1)}{F(z_2)} \left[-\frac{1}{4 \cos x_*} - \frac{F(2z_1)}{32F(z_1)} \right].$$

The equal signed-base pass gives

$$\cos x_* = -\frac{1}{4(F(2z_1)/(32F(z_1)) + F(z_2)/F(z_1))}.$$

At $z_1 = z_2 = 0$, the result is $\cos x_* = -8/33$. Since

$$F'(z) = -\frac{1}{3}e^{-z}z(1+z) \leq 0,$$

the equal signed-base pass with $D_1 > 0$ has a simple threshold domain:

$$F(z_2) > \frac{1}{4}F(z_1) - \frac{1}{32}F(2z_1).$$

For $z_1 = 0$, this gives

$$F(z_2) > \frac{7}{32}, \quad z_2 < 3.773046018411012 \dots$$

For common thresholds $z_2 = z_1$, the pass condition is automatic because

$$1 + \frac{F(2z_1)}{32F(z_1)} > \frac{1}{4}.$$

The Borel–Weil sectors themselves give a distinguished zero-mode pass. For $n \geq 0$,

$$H^0(\mathbb{CP}^1, \mathcal{O}(n)) = \ker \bar{\partial}_{\mathcal{O}(n)}.$$

Thus the charge pair $q = \pm 1$ in $\mathcal{O}(1)$ and the charge pair $q = \pm 2$ in $\mathcal{O}(2)$ lie in holomorphic zero-eigenvalue sectors. The localized determinant then has

$$z_1 = z_2 = 0, \quad F(z_1) = F(z_2) = F(2z_1) = 1.$$

The $q = 0$ middle state of $\mathcal{O}(2)$ is independent of θ_H . With common signed oscillator data,

$$\sigma_2 d_2 = \sigma_1 d_1,$$

one obtains

$$c_1 = D_1, \quad c_2 = \frac{33}{32}D_1, \quad \frac{c_1}{c_2} = \frac{32}{33}.$$

Therefore

$$\cos x_* = -\frac{8}{33}, \quad \frac{f_H}{v_{\text{EW}}} = 0.389448856446748 \dots w_H.$$

In homogeneous coordinates $[u : v]$ on $\mathbb{CP}^1$, the charged bases are

$$H^0(\mathbb{CP}^1, \mathcal{O}(1)) : \quad u \ (q = +1), \quad v \ (q = -1),$$

and

$$H^0(\mathbb{CP}^1, \mathcal{O}(2)) : \quad u^2 \ (q = +2), \quad uv \ (q = 0), \quad v^2 \ (q = -2).$$

Thus each charged pair has one positive-charge and one negative-charge complex mode, and the uv mode is neutral for the Wilson angle. For a common localized boundary complex $\mathcal{B}_\Gamma$ whose oscillator operator acts with the same sign and degeneracy on (u, v) and (u^2, v^2) ,

$$\sigma_2 = \sigma_1, \quad d_2 = d_1, \quad D_2 = D_1.$$

The holomorphic zero-mode pass then yields the coefficient ratio 32/33. Equivalently, define charged projectors by

$$P_1^\chi : H^0(\mathbb{CP}^1, \mathcal{O}(1)) \rightarrow \text{span}\{u, v\}, \quad P_2^\chi : H^0(\mathbb{CP}^1, \mathcal{O}(2)) \rightarrow \text{span}\{u^2, v^2\}.$$

Then

$$\text{Tr } P_1^\chi = \text{Tr } P_2^\chi = 2, \quad P_2^\chi = 1 - |uv\rangle\langle uv|.$$

For a common localized charged oscillator block

$$\mathcal{B}_\Gamma^\chi = B_0 \otimes P_r^\chi,$$

the signed base weights are

$$D_r = \mathcal{C}_H \sigma \text{Tr}_{P_r^\chi} \mathbf{1} = 2\mathcal{C}_H \sigma.$$

Thus

$$D_2 = D_1, \quad V_H(x) = D_1 \cos x + \frac{33}{32} D_1 \cos(2x).$$

The same equality can be expressed as an equivariant index calculation. For the S^1 action generated by $x = w_H \theta_H$,

$$\text{Ind}_{S^1} \bar{\partial}_{\mathcal{O}(n)}(x) = \sum_{k=0}^n \exp(i(n-2k)x), \quad n \geq 0,$$

because $H^1(\mathbb{CP}^1, \mathcal{O}(n)) = 0$. Consequently

$$I_1(x) = 2 \cos x, \quad I_2(x) = 1 + 2 \cos(2x).$$

The charged endpoint trace is

$$I_\chi(x) = I_1(x) + I_2(x) - 1 = 2 \cos x + 2 \cos(2x).$$

The 1 in I_2 is the Wilson-neutral uv mode. The charged character coefficients for the $\mathcal{O}(1)$ and $\mathcal{O}(2)$ sectors are equal. Payen's boundary path integral fixes the trace convention [9]: the discretized endpoint integral uses

$$\int N_R Dg R(g) v_\kappa v_\kappa^\dagger R^\dagger(g) = \mathbf{1}_{V_R},$$

and the external Yang–Mills coupling reduces to

$$\text{Tr}_{V_R} \mathcal{T}_{\partial\Sigma} \exp \left(i \int_{\partial\Sigma} d\tau R(A_\tau(\tau)) \right).$$

The endpoint determinant therefore uses the ordinary finite representation trace with no dimension divisor. In the charged Borel–Weil subspaces,

$$\text{Tr}_{P_1^\chi} \mathbf{1} = \text{Tr}_{P_2^\chi} \mathbf{1} = 2,$$

so the source-backed endpoint trace gives $d_1 = d_2 = d_\chi$ before oscillator thresholds. An endpoint determinant with common sign and localized density on these Dolbeault zero-mode sectors gives $A_1 = A_2$, hence

$$c_1 = A_1, \quad c_2 = \frac{A_1}{32} + A_2 = \frac{33}{32} A_1, \quad \frac{c_1}{c_2} = \frac{32}{33}.$$

In pair-normalized form this contribution is

$$V_{\text{BW}}(x) = D \sum_{m \geq 1} \frac{\cos(mx) + \cos(2mx)}{m^5},$$

so that

$$c_1 = D, \quad c_2 = \frac{33}{32}D.$$

At the two-harmonic level this gives $\cos x_* = -8/33$. The full massless character has stationary equation

$$\text{Im Li}_4(e^{ix}) + 2 \text{Im Li}_4(e^{2ix}) = 0,$$

with interior solution

$$x_{\text{full}} = 1.837672606049940 \dots, \quad \frac{f_H}{v_{\text{EW}}} = \frac{w_H}{\sqrt{2} x_{\text{full}}} = 0.384783872197163 \dots w_H.$$

At this point

$$K_{\text{full}} = -\text{Re Li}_3(e^{ix_{\text{full}}}) - 4\text{Re Li}_3(e^{2ix_{\text{full}}}) = 3.557941701817797 \dots,$$

and matching $m_\theta^2 = M_H^2 = 2\lambda_F v_{\text{EW}}^2$ gives

$$\frac{D}{v_{\text{EW}}^4} = \frac{\lambda_F}{x_{\text{full}}^2 K_{\text{full}}} = 0.083227124394645 \dots \lambda_F.$$

With a common threshold z , the same character gives

$$V_z(x) = D \sum_{m \geq 1} F(mz) \frac{\cos(mx) + \cos(2mx)}{m^5},$$

and the stationary equation

$$\sum_{m \geq 1} F(mz) \frac{\sin(mx) + 2 \sin(2mx)}{m^4} = 0.$$

The root moves from $x_0 = 1.837672606 \dots$ to

$$x_\infty = \arccos(-1/4) = 1.8234765819 \dots$$

as higher wrappings are suppressed. Correspondingly

$$\frac{f_H}{w_H v_{\text{EW}}} = 0.384783872 \dots \quad (z = 0), \quad 0.387779469 \dots \quad (z = \infty).$$

The corresponding D10 coefficient target is the determinant amplitude. Write

$$\mathcal{N}_\Gamma^\chi = \text{Str}_\chi \zeta_\alpha$$

for the signed charged endpoint supertrace after canonical CP2 and boundary zero-mode normalization. The weights ζ_α are dimensionless and may include threshold factors. For a tower with quadratic density Z_α ,

$$\frac{1}{2} \text{Tr} \log[Z_\alpha \Delta_\alpha(x)] = \frac{1}{2} \text{Tr} \log \Delta_\alpha(x) + \frac{1}{2} \text{Tr} \log Z_\alpha.$$

The last term is angle-independent and is absorbed into the vacuum constant. Thus the angle-dependent determinant uses the canonical signed supertrace. The DBI/open-boundary density enters f_H^2 and the threshold parameters. Then

$$D = \frac{3}{4\pi^2} \frac{\mathcal{N}_\Gamma^\chi}{L_H^4}.$$

The coefficient follows from the standard circle determinant

$$\frac{1}{2} \sum_{n \in \mathbb{Z}} \int \frac{d^4 p_E}{(2\pi)^4} \log \left[p_E^2 + \left(\frac{2\pi n + x}{L_H} \right)^2 \right]$$

after Schwinger parameterization and Poisson resummation; the tower sign is included in $\mathcal{N}_\Gamma^\chi$. For the full massless character, the Higgs-curvature match is

$$\frac{3}{4\pi^2} \frac{\mathcal{N}_\Gamma^\chi}{L_H^4 v_{\text{EW}}^4} = 0.083227124394645 \dots \lambda_F.$$

Equivalently,

$$\frac{\mathcal{N}_\Gamma^\chi}{L_H^4 v_{\text{EW}}^4} = 1.095225057620535 \dots \lambda_F.$$

The same loop length enters the scalar kinetic term. With arclength s on γ_H ,

$$\omega_H = \frac{ds}{L_H}, \quad \int_{\gamma_H} \omega_H = 1, \quad \int_{\gamma_H} |\omega_H|^2 ds = \frac{1}{L_H}.$$

Writing $Z_{\Gamma,H}$ for the localized density gives

$$f_H^2 = \frac{Z_{\Gamma,H}}{L_H}.$$

Since $f_H \theta_{H,*} = v_{\text{EW}}/\sqrt{2}$ and $x_* = w_H \theta_{H,*}$,

$$L_H = \frac{2x_*^2 Z_{\Gamma,H}}{w_H^2 v_{\text{EW}}^2}.$$

For the full massless-character root, the amplitude target becomes

$$\frac{\mathcal{N}_\Gamma^\chi w_H^8 v_{\text{EW}}^4}{16x_*^8 Z_{\Gamma,H}^4} = 1.095225057620535 \dots \lambda_F.$$

The point-projector current metric in the transverse weak channel gives

$$f_H^2 = \frac{Q}{4} f_\Gamma^2.$$

Combining this metric with $x_{\text{full}} = 1.837672606049940 \dots$ gives

$$\frac{f_H}{v_{\text{EW}}} = \frac{w_H}{\sqrt{2} x_{\text{full}}} = 0.384783872197163 \dots w_H,$$

and hence

$$\frac{f_\Gamma}{v_{\text{EW}}} = \frac{\sqrt{2}}{\sqrt{Q} x_{\text{full}}} w_H.$$

For the $Q = 7$ boundary stack,

$$\frac{f_\Gamma}{v_{\text{EW}}} = 0.290869266954901 \dots w_H.$$

The loop and point-projector normalizations meet through

$$\frac{Z_{\Gamma,H}}{L_H} = \frac{Q}{4} f_\Gamma^2.$$

Eliminating L_H gives the localized density target

$$\frac{\mathcal{N}_\Gamma^\chi}{Z_{\Gamma,H}^4} = \frac{2279.126786313921 \dots}{w_H^8 v_{\text{EW}}^4} \lambda_F$$

at the full massless-character root. The kinetic density in this equation comes from the localized Yang–Mills term

$$S_{\text{YM}}^\Gamma = -\frac{T_8(2\pi\alpha')^2}{4} \int_{\Gamma_{\text{EW}}} e^{-\phi} \sqrt{\gamma_\Gamma} \text{tr}_{\text{end}}(F_{MN} F^{MN}).$$

For the Wilson ansatz

$$A_s(x, y) = \theta_H(x) \omega_{H,s}(y) n_H^a T_a, \quad F_{\mu s} = (\partial_\mu \theta_H) \omega_{H,s} n_H^a T_a,$$

one obtains

$$f_H^2 = T_8(2\pi\alpha')^2 \int_{\Gamma_{\text{EW},\text{int}}} e^{-\phi} \sqrt{\gamma_\Gamma} g^{ss} \omega_{H,s} \omega_{H,s} \mathcal{T}_H,$$

with

$$\mathcal{T}_H = \text{tr}_{\text{end}}(n_H^a T_a n_H^b T_b) \Pi_{ab}^{\text{Schur}}.$$

For $\omega_H = ds/L_H$ and uniform transverse density,

$$Z_{\Gamma,H} = T_8(2\pi\alpha')^2 \int_{\mathbb{CP}^2} e^{-\phi} \sqrt{\gamma_\perp} \mathcal{T}_H.$$

The determinant factor is the canonical charged-kernel supertrace

$$\mathcal{N}_\Gamma^\chi = \sum_{\alpha \in \ker \mathcal{B}_\Gamma^\chi} \sigma_\alpha d_\alpha \tau_\alpha.$$

In the equal-amplitude Borel–Weil pass,

$$\mathcal{N}_{\Gamma,1}^\chi = \mathcal{N}_{\Gamma,2}^\chi = \mathcal{N}_\Gamma^\chi,$$

so the total charged supertrace over $\mathcal{O}(1) \oplus \mathcal{O}(2)$ is

$$\mathcal{N}_{\Gamma,\text{tot}}^\chi = 2\mathcal{N}_\Gamma^\chi.$$

The curvature target uses the per-pair amplitude $\mathcal{N}_\Gamma^\chi$, multiplying $\cos(mx) + \cos(2mx)$ with a common coefficient. For the zero-threshold endpoint trial, the charged kernels are

$$H^0(\mathbb{CP}^1, \mathcal{O}(1))^\chi = \text{span}\{u, v\}, \quad H^0(\mathbb{CP}^1, \mathcal{O}(2))^\chi = \text{span}\{u^2, v^2\}.$$

The uv state is neutral under the Wilson angle. With common oscillator multiplicity d_χ and Dolbeault index sign σ_χ ,

$$\mathcal{N}_\Gamma^\chi = 2\sigma_\chi d_\chi.$$

The Higgs-curvature match has $D > 0$ for $\lambda_F > 0$, hence

$$\sigma_\chi d_\chi > 0.$$

For the minimal even endpoint determinant,

$$d_\chi = 1, \quad \sigma_\chi = +1, \quad \mathcal{N}_\Gamma^\chi = 2.$$

With threshold parameters $z_r = M_r L_H$, the charged-pair amplitudes become

$$\mathcal{N}_{\Gamma,1}^\chi = 2\sigma_\chi d_1 F(z_1), \quad \mathcal{N}_{\Gamma,2}^\chi = 2\sigma_\chi d_2 F(z_2),$$

where

$$F(z) = e^{-z} \left(1 + z + \frac{z^2}{3} \right).$$

The equal-amplitude Borel–Weil shape requires

$$d_1 F(z_1) = d_2 F(z_2).$$

A sufficient local source for this equality is a product Dolbeault boundary complex

$$\mathcal{D}_\Gamma^\chi = \left(\bar{\partial}_{\mathcal{O}(1)}^\chi \oplus \bar{\partial}_{\mathcal{O}(2)}^\chi \right) \otimes \mathbf{1}_{E_\chi} + \mathbf{1} \otimes \mathcal{D}_{E,\chi}.$$

On the product zero-mode reduction,

$$(\mathcal{D}_\Gamma^\chi)^\dagger \mathcal{D}_\Gamma^\chi = \Delta_{\bar{\partial},\chi} \otimes \mathbf{1}_{E_\chi} + \mathbf{1} \otimes \Delta_{E,\chi}.$$

For $n \geq 0$,

$$H^0(\mathbb{CP}^1, \mathcal{O}(n)) = \ker \bar{\partial}_{\mathcal{O}(n)}, \quad H^1(\mathbb{CP}^1, \mathcal{O}(n)) = 0.$$

Thus $\Delta_{\bar{\partial},\chi}$ vanishes on the charged projectors

$$P_1^\chi = \text{span}\{u, v\}, \quad P_2^\chi = \text{span}\{u^2, v^2\}.$$

The charged-pair threshold spectrum is then supplied by the common oscillator block $\Delta_{E,\chi}$, so a single eigenvalue M^2 gives

$$z_1 = z_2 = ML_H, \quad d_1 = d_2 = d_\chi.$$

Degree-dependent rough-Laplacian, Casimir, or curvature terms give the splitting parameters in the general equation above. The minimal common-threshold pass has

$$d_1 = d_2 = 1, \quad z_1 = z_2 = z,$$

and gives

$$\mathcal{N}_\Gamma^\chi = 2F(z)$$

for positive endpoint parity. The point-projector endpoint trace evaluates the Schur-channel trace factor:

$$G_{ab}^{(p)} = \text{Re} \langle 0 | T_a K_p T_b | 0 \rangle = \frac{Q}{4} \text{diag}(1, 1, 0)_{ab}.$$

Writing

$$Z_{\Gamma,0} = T_8 (2\pi\alpha')^2 \int_{\mathbb{CP}^2} e^{-\phi} \sqrt{\gamma_\perp} \mathcal{Z}_{\text{end}},$$

one gets

$$Z_{\Gamma,H} = \frac{Q}{4} Z_{\Gamma,0}, \quad f_\Gamma^2 = \frac{Z_{\Gamma,0}}{L_H}.$$

The same localized measure defines the charged-kernel inner product. For an orthonormal determinant basis

$$\psi_\alpha \in L^2(\Gamma_{\text{EW,int}}, d\nu_\Gamma) \otimes E_\chi, \quad \int_{\Gamma_{\text{EW,int}}} d\nu_\Gamma |\psi_\alpha|^2 = 1,$$

the angle-dependent Wilson determinant has the finite coefficient

$$\mathcal{N}_\Gamma^\chi = \sum_{\alpha \in \ker \mathcal{B}_\Gamma^\chi} \sigma_\alpha d_\alpha \tau_\alpha.$$

Consequently the local density fixes the Hilbert normalization and the kinetic scale $Z_{\Gamma,0}$. The Wilson amplitude is the finite supertrace over the normalized kernel. For $Q = 7$, the same determinant target becomes

$$\frac{\mathcal{N}_{\Gamma}^{\chi}}{Z_{\Gamma,0}^4} = \frac{21375.716460702053\dots}{w_H^8 v_{\text{EW}}^4} \lambda_F.$$

In the minimal even endpoint trial the equivalent form is

$$\frac{Z_{\Gamma,0}}{v_{\text{EW}}} = \left(\frac{2}{21375.716460702053\dots \lambda_F} \right)^{1/4} w_H^2.$$

With a common threshold z , this becomes

$$\frac{Z_{\Gamma,0}}{v_{\text{EW}}} = \left(\frac{2F(z)}{21375.716460702053\dots \lambda_F} \right)^{1/4} w_H^2.$$

Since

$$F'(z) = -\frac{1}{3}e^{-z}z(1+z),$$

one has $0 < F(z) \leq 1$ for $z \geq 0$. Define

$$C_0 = \left(\frac{2}{21375.716460702053\dots} \right)^{1/4} = 0.0983506713707568\dots$$

Then every minimal positive endpoint realization satisfies the density window

$$0 < \frac{Z_{\Gamma,0}}{w_H^2 v_{\text{EW}}} \leq C_0 \lambda_F^{-1/4}.$$

Equivalently, a localized D8/O8 calculation passes this row exactly when

$$0 < \frac{21375.716460702053\dots \lambda_F}{2w_H^8} \left(\frac{Z_{\Gamma,0}}{v_{\text{EW}}} \right)^4 \leq 1,$$

with z then fixed by the monotone equation for $F(z)$. It is useful to name the left-hand side

$$\Xi_{\Gamma} = \frac{21375.716460702053\dots \lambda_F}{2w_H^8} \left(\frac{Z_{\Gamma,0}}{v_{\text{EW}}} \right)^4.$$

The value $\Xi_{\Gamma} = 1$ gives the massless endpoint threshold $z = 0$. Values $0 < \Xi_{\Gamma} < 1$ give a unique $z > 0$ through

$$e^{-z} \left(1 + z + \frac{z^2}{3} \right) = \Xi_{\Gamma}.$$

Values $\Xi_{\Gamma} > 1$ reject the minimal positive endpoint row and require additional localized spectrum data. Equivalently,

$$R_{\Gamma} = \Xi_{\Gamma}^{1/4} = \frac{Z_{\Gamma,0}}{C_0 \lambda_F^{-1/4} w_H^2 v_{\text{EW}}} = F(z)^{1/4}.$$

Thus R_{Γ} is the direct threshold-suppression coordinate for this localized row. An endpoint trace metric therefore gives

$$w_y : w_x = Q : 1,$$

and for $Q = 7$ the center is $t_Q = -7/8$. The same filtration gives

$$\dim H^0(\mathbb{CP}^1, \mathcal{O}(Q)) = Q + 1,$$

the Chan–Paton half-rank used above. Substitution of t_Q in the neutral pass equation yields

$$9q(q+2) = 2R_{\text{dV}} \left(q \frac{5Q^2 + 18Q + 29}{(Q+1)^2} + 8 \right) \left(q \frac{Q^2 + 1}{(Q+1)^2} + 1 \right).$$

For $Q = 7$, the equation becomes

$$576q(q+2) = R_{\text{dV}}(25q+32)^2,$$

with positive root

$$q_c = 3.511614004347\dots, \quad \lambda_c = 1.873930095907\dots$$

The same point-evaluation specialization has an explicit photon length. With the primitive relation $P = Q - 1$,

$$\omega_T = \eta(t_Q), \quad \omega_Y = (Q-1)h + Q\eta(t_Q), \quad \omega_Q = (Q-1)h + (Q+1)\eta(t_Q).$$

For $t_Q = -Q/(Q+1)$,

$$A = 2q, \quad B = -\frac{Q-1}{Q+1}q, \quad C = \frac{Q^2+1}{(Q+1)^2}q + 1.$$

In the weak-current specialization $L_2^2 = C$, one obtains

$$L_Q^2 = (Q-1)^2A + 2(Q-1)(Q+1)B + (Q+1)^2C = (Q+1)^2L_2^2.$$

For $Q = 7$, this gives $L_Q^2 = 64L_2^2$. The neutral determinant is

$$\det \mathcal{L}_\Gamma = (Q-1)^2q(q+2),$$

so the equation

$$\det \mathcal{L}_\Gamma = R_{\text{dV}}L_2^2L_Q^2$$

recovers the reduced Q -row above. On this neutral pass surface,

$$G_Z^\Gamma = \frac{L_Q^2}{\det \mathcal{L}_\Gamma} = \frac{1}{R_{\text{dV}}L_2^2}.$$

Writing C_W for the Weinberg length-normalization constant in this local calculation, the charged and neutral source masses read

$$M_W^2 = \frac{C_W^2 v_{\text{EW}}^2}{4L_2^2}, \quad M_Z^2 = \frac{C_W^2 v_{\text{EW}}^2}{4R_{\text{dV}}L_2^2}.$$

Since $R_{\text{dV}} = X_+(3/4)/X_+(2)$, the Schur coefficient becomes

$$\mu_\Gamma^2 = \frac{M_Z^2}{X_+(2)} = \frac{M_W^2}{X_+(3/4)} = \frac{C_W^2 v_{\text{EW}}^2}{4X_+(3/4)L_2^2}.$$

The electromagnetic normalization in the same specialization is

$$\alpha_\Gamma = \frac{C_W^2}{4\pi L_Q^2} = \frac{C_W^2}{4\pi(Q+1)^2L_2^2} = \frac{M_W^2}{\pi(Q+1)^2v_{\text{EW}}^2}.$$

For $Q = 7$, this specialization gives

$$\alpha_\Gamma = \frac{M_W^2}{64\pi v_{\text{EW}}^2}.$$

Using $M_W = 80.377$ GeV and $v_{EW} = 246.21965$ GeV as representative electroweak inputs,

$$\alpha_\Gamma^{-1} = 1886.738366864 \dots$$

The physical electromagnetic input $\alpha^{-1} = 137.035999084 \dots$ corresponds to

$$k_\alpha = \frac{M_W^2}{\pi \alpha v_{EW}^2} = 4.648394337766 \dots, \quad \frac{64}{k_\alpha} = 13.768195069001 \dots$$

Thus the point-evaluation Hodge row is a neutral-matrix filter. Full electromagnetic normalization requires an additional localized contribution to the Q -current norm. Equivalently, set $L_2^2 = 1$, $k = k_\alpha$, and write

$$\widehat{\mathcal{L}}_\Gamma = \begin{pmatrix} 1 & \varepsilon \\ \varepsilon & y \end{pmatrix}.$$

The alpha-closed neutral matrix obeys

$$1 + y + 2\varepsilon = k, \quad y - \varepsilon^2 = R_{dV}k,$$

hence

$$\varepsilon_\pm = -1 \pm \sqrt{k(1 - R_{dV})}, \quad y_\pm = k + 1 \mp 2\sqrt{k(1 - R_{dV})}.$$

For the numerical inputs above,

branch	ε	y	$\varepsilon/\sqrt{y}$
+	0.018362864274...	3.611668609219...	0.009662432587...
-	-2.018362864274...	7.685120066313...	-0.728071311237....

At the point-evaluation neutral root, the corresponding normalized matrix is

$$\widehat{\mathcal{L}}_{\text{pt}} = \begin{pmatrix} 1 & 2.778688215142 \dots \\ 2.778688215142 \dots & 57.442623569715 \dots \end{pmatrix}.$$

A sharper decomposition writes each alpha-closed branch $T_\pm$ as

$$T_\pm = \frac{\widehat{\mathcal{L}}_{\text{pt}} - D_\pm}{s_\pm}, \quad D_\pm \succeq 0, \quad \text{rank } D_\pm = 1.$$

The two decompositions are

branch	s	nonzero eigenvalue of D	slope of the image line
+	0.859516413258733...	54.478818707582...	19.667102015222...
-	0.672682628439277...	52.600294175197...	12.637293988103....

Thus the localized action data are a rank-one subtraction in the Q -current plane together with a common current-frame rescaling. The local coefficient row is the extraction of $H_{hh}^\Gamma, H_{h\eta}^\Gamma, H_{\eta\eta}^\Gamma$ and L_2^2 from the same $d\nu_\Gamma$, endpoint image data, and DBI/CS pullback forms. A single boundary auxiliary gives the required algebra. For $J = (J_3, J_Y)$, write

$$S_{\text{img}}^{(2)} = \frac{1}{2} J^T \widehat{\mathcal{L}}_{\text{pt}} J + r_{\text{img}} \ell_\pm^T J + \frac{1}{2} m_\pm r_{\text{img}}^2.$$

Eliminating r_{img} gives

$$T_\pm = s_\pm^{-1} \left(\widehat{\mathcal{L}}_{\text{pt}} - \frac{\ell_\pm \ell_\pm^T}{m_\pm} \right), \quad \frac{\ell_\pm \ell_\pm^T}{m_\pm} = \delta_\pm \begin{pmatrix} 1 \\ \rho_\pm \end{pmatrix} \begin{pmatrix} 1 & \rho_\pm \end{pmatrix}.$$

The coefficient extraction target is therefore

$$\begin{aligned}
+ : \quad & s_+ = 0.859516413258733 \dots, \quad s_+^{-1} = 1.163444914575446 \dots, \\
& \delta_+ = 0.140483586741 \dots, \quad \rho_+ = 19.667102015222 \dots, \\
- : \quad & s_- = 0.672682628439277 \dots, \quad s_-^{-1} = 1.486585140930646 \dots, \\
& \delta_- = 0.327317371561 \dots, \quad \rho_- = 12.637293988103 \dots
\end{aligned}$$

Equivalently, for $c_\rho = (1, \rho)^T$ and

$$M_\rho = \widehat{\mathcal{L}}_{\text{pt}} - \delta(\rho) c_\rho c_\rho^T,$$

the equations

$$(1, 1) M_\rho \begin{pmatrix} 1 \\ 1 \end{pmatrix} = k_\alpha (M_\rho)_{33}, \quad \det M_\rho = R_{\text{dV}} k_\alpha (M_\rho)_{33}^2$$

give

$$\begin{aligned}
0 = P_\rho(\rho) = & \rho^4 - 29.874189325032 \dots \rho^3 + 170.407256725780 \dots \rho^2 \\
& + 591.897671281551 \dots \rho + 93.118987613750 \dots
\end{aligned}$$

Positive Schur subtraction selects the roots

$$\rho_- = 12.637293988103 \dots, \quad \rho_+ = 19.667102015221 \dots$$

For positive image mass $m > 0$, the coefficient extraction reads

$$\frac{\ell_3^2}{m} = \delta, \quad \frac{\ell_Y}{\ell_3} = \rho, \quad s = (M_\rho)_{33} = 1 - \delta.$$

Thus a real localized image auxiliary realizes either

$$\delta_+ = 0.140483586741 \dots, \quad \rho_+ = 19.667102015221 \dots,$$

or

$$\delta_- = 0.327317371561 \dots, \quad \rho_- = 12.637293988103 \dots$$

In local variables, choose a normalized image eigenmode ψ with

$$\int_{\Gamma_{\text{EW}, \text{int}}} d\nu_\Gamma |\psi|^2 = 1, \quad m_\psi = \langle \psi, \mathcal{M}_{\text{img}} \psi \rangle_\Gamma.$$

For image current profile Θ_{img} , define

$$\mathcal{J}_A = \gamma_\Gamma^{-1}(\theta_\Gamma^A, \Theta_{\text{img}}), \quad \ell_A[\psi] = \int_{\Gamma_{\text{EW}, \text{int}}} d\nu_\Gamma \psi \mathcal{J}_A, \quad A = 3, Y.$$

The extraction invariants are

$$\delta_\psi = \frac{\ell_3[\psi]^2}{m_\psi L_2^2}, \quad \rho_\psi = \frac{\ell_Y[\psi]}{\ell_3[\psi]}, \quad s_\psi = 1 - \delta_\psi.$$

Equivalently, the full image sector gives the Green-kernel correction

$$\mathcal{D}_{AB} = \frac{1}{L_2^2} \left\langle \mathcal{J}_A, \mathcal{M}_{\text{img}}^{-1} \mathcal{J}_B \right\rangle_\Gamma, \quad A, B = 3, Y.$$

The one-field Schur row is the rank-one case

$$\mathcal{D}_{33} \mathcal{D}_{YY} - \mathcal{D}_{3Y}^2 = 0, \quad \mathcal{D}_{33} > 0.$$

A spectral resolution makes this condition computable from the localized image spectrum. For

$$\mathcal{M}_{\text{img}}\psi_n = m_n\psi_n, \quad m_n > 0,$$

set

$$\ell_A^{(n)} = \int_{\Gamma_{\text{EW,int}}} d\nu_{\Gamma} \psi_n \mathcal{J}_A.$$

Then

$$\mathcal{D}_{AB} = \sum_n \frac{\ell_A^{(n)} \ell_B^{(n)}}{m_n L_2^2}.$$

Writing

$$u_n = \frac{1}{L_2 \sqrt{m_n}} \begin{pmatrix} \ell_3^{(n)} \\ \ell_Y^{(n)} \end{pmatrix},$$

one has

$$\mathcal{D} = \sum_n u_n u_n^T, \quad \Delta_{\text{rank}} = \mathcal{D}_{33} \mathcal{D}_{YY} - \mathcal{D}_{3Y}^2 = \sum_{n < m} (u_n^3 u_m^Y - u_n^Y u_m^3)^2.$$

Thus rank one is the collinearity condition for the sourced overlap vectors. For $\mathcal{D}_{33} > 0$, put

$$\delta_D = \mathcal{D}_{33}, \quad \rho_D = \frac{\mathcal{D}_{3Y}}{\mathcal{D}_{33}}.$$

The two Schur branches are selected by

$$\Delta_{\text{rank}} = 0, \quad (\delta_D, \rho_D) \in \{(\delta_+, \rho_+), (\delta_-, \rho_-)\}.$$

Several localized image modes can contribute to one branch when the source-slope ratio is common and the strengths add to $\delta_{\pm}$. A Green-metric form records the same condition directly on the source profiles. Define

$$(F, G)_G = \frac{1}{L_2^2} \langle F, \mathcal{M}_{\text{img}}^{-1} G \rangle_{\Gamma}.$$

Then $\mathcal{D}_{AB} = (\mathcal{J}_A, \mathcal{J}_B)_G$. For $\sigma \in \{+, -\}$, the target branch is

$$\mathcal{D}^{(\sigma)} = \delta_{\sigma} \begin{pmatrix} 1 \\ \rho_{\sigma} \end{pmatrix} \begin{pmatrix} 1 & \rho_{\sigma} \end{pmatrix},$$

equivalently

$$\|\mathcal{J}_Y - \rho_{\sigma} \mathcal{J}_3\|_G^2 = 0, \quad \|\mathcal{J}_3\|_G^2 = \delta_{\sigma}.$$

For a positive image Green metric, the local calculation reduces to source proportionality in the sourced image quotient plus one norm calculation. If the image source is the pure weak-period line

$$\mathcal{J}_Y = \rho_Q \mathcal{J}_3, \quad \rho_Q = 2Q + 1,$$

then

$$\mathcal{D} = \delta_Q \begin{pmatrix} 1 \\ \rho_Q \end{pmatrix} \begin{pmatrix} 1 & \rho_Q \end{pmatrix}.$$

For $Q = 7$, $\rho_Q = 15$. The Schur branch equations require

$$\rho_D \in \{12.637293988103\dots, 19.667102015221\dots\}.$$

The pure weak-period line is therefore excluded as the final image source. With the alpha-length value,

$$\delta_Q = \frac{64 - k_{\alpha}}{(1 + \rho_Q)^2 - k_{\alpha}} = 0.236129805122\dots, \quad \mathcal{R}(15) = 0.999999862211\dots$$

Thus the localized action must deform the source line to one Schur slope or project the weak-period source away from the Schur image channel. Using δ_Q as the alpha-normalized reference strength, a coframe deformation in the sourced quotient may be parameterized by

$$\mathcal{J}_3^{(\sigma)} = a_\sigma \mathcal{J}_3^{(Q)}, \quad \mathcal{J}_Y^{(\sigma)} = a_\sigma (\rho_Q + \zeta_\sigma) \mathcal{J}_3^{(Q)}.$$

Branch matching gives

$$\zeta_\sigma = \rho_\sigma - \rho_Q, \quad a_\sigma^2 = \frac{\delta_\sigma}{\delta_Q}.$$

For $Q = 7$,

branch	ζ_σ	a_σ^2	a_σ
+	4.667102015221...	0.594942204219...	0.771324966677...
-	-2.362706011897...	1.386175588431...	1.177359583318....

These two rows are the local coefficient targets for a sourced D8/O8 coframe deformation. Equivalently, let

$$\mathcal{C}_Q = \begin{pmatrix} \mathcal{J}_3^{(Q)} \\ \mathcal{J}_Y^{(Q)} \end{pmatrix}, \quad \mathcal{J}_Y^{(Q)} = \rho_Q \mathcal{J}_3^{(Q)}.$$

The deformation can be placed in a lower-triangular source-coframe map

$$\mathcal{C}_\sigma = R_\sigma \mathcal{C}_Q, \quad R_\sigma = a_\sigma \begin{pmatrix} 1 & 0 \\ \zeta_\sigma & 1 \end{pmatrix}.$$

It satisfies

$$R_\sigma \begin{pmatrix} 1 \\ \rho_Q \end{pmatrix} = a_\sigma \begin{pmatrix} 1 \\ \rho_\sigma \end{pmatrix}, \quad R_\sigma \mathcal{D}_Q R_\sigma^T = \mathcal{D}_\sigma.$$

For $Q = 7$,

$$R_+ = \begin{pmatrix} 0.771324966677... & 0 \\ 3.599852306368... & 0.771324966677... \end{pmatrix}$$

and

$$R_- = \begin{pmatrix} 1.177359583318... & 0 \\ -2.781754565670... & 1.177359583318... \end{pmatrix}.$$

The diagonal entry is the Green-norm rescaling, and the lower-left entry is the localized T_3 -to- Y source mixing required from the D8/O8 pullback. The equivalent pullback condition on localized source coframes is

$$\theta_{\Gamma,\sigma}^3 = a_\sigma \theta_{\Gamma,Q}^3, \quad \theta_{\Gamma,\sigma}^Y = a_\sigma (\theta_{\Gamma,Q}^Y + \zeta_\sigma \theta_{\Gamma,Q}^3)$$

inside the image source quotient. Thus the localized action returns

$$Z_{\text{img}}^{(\sigma)} = a_\sigma^2 Z_{\text{img}}^{(Q)}, \quad \beta_{Y3}^{(\sigma)} = \zeta_\sigma.$$

For $Q = 7$,

branch	$Z_{\text{img}}^{(\sigma)} / Z_{\text{img}}^{(Q)}$	$\beta_{Y3}^{(\sigma)}$
+	0.594942204219...	4.667102015221...
-	1.386175588431...	-2.362706011897....

For a computed localized action, define

$$\hat{\delta}_G = (\mathcal{J}_3, \mathcal{J}_3)_G = \mathcal{D}_{33}, \quad \hat{\rho}_G = \frac{(\mathcal{J}_3, \mathcal{J}_Y)_G}{(\mathcal{J}_3, \mathcal{J}_3)_G} = \frac{\mathcal{D}_{3Y}}{\mathcal{D}_{33}},$$

and

$$\hat{\epsilon}_G^2 = \frac{\|\mathcal{J}_Y - \hat{\rho}_G \mathcal{J}_3\|_G^2}{\hat{\delta}_G} = \frac{\mathcal{D}_{33} \mathcal{D}_{YY} - \mathcal{D}_{3Y}^2}{\mathcal{D}_{33}^2}.$$

The Schur-branch extraction condition is

$$(\hat{\delta}_G, \hat{\rho}_G, \hat{\epsilon}_G^2) = (\delta_\sigma, \rho_\sigma, 0), \quad \sigma \in \{+, -\}.$$

The deformation coefficients are then

$$a_\sigma^2 = \frac{\hat{\delta}_G}{\hat{\delta}_Q}, \quad \zeta_\sigma = \hat{\rho}_G - \rho_Q.$$

In the pullback notation the direct readout is

$$\begin{aligned} \hat{Z}_{\text{img}} &= \mathcal{D}_{33}, & \hat{\beta}_{Y3} &= \frac{\mathcal{D}_{3Y}}{\mathcal{D}_{33}} - \rho_Q, \\ \hat{E}_\perp &= \mathcal{D}_{YY} - \frac{\mathcal{D}_{3Y}^2}{\mathcal{D}_{33}}. \end{aligned}$$

The branch extraction condition becomes

$$\frac{\hat{Z}_{\text{img}}}{Z_{\text{img}}^{(Q)}} = a_\sigma^2, \quad \hat{\beta}_{Y3} = \zeta_\sigma, \quad \hat{E}_\perp = 0.$$

A projected-image realization uses a G -self-adjoint idempotent Π_σ satisfying

$$\begin{aligned} \Pi_\sigma^2 &= \Pi_\sigma, & (\Pi_\sigma F, G)_G &= (F, \Pi_\sigma G)_G, \\ \Pi_\sigma \mathcal{J}_Y &= \rho_\sigma \Pi_\sigma \mathcal{J}_3, & \|\Pi_\sigma \mathcal{J}_3\|_G^2 &= \delta_\sigma. \end{aligned}$$

The coframe map also defines a symmetric source-index block

$$\mathcal{H}_\sigma = R_\sigma R_\sigma^T = a_\sigma^2 \begin{pmatrix} 1 & \zeta_\sigma \\ \zeta_\sigma & 1 + \zeta_\sigma^2 \end{pmatrix}.$$

It has

$$\det \mathcal{H}_\sigma = a_\sigma^4, \quad \chi_\sigma = \frac{\zeta_\sigma}{\sqrt{1 + \zeta_\sigma^2}}.$$

For $Q = 7$,

$$\mathcal{H}_+ = \begin{pmatrix} 0.594942204219 \dots & 2.776655960251 \dots \\ 2.776655960251 \dots & 13.553878831880 \dots \end{pmatrix}, \quad \chi_+ = 0.977806418243 \dots,$$

and

$$\mathcal{H}_- = \begin{pmatrix} 1.386175588431 \dots & -3.275125396331 \dots \\ -3.275125396331 \dots & 9.124334052058 \dots \end{pmatrix}, \quad \chi_- = -0.920912188652 \dots$$

The same rank-one image subtraction can be written with one real localized field r_σ :

$$S_{\text{img}, \sigma}^{(2)} = \frac{1}{2} m_\sigma r_\sigma^2 + L_2 \sqrt{m_\sigma} r_\sigma g_\sigma^T J, \quad J = \begin{pmatrix} J_3 \\ J_Y \end{pmatrix}.$$

With

$$g_\sigma = \sqrt{\delta_\sigma} \begin{pmatrix} 1 \\ \rho_\sigma \end{pmatrix},$$

eliminating r_σ gives

$$-\frac{1}{2}L_2^2 J^T g_\sigma g_\sigma^T J, \quad \mathcal{D}_\sigma = g_\sigma g_\sigma^T.$$

For the two branches,

branch	$(g_\sigma)_3$	$(g_\sigma)_Y$
+	0.374811401563...	7.371454070999...
-	0.572116571654...	7.230005311453....

Return to the local source definition

$$\mathcal{J}_A = \gamma_\Gamma^{-1}(\theta_\Gamma^A, \Theta_{\text{img}}), \quad A = 3, Y.$$

For a normalized image eigenmode ψ_σ ,

$$\int_{\Gamma_{\text{EW,int}}} d\nu_\Gamma |\psi_\sigma|^2 = 1, \quad \mathcal{M}_{\text{img}} \psi_\sigma = m_\sigma \psi_\sigma,$$

the one-field packet is equivalent to

$$\int_{\Gamma_{\text{EW,int}}} d\nu_\Gamma \psi_\sigma \gamma_\Gamma^{-1}(\theta_\Gamma^A, \Theta_{\text{img},\sigma}) = L_2 \sqrt{m_\sigma} (g_\sigma)_A, \quad A = 3, Y.$$

With $\Xi_\sigma = \Theta_{\text{img},\sigma} / \sqrt{m_\sigma}$, this reads

$$\frac{1}{L_2} \int_{\Gamma_{\text{EW,int}}} d\nu_\Gamma \psi_\sigma \gamma_\Gamma^{-1}(\theta_\Gamma^A, \Xi_\sigma) = (g_\sigma)_A.$$

A sufficient local profile is obtained from

$$\mathcal{G}_{AB}^{(\psi)} = \int_{\Gamma_{\text{EW,int}}} d\nu_\Gamma \psi_\sigma^2 \gamma_\Gamma^{-1}(\theta_\Gamma^A, \theta_\Gamma^B).$$

When this matrix is invertible,

$$\Xi_\sigma = L_2 \psi_\sigma c_\sigma^B \theta_{\Gamma B}, \quad c_\sigma^B = \left(\mathcal{G}^{(\psi)-1} \right)^{BA} (g_\sigma)_A$$

solves the two overlap equations. The D8/O8 source row is therefore tested by $\mathcal{G}^{(\psi)}$ and the Gram-dual current combination in the sourced image quotient. Writing

$$\mathcal{G}^{(\psi)} = \begin{pmatrix} A_\psi & B_\psi \\ B_\psi & C_\psi \end{pmatrix}, \quad \Delta_\psi = A_\psi C_\psi - B_\psi^2,$$

the coefficients are

$$c_\sigma^3 = \frac{C_\psi(g_\sigma)_3 - B_\psi(g_\sigma)_Y}{\Delta_\psi}, \quad c_\sigma^Y = \frac{-B_\psi(g_\sigma)_3 + A_\psi(g_\sigma)_Y}{\Delta_\psi}.$$

In an orthonormal weighted current frame, $\mathcal{G}^{(\psi)} = \mathbf{1}$, hence

$$\Xi_\sigma = L_2 \psi_\sigma \left((g_\sigma)_3 \theta_\Gamma^3 + (g_\sigma)_Y \theta_\Gamma^Y \right).$$

The general solution splits as

$$\Xi_\sigma = \Xi_{\sigma,\text{min}} + \Xi_{\sigma,\perp}, \quad \int_{\Gamma_{\text{EW,int}}} d\nu_\Gamma \psi_\sigma \gamma_\Gamma^{-1}(\theta_\Gamma^A, \Xi_{\sigma,\perp}) = 0.$$

The forced profile has norm

$$\frac{\|\Xi_{\sigma,\min}\|_{\Gamma}^2}{L_2^2} = g_{\sigma}^T \mathcal{G}^{(\psi)-1} g_{\sigma}.$$

In an orthonormal weighted current frame this gives

branch	$\ \Xi_{\sigma,\min}\ _{\Gamma}^2/L_2^2$
+	54.478818707582...
-	52.600294175197...

Any nonzero $\Xi_{\sigma,\perp}$ raises the image-profile norm without changing the Schur matrix. Define the profile residual

$$\mathcal{R}_{\Xi,\sigma} = \frac{\|\Xi_{\sigma,\perp}\|_{\Gamma}^2}{\|\Xi_{\sigma,\min}\|_{\Gamma}^2}.$$

The minimal image-source realization has

$$\mathcal{R}_{\Xi,\sigma} = 0.$$

A positive value records extra localized image content that leaves $\mathcal{D}_{\sigma}$ invariant and belongs to the D8/O8 spectrum.

Proposition 3 (One-mode image-source closure). *Fix $\sigma \in \{+, -\}$ and fix the orientation of the real image field r_{σ} . A positive one-mode image sector over Γ_{EW} realizes the Schur branch*

$$\mathcal{D}_{\sigma} = \delta_{\sigma} \begin{pmatrix} 1 \\ \rho_{\sigma} \end{pmatrix} \begin{pmatrix} 1 & \rho_{\sigma} \end{pmatrix}$$

precisely when the mass-normalized image profile satisfies

$$\frac{1}{L_2} \int_{\Gamma_{\text{EW,int}}} d\nu_{\Gamma} \psi_{\sigma} \gamma_{\Gamma}^{-1}(\theta_{\Gamma}^A, \Xi_{\sigma}) = (g_{\sigma})_A, \quad A = 3, Y.$$

Proof. The overlap vector is

$$\ell_A = \int_{\Gamma_{\text{EW,int}}} d\nu_{\Gamma} \psi_{\sigma} \gamma_{\Gamma}^{-1}(\theta_{\Gamma}^A, \Theta_{\text{img},\sigma}) = L_2 \sqrt{m_{\sigma}} (g_{\sigma})_A.$$

Integrating out r_{σ} gives

$$\mathcal{D}_{AB} = \frac{\ell_A \ell_B}{m_{\sigma} L_2^2} = (g_{\sigma})_A (g_{\sigma})_B.$$

Since $g_{\sigma} = \sqrt{\delta_{\sigma}}(1, \rho_{\sigma})^T$, the displayed branch matrix follows. The reverse implication follows by choosing the orientation of r_{σ} so that $\ell_3 > 0$. $\square$

The localized realization data are the packet

$$\mathcal{P}_{\Gamma,\sigma} = \left(d\nu_{\Gamma}, \psi_{\sigma}, m_{\sigma}, \Xi_{\sigma}, \mathcal{G}^{(\psi)}, Z_{\Gamma,0}, \mathcal{N}_{\Gamma}^{\chi} \right)$$

with

$$\begin{aligned} \int_{\Gamma_{\text{EW,int}}} d\nu_{\Gamma} |\psi_{\sigma}|^2 &= 1, & \mathcal{M}_{\text{img}} \psi_{\sigma} &= m_{\sigma} \psi_{\sigma}, & m_{\sigma} &> 0, \\ L_2^{-1} \int_{\Gamma_{\text{EW,int}}} d\nu_{\Gamma} \psi_{\sigma} \gamma_{\Gamma}^{-1}(\theta_{\Gamma}^A, \Xi_{\sigma}) &= (g_{\sigma})_A, & A &= 3, Y, \\ \mathcal{R}_{\Xi,\sigma} &= 0, & Z_{\Gamma,H} &= \frac{Q}{4} Z_{\Gamma,0}, & \mathcal{N}_{\Gamma}^{\chi} &= 2\sigma_{\chi} d_{\chi} F(z). \end{aligned}$$

7.2 Localized Quadratic Master Form

Let

$$\Gamma_{\text{int}} = \mathbb{CP}^2 \times S_Q^1.$$

Let $\mathcal{S}_\Gamma$ label the D8 image sectors retained by the O8 projection, and set

$$\kappa_8 = T_8(2\pi\alpha')^2.$$

The string-frame DBI expansion on sector s ,

$$S_{\text{DBI},s} = -T_8 \int e^{-\phi_s} \text{Str}_{E_s} \sqrt{-\det(P_s[G+B] + 2\pi\alpha' F_s)},$$

has quadratic Yang–Mills density

$$d\nu_{\Gamma,s} = \kappa_8 \omega_s e^{-\phi_s} \sqrt{\gamma_{\Gamma,s}} d^5\sigma,$$

where ω_s is the orientifold and endpoint multiplicity assigned to the retained sector. The common Γ -inner product is

$$\langle X, Y \rangle_\Gamma = \sum_{s \in \mathcal{S}_\Gamma} \int_{\Gamma_{\text{int}}^{(s)}} d\nu_{\Gamma,s} \gamma_s^{mn} \text{tr}_{E_s} (\Pi_s X_m \Pi_s Y_n).$$

The scalar density $d\nu_\Gamma$ used below denotes this sector-sum measure together with its finite trace projectors. The immediate DBI readouts are

$$\mathcal{K}_{AB}^\Gamma = \langle \theta_\Gamma^A, \theta_\Gamma^B \rangle_\Gamma, \quad \mathcal{G}_{AB}^{(\psi)} = \langle \psi_\sigma \theta_\Gamma^A, \psi_\sigma \theta_\Gamma^B \rangle_\Gamma,$$

$$Z_{\Gamma,0} = \langle \eta_0, \eta_0 \rangle_\Gamma, \quad Z_{\Gamma,H} = \langle \omega_H n_H^a T_a \eta_0, \omega_H n_H^b T_b \eta_0 \rangle_\Gamma.$$

The image operator is an adjoint square,

$$\mathcal{M}_{\text{img}} = \mathcal{Q}_{\Gamma,\sigma}^\dagger \mathcal{Q}_{\Gamma,\sigma}, \quad \dagger \text{ taken with } \langle, \rangle_\Gamma.$$

The weak-period Chan–Paton half-stack is

$$W_Q = H^0(\mathbb{CP}^1, \mathcal{O}(Q)), \quad \dim W_Q = Q + 1.$$

The local D8 endpoint space is the orientifold-real double

$$E_Q = (W_Q \oplus W_Q^\vee)^\Omega, \quad \text{rk}_\mathbb{C}(E_Q^\mathbb{C}) = 2(Q + 1).$$

At the active value $Q = 7$,

$$\text{rk}_\mathbb{C}(E_Q^\mathbb{C}) = 16, \quad q_8^{\text{net}} = 16 - 16 = 0, \quad k = \frac{16}{2} - 8 = 0.$$

Thus the retained endpoint is the locally neutral D8/O8 row in the doubled type-I' convention.

Introduce row projectors

$$\Pi_Q = P_\Omega P_{E_Q}, \quad \Pi_\sigma = |e_\sigma\rangle\langle e_\sigma|, \quad \Pi_\chi = P_1^\chi + P_2^\chi,$$

with

$$P_1^\chi = \text{proj}_{\text{span}\{u,v\}}, \quad P_2^\chi = \text{proj}_{\text{span}\{u^2,v^2\}}.$$

For a row projector Π , define

$$\langle X, Y \rangle_{\Gamma,\Pi} = \int_{\Gamma_{\text{int}}} d\mu_\Gamma \gamma_\Gamma^{mn} \text{tr}_{\mathcal{H}_\Gamma^{\text{loc}}} (\Pi X_m \Pi Y_n), \quad d\mu_\Gamma = \kappa_8 e^{-\phi} \sqrt{\gamma_\Gamma} d^5\sigma.$$

The row readouts become

$$\begin{aligned}\mathcal{K}_{AB}^\Gamma &= \langle \theta_\Gamma^A, \theta_\Gamma^B \rangle_{\Gamma, \Pi_Q}, & \mathcal{G}_{AB}^{(\psi)} &= \langle \psi_\sigma \theta_\Gamma^A, \psi_\sigma \theta_\Gamma^B \rangle_{\Gamma, \Pi_\sigma}, \\ Z_{\Gamma,0} &= \langle \eta_0, \eta_0 \rangle_{\Gamma, \Pi_Q}, & Z_{\Gamma,H} &= \langle \omega_H n_H^a T_a \eta_0, \omega_H n_H^b T_b \eta_0 \rangle_{\Gamma, \Pi_Q},\end{aligned}$$

and

$$\mathcal{N}_\Gamma^\chi = \text{Str}_{\Pi_\chi \ker \mathcal{B}_\Gamma^\chi}(\tau_\alpha).$$

Let

$$\mathcal{H}_n = H^0(\mathbb{CP}^1, \mathcal{O}(n)), \quad e_k = u^{n-k}v^k, \quad k = 0, \dots, n.$$

The $SU(2)$ -equivariant antiunitary

$$J_n e_k = (-1)^k e_{n-k}$$

satisfies

$$J_n^2 = (-1)^n.$$

Thus $n = 1$ gives the pseudoreal doublet and $n = 2$ gives the real triplet. Using the invariant pairing to identify $\mathcal{H}_n^\vee$ with $\mathcal{H}_n$, the doubled endpoint involution is

$$\Omega_n(s, \lambda) = (J_n^{-1} \lambda, J_n s), \quad \Omega_n^2 = 1, \quad P_{\Omega, n} = \frac{1}{2}(1 + \Omega_n).$$

On the charged bases,

$$J_1 u = v, \quad J_1 v = -u,$$

and

$$J_2 u^2 = v^2, \quad J_2 uv = -uv, \quad J_2 v^2 = u^2.$$

Consequently $P_{\Omega,1}$ preserves the (u, v) charged doublet, and $P_{\Omega,2}$ preserves the charged pair (u^2, v^2) after applying $P_2^\chi = 1 - |uv\rangle\langle uv|$. The trace readout is

$$\text{Tr } P_{\Omega,1} P_1^\chi = 2, \quad \text{Tr } P_{\Omega,2} P_2^\chi = 2.$$

The orientifold row therefore contributes

$$\alpha_{\text{parity}} = 0, \quad d_1 = d_2 = d_\chi.$$

The determinant sign σ_χ and the common threshold M_χ are properties of the localized oscillator $\mathcal{B}_\Gamma^\chi$. Let $x = w_H \theta_H$. On the charged Borel–Weil pair with charges

$$r = 1, \quad r = 2,$$

take the localized oscillator block to have eigenvalues

$$\lambda_{\ell,r}(x) = p_E^2 + M_\chi^2 + \frac{(2\pi\ell + rx)^2}{L_H^2}, \quad \ell \in \mathbb{Z},$$

on $P_r^\chi \otimes E_\chi$. The Euclidean complex-boson Gaussian gives the positive one-loop sign

$$\sigma_\chi = +1.$$

After Schwinger parameterization and Poisson resummation, the angle-dependent term is

$$V_\chi(x) = \frac{3}{4\pi^2 L_H^4} 2d_\chi \sum_{m \geq 1} \frac{F(mz_\chi)}{m^5} (\cos(mx) + \cos(2mx)), \quad z_\chi = M_\chi L_H,$$

with

$$F(z) = e^{-z} \left(1 + z + \frac{z^2}{3} \right).$$

Thus the first-wrapping determinant density is

$$\mathcal{N}_\Gamma^\chi = 2d_\chi F(z_\chi).$$

For $d_\chi = 1$,

$$\mathcal{N}_\Gamma^\chi = 2F(z_\chi), \quad M_\chi = L_H^{-1} F^{-1} \left(\frac{\mathcal{N}_\Gamma^\chi}{2} \right),$$

with F^{-1} taken on $0 < F \leq 1$. Let Π_A denote projection of the pulled-back RR one-form to the retained current generator $A \in \{3, Y\}$. Split the localized coframe as

$$\theta_\Gamma^A = \theta_{\Gamma, \text{CP}}^A + q_A^{-1} \Pi_A \iota_\Gamma^* C_1, \quad A = 3, Y,$$

with curvature check

$$d_\Gamma \theta_\Gamma^A = d_\Gamma \theta_{\Gamma, \text{CP}}^A + q_A^{-1} \Pi_A \iota_\Gamma^* F_2.$$

The weak triplet is the equivariant completion of θ_Γ^3 , and the unit-current pass is

$$\langle \theta_\Gamma^a, \theta_\Gamma^b \rangle_{\Gamma, \Pi_Q} = L_{2, \Gamma}^2 \delta^{ab}, \quad e_\Gamma^a = L_{2, \Gamma}^{-1} \theta_\Gamma^a.$$

The neutral current matrix is

$$\mathcal{L}_\Gamma = \begin{pmatrix} K_{33}^\Gamma & K_{3Y}^\Gamma \\ K_{3Y}^\Gamma & K_{YY}^\Gamma \end{pmatrix}, \quad K_{AB}^\Gamma = \langle \theta_\Gamma^A, \theta_\Gamma^B \rangle_{\Gamma, \Pi_Q}.$$

The photon coframe and length are

$$\theta_\Gamma^Q = \theta_\Gamma^3 + \theta_\Gamma^Y, \quad L_Q^2 = (1, 1) \mathcal{L}_\Gamma \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

The neutral branch test becomes

$$\det \mathcal{L}_\Gamma = R_{\text{dV}} K_{33}^\Gamma L_Q^2, \quad R_{\text{dV}} = \frac{X_+(3/4)}{X_+(2)}.$$

In this row the vertical contribution is read directly as

$$L_{2, \text{vert}}^2 = q_3^{-2} \langle \Pi_3 \iota_\Gamma^* C_1, \Pi_3 \iota_\Gamma^* C_1 \rangle_{\Gamma, \Pi_Q}.$$

The Chern–Simons source for this one-form is

$$S_{\text{CS}}^\Gamma = \mu_8 \sum_{s \in \mathcal{S}_\Gamma} \epsilon_s \int_{\Gamma_{\text{EW}}^{(s)}} \sum_r C_{9-2r} \wedge [\mathcal{Y}_{\Gamma, s}]_{2r},$$

with

$$\mathcal{Y}_{\Gamma, s} = \text{ch}(\mathcal{F}_s) \sqrt{\frac{\widehat{A}(T\Gamma_s)}{\widehat{A}(N\Gamma_s)}}.$$

On the $\text{spin}^c \mathbb{CP}^2$ boundary row,

$$\mathcal{F}_Q / (2\pi) = f_m h_Q, \quad f_m = m + \frac{13}{2}, \quad \int_{\mathbb{CP}^2} h_Q^2 = 1,$$

and

$$\mathcal{Y}_\Gamma = N_8 \left(1 + f_m h_Q + \left(\frac{f_m^2}{2} - \frac{1}{16} \right) h_Q^2 \right).$$

Hence the C_1 variation gives the boundary source coefficient

$$\mathfrak{q}_{C_1} = N_8 \left(\frac{f_m^2}{2} - \frac{1}{16} \right), \quad \delta_{C_1} S_{\text{CS}}^\Gamma = \mu_8 \int_{M_4 \times S_Q^1} \delta C_{1,Q} \mathfrak{q}_{C_1}.$$

For the minimal spin^c local stack $m = 0$, $N_8 = 16$,

$$\mathfrak{q}_{C_1} = 337.$$

Define the total projected one-form source by adding the orientifold and bulk counterterms:

$$\mathcal{I}_{1,A}^\Gamma = \Pi_A(\mathfrak{q}_{C_1} ds_Q) + \mathcal{I}_{1,A}^{O8} + \mathcal{I}_{1,A}^{\text{bulk}}.$$

Compact solvability imposes the zero-mode equation

$$\langle \mathcal{I}_{1,A}^\Gamma, \omega_Q \rangle_\Gamma = 0, \quad \omega_Q = L_Q^{-1} ds_Q.$$

The Green operator acts on

$$\mathcal{I}_{1,A}^\perp = \mathcal{I}_{1,A}^\Gamma - \frac{\langle \mathcal{I}_{1,A}^\Gamma, \omega_Q \rangle_\Gamma}{\langle \omega_Q, \omega_Q \rangle_\Gamma} \omega_Q.$$

The projected Green solve is

$$\Pi_A \iota_\Gamma^* C_1 = \Pi_A C_{1,\text{harm}} + \mu_8 G_{1,\Gamma} \mathcal{I}_{1,A}^\perp, \quad A = 3, Y,$$

and therefore

$$L_{2,\text{vert}}^2 = q_3^{-2} \left\| \Pi_3 C_{1,\text{harm}} + \mu_8 G_{1,\Gamma} \mathcal{I}_{1,3}^\perp \right\|_{\Gamma, \Pi_Q}^2.$$

Choose a Γ -orthonormal eigenbasis $\{\alpha_i\}_{i \geq 1}$ of the zero-mode-orthogonal retained one-form sector:

$$\langle \alpha_i, \alpha_j \rangle_{\Gamma, \Pi_Q} = \delta_{ij}, \quad \Delta_{1,\Gamma} \alpha_i = \lambda_i \alpha_i, \quad \lambda_i > 0.$$

Set

$$B_{Ai} = \langle \alpha_i, \mathcal{I}_{1,A}^\perp \rangle_{\Gamma, \Pi_Q}, \quad P_{Ai} = \langle \alpha_i, \theta_{\Gamma, \text{CP}}^A \rangle_{\Gamma, \Pi_Q},$$

and

$$\Pi_A C_{1,\text{harm}} = h_A \omega_Q, \quad \Omega_Q = \langle \omega_Q, \omega_Q \rangle_{\Gamma, \Pi_Q}, \quad p_A = \langle \theta_{\Gamma, \text{CP}}^A, \omega_Q \rangle_{\Gamma, \Pi_Q}.$$

Then

$$G_{1,\Gamma} \mathcal{I}_{1,A}^\perp = \sum_{i \geq 1} \frac{B_{Ai}}{\lambda_i} \alpha_i,$$

and the retained coframe is

$$\theta_\Gamma^A = \theta_{\Gamma, \text{CP}}^A + q_A^{-1} h_A \omega_Q + q_A^{-1} \mu_8 \sum_{i \geq 1} \frac{B_{Ai}}{\lambda_i} \alpha_i.$$

Thus the current matrix entries are

$$\begin{aligned} K_{AB}^\Gamma &= K_{AB}^{\text{CP}} + q_A^{-1} h_A p_B + q_B^{-1} h_B p_A + q_A^{-1} q_B^{-1} h_A h_B \Omega_Q \\ &\quad + \mu_8 q_A^{-1} \sum_i \frac{B_{Ai} P_{Bi}}{\lambda_i} + \mu_8 q_B^{-1} \sum_i \frac{B_{Bi} P_{Ai}}{\lambda_i} + \mu_8^2 q_A^{-1} q_B^{-1} \sum_i \frac{B_{Ai} B_{Bi}}{\lambda_i^2}, \end{aligned}$$

where

$$K_{AB}^{\text{CP}} = \langle \theta_{\Gamma, \text{CP}}^A, \theta_{\Gamma, \text{CP}}^B \rangle_{\Gamma, \Pi_Q}.$$

On the product zero-mode reduction of the boundary circle, let $\varphi \in [0, 2\pi]$ parameterize S_Q^1 , with

$$ds_Q = R_Q d\varphi, \quad L_Q = 2\pi R_Q.$$

After the $\mathbb{CP}^2$ zero-mode projection, write

$$\mathcal{I}_{1,A}^\Gamma = \rho_A(\varphi) ds_Q.$$

The compact condition removes the average

$$\bar{\rho}_A = \frac{1}{2\pi} \int_0^{2\pi} \rho_A(\varphi) d\varphi, \quad \rho_A^\perp(\varphi) = \rho_A(\varphi) - \bar{\rho}_A.$$

With

$$\rho_A^\perp(\varphi) = \sum_{n \geq 1} (a_{A,n} \cos n\varphi + b_{A,n} \sin n\varphi),$$

the retained one-form eigenvalues are

$$\lambda_n = \frac{n^2}{R_Q^2} = \left(\frac{2\pi n}{L_Q} \right)^2.$$

In an orthonormal Fourier basis, the Green part of the current matrix is

$$K_{AB}^{\text{Green}} = \mu_8^2 q_A^{-1} q_B^{-1} \sum_{n \geq 1} \frac{a_{A,n} a_{B,n} + b_{A,n} b_{B,n}}{\lambda_n^2}.$$

The constant spin^c stack source

$$\rho_A(\varphi) = \Pi_A \mathfrak{q}_{C_1}$$

has $\rho_A^\perp = 0$ and contributes through the tadpole equation. Its Green-current contribution is

$$K_{AB}^{\text{Green}} = 0.$$

Thus a nonzero sourced vertical current from the CS row requires Fourier content in

$$\mathcal{I}_{1,A}^{\text{O8}} + \mathcal{I}_{1,A}^{\text{bulk}}$$

or a harmonic coefficient h_A in $\Pi_A C_{1,\text{harm}}$. Let

$$v_A = q_A^{-1} h_A, \quad A = 3, Y,$$

and collect the zero-mode-orthogonal part into

$$\widehat{C}_{AB} = K_{AB}^{\text{CP}} + K_{AB}^{\text{Green}} + K_{AB}^{\text{CP-Green}}.$$

With

$$p_A = \langle \theta_{\Gamma, \text{CP}}^A, \omega_Q \rangle_{\Gamma, \Pi_Q}, \quad \Omega_Q = \langle \omega_Q, \omega_Q \rangle_{\Gamma, \Pi_Q},$$

the neutral current matrix is

$$\mathcal{L}_{AB}^\Gamma = \widehat{C}_{AB} + v_A p_B + v_B p_A + \Omega_Q v_A v_B.$$

Set

$$C_Q = (1, 1) \widehat{C} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad p_Q = p_3 + p_Y, \quad v_Q = v_3 + v_Y.$$

Then the branch-current closure condition becomes

$$\det \mathcal{L}^\Gamma = R_{\text{dV}} \left(\widehat{C}_{33} + 2v_3 p_3 + \Omega_Q v_3^2 \right) (C_Q + 2v_Q p_Q + \Omega_Q v_Q^2).$$

For the product specialization $p_A = 0$ with vanishing perpendicular Green contribution,

$$\mathcal{L}^\Gamma = C + \Omega_Q v v^T,$$

and therefore

$$\begin{aligned} \det C + \Omega_Q (C_{YY}v_3^2 - 2C_{3Y}v_3v_Y + C_{33}v_Y^2) \\ = R_{\text{dV}} (C_{33} + \Omega_Q v_3^2) (C_Q + \Omega_Q (v_3 + v_Y)^2). \end{aligned}$$

This specialization gives

$$L_{2,\text{vert}}^2 = \Omega_Q v_3^2.$$

For a fixed harmonic direction

$$t = \frac{v_Y}{v_3}, \quad u = \Omega_Q v_3^2,$$

define

$$\Delta_C = \det C, \quad C_Q = C_{33} + 2C_{3Y} + C_{YY}, \quad C_N = C_{33} - 2C_{3Y} + C_{YY}.$$

The same branch equation becomes

$$A_t u^2 + B_t u + D_C = 0,$$

where

$$\begin{aligned} A_t &= R_{\text{dV}}(1+t)^2, \\ B_t &= R_{\text{dV}} (C_{33}(1+t)^2 + C_Q) - (C_{YY} - 2tC_{3Y} + t^2C_{33}), \end{aligned}$$

and

$$D_C = R_{\text{dV}} C_{33} C_Q - \Delta_C.$$

For $t \neq -1$,

$$u_{\pm}(t) = \frac{-B_t \pm \sqrt{B_t^2 - 4A_t D_C}}{2A_t}.$$

The massive-neutral harmonic alignment $v_Y = -v_3$ has $v_Q = 0$ and

$$L_Q^2 = C_Q, \quad u_N = \frac{D_C}{(1 - R_{\text{dV}})C_Q}.$$

The Q -parallel alignment $v_Y = v_3$ obeys

$$4R_{\text{dV}}u^2 + (R_{\text{dV}}(4C_{33} + C_Q) - C_N)u + D_C = 0.$$

Normalize the circle one-form by

$$\int_{S_Q^1} \omega_Q = 1.$$

The harmonic RR potential is parametrized by Wilson-line variables

$$\Pi_A C_{1,\text{harm}} = 2\pi \vartheta_A \omega_Q, \quad \vartheta_A \in \mathbb{R}/\mathbb{Z}, \quad A = 3, Y.$$

Consequently

$$v_A = 2\pi q_A^{-1} \vartheta_A,$$

and, for $v_3 \neq 0$,

$$t = \frac{q_3 \vartheta_Y}{q_Y \vartheta_3}, \quad u = (2\pi)^2 \Omega_Q q_3^{-2} \vartheta_3^2.$$

The massive-neutral harmonic row is

$$q_3^{-1} \vartheta_3 + q_Y^{-1} \vartheta_Y = 0,$$

and the Q -parallel harmonic row is

$$q_3^{-1} \vartheta_3 - q_Y^{-1} \vartheta_Y = 0.$$

Retaining the CP-circle cross vector $p = (p_3, p_Y)$ gives a closed photon-preserving row. Put

$$v = a \begin{pmatrix} 1 \\ -1 \end{pmatrix}, \quad C = \begin{pmatrix} A & B \\ B & D \end{pmatrix}, \quad C_Q = A + 2B + D, \quad p_Q = p_3 + p_Y,$$

and

$$S_p = p_3(D + B) - p_Y(A + B).$$

Then

$$\det(C + vp^T + pv^T + \Omega_Q vv^T) = \Delta_C + 2aS_p + a^2(\Omega_Q C_Q - p_Q^2).$$

Since $L_Q^2 = C_Q$, branch closure becomes

$$((1 - R_{\text{dV}})\Omega_Q C_Q - p_Q^2) a^2 + 2(S_p - R_{\text{dV}} p_3 C_Q) a - D_C = 0.$$

Equivalently, with $u = \Omega_Q a^2$, $\epsilon = \text{sign}(a)$, and

$$A_N = (1 - R_{\text{dV}})C_Q - \frac{p_Q^2}{\Omega_Q}, \quad B_N = \frac{S_p - R_{\text{dV}} p_3 C_Q}{\sqrt{\Omega_Q}},$$

the positive square-root variable $y = \sqrt{u}$ obeys

$$A_N y^2 + 2\epsilon B_N y - D_C = 0.$$

Thus the harmonic square is

$$u_\epsilon = \left(\frac{-\epsilon B_N + \sqrt{B_N^2 + A_N D_C}}{A_N} \right)^2, \quad A_N \neq 0,$$

with affine limit $u_\epsilon = D_C^2/(4B_N^2)$ when $A_N = 0$ and $\epsilon B_N > 0$. For $p_A = 0$, the row gives

$$u_N = \frac{D_C}{(1 - R_{\text{dV}})C_Q}.$$

The cross vector has a direct DBI readout. Let s_Q be arclength on the Q -circle, $0 \leq s_Q < L_Q$, and take

$$\theta_{\Gamma, \text{CP}}^A = \theta_i^A(y, \sigma) dy^i, \quad \omega_Q = \frac{ds_Q}{L_Q}.$$

With sector density

$$d\nu_{\Gamma, \sigma} = \kappa_8 \omega_\sigma e^{-\phi_\sigma} \sqrt{\gamma_{\Gamma, \sigma}} d^4 y ds_Q,$$

one obtains

$$p_A = \frac{1}{L_Q} \sum_{\sigma \in \mathcal{S}_\Gamma} \int_{\Gamma_{\text{int}, \sigma}} d\nu_{\Gamma, \sigma} \gamma_\sigma^{is_Q} \text{tr}_{E_\sigma} (\Pi_Q \theta_i^A \Pi_Q),$$

and

$$\Omega_Q = \frac{1}{L_Q^2} \sum_{\sigma \in \mathcal{S}_\Gamma} \int_{\Gamma_{\text{int}, \sigma}} d\nu_{\Gamma, \sigma} \gamma_\sigma^{s_Q s_Q} \text{tr}_{E_\sigma} \Pi_Q.$$

For a block-product internal metric,

$$\gamma_\sigma^{is_Q} = 0, \quad \gamma_\sigma^{s_Q s_Q} = 1,$$

hence

$$p_3 = p_Y = 0, \quad \Omega_Q = \frac{Z_{\Gamma, Q}}{L_Q}.$$

A nonzero p_A records vertical leakage of the retained CP coframe or an off-diagonal frame contribution after choosing the adapted Q -circle one-form. Equivalently, for

$$\chi_Q = ds_Q + \mathcal{A}_i(y)dy^i, \quad \omega_Q = \frac{\chi_Q}{L_Q},$$

decompose

$$\theta_{\Gamma, \text{CP}}^A = \theta_H^A + \ell_A \chi_Q, \quad \gamma_{\Gamma}^{-1}(\theta_H^A, \chi_Q) = 0.$$

Then

$$p_A = \frac{1}{L_Q} \sum_{\sigma \in \mathcal{S}_{\Gamma}} \int_{\Gamma_{\text{int}, \sigma}} d\nu_{\Gamma, \sigma} \ell_A \text{tr}_{E_{\sigma}} \Pi_Q.$$

Thus the horizontal adapted row has $p_A = 0$, and the leakage functions ℓ_3, ℓ_Y are the localized sources for the cross-vector branch. Pointwise,

$$\ell_A = \frac{\gamma_{\Gamma}^{-1}(\theta_{\Gamma, \text{CP}}^A, \chi_Q)}{\gamma_{\Gamma}^{-1}(\chi_Q, \chi_Q)}.$$

For the product representative

$$\pi_{\mathbb{CP}^2} : \mathbb{CP}^2 \times S_Q^1 \longrightarrow \mathbb{CP}^2, \quad \theta_{\Gamma, \text{CP}}^A = \pi_{\mathbb{CP}^2}^* \alpha_A,$$

and adapted metric

$$ds_{\Gamma}^2 = g_{ij}(y)dy^i dy^j + L_Q^2 \chi_Q^2,$$

one obtains

$$\gamma_{\Gamma}^{-1}(\pi_{\mathbb{CP}^2}^* \alpha_A, \chi_Q) = 0,$$

hence

$$\ell_3 = \ell_Y = 0, \quad p_3 = p_Y = 0, \quad \Omega_Q = \frac{Z_{\Gamma, Q}}{L_Q}.$$

A cross-vector branch with $p_A \neq 0$ is a separate leakage realization in which the retained CP coframe has a χ_Q -component after adapted projection. For the product $\mathbb{CP}^2 \times S_Q^1$ row, take

$$\omega_Q = \frac{ds_Q}{L_Q}, \quad \int_{S_Q^1} \omega_Q = 1.$$

The CP2 zero-mode DBI density is

$$Z_{\Gamma, Q} = \sum_s \int_{\mathbb{CP}_s^2} \kappa_8 \omega_s e^{-\phi_s} \sqrt{\gamma_{\mathbb{CP}^2, s}} \text{tr}_{E_s} \Pi_Q.$$

Hence

$$\Omega_Q = \langle \omega_Q, \omega_Q \rangle_{\Gamma, \Pi_Q} = \frac{Z_{\Gamma, Q}}{L_Q},$$

and the harmonic weak vertical length is

$$L_{2, \text{vert}}^2 = (2\pi)^2 \frac{Z_{\Gamma, Q}}{L_Q} q_3^{-2} \vartheta_3^2.$$

Define the finite trace ratio

$$\tau_Q = \frac{Z_{\Gamma, Q}}{Z_{\Gamma, 0}}.$$

Then

$$L_{2, \text{vert}}^2 = (2\pi)^2 \frac{\tau_Q Z_{\Gamma, 0}}{L_Q} q_3^{-2} \vartheta_3^2.$$

For a shared CP2 density profile and normalized endpoint vector η_0 ,

$$\tau_Q = \text{tr}_{E_Q} \Pi_Q.$$

At $Q = 7$, the full retained E_Q trace gives

$$\tau_Q = 16.$$

Using the integer residual convention

$$q_3 = 3Q,$$

and the full retained trace $\tau_Q = 2(Q + 1)$, this becomes

$$L_{2,\text{vert}}^2 = \frac{8\pi^2(Q+1)}{9Q^2} \frac{Z_{\Gamma,0}}{L_Q} \delta\vartheta_3^2,$$

where $\delta\vartheta_3$ is the shortest representative of $\vartheta_3 \in \mathbb{R}/\mathbb{Z}$. For $Q = 7$,

$$L_{2,\text{vert}}^2 = \frac{64\pi^2}{441} \frac{Z_{\Gamma,0}}{L_Q} \delta\vartheta_3^2.$$

Equivalently,

$$\delta\vartheta_3 = \frac{21}{8\pi} \sqrt{\frac{L_Q L_{2,\text{vert}}^2}{Z_{\Gamma,0}}}.$$

The Wilson chamber $|\delta\vartheta_3| \leq \frac{1}{2}$ gives

$$L_{2,\text{vert}}^2 \leq \frac{16\pi^2}{441} \frac{Z_{\Gamma,0}}{L_Q}.$$

For a nonzero harmonic representative, define

$$\Xi_Q^{\text{harm}} = \frac{Z_{\Gamma,0}}{L_Q L_{2,\text{vert}}^2}.$$

Then the full-trace row gives

$$\Xi_Q^{\text{harm}} = \frac{9Q^2}{8\pi^2(Q+1)} \frac{1}{\delta\vartheta_3^2},$$

and, at $Q = 7$,

$$\Xi_7^{\text{harm}} = \frac{441}{64\pi^2} \frac{1}{\delta\vartheta_3^2}.$$

The fundamental Wilson chamber gives

$$\Xi_7^{\text{harm}} \geq \frac{441}{16\pi^2} = 2.792665124141 \dots$$

The D9 global-form descent fixes $q_3 = 3Q$. For the retained $Q = 7$ stack, $q_3 = 21$, so

$$\Theta_N(21, q_Y) = \frac{1}{2} \min \left(1, \left| \frac{21}{q_Y} \right| \right).$$

Equivalently,

$$\Theta_N(21, q_Y) = \frac{1}{2} \quad (|q_Y| \leq 21),$$

and

$$\Theta_N(21, q_Y) = \frac{21}{2|q_Y|} \quad (|q_Y| > 21).$$

For the endpoint character in the same retained $Q = 7$ row,

$$c_1(L_Y) = 6h + Q\eta, \quad c_1(L_{T_3}) = \eta, \quad c_1(L_Y) - Q c_1(L_{T_3}) = 6h.$$

The quotient line operator

$$\mathcal{O}_Y(\gamma_Q) = \text{Hol}_{L_Y}(\gamma_Q) \text{Hol}_{L_{T_3}}(\gamma_Q)^{-Q}$$

has Chern class $6h$. Since h pulls back from $\mathbb{CP}^2$, this operator has unit S_Q^1 character. The hypercharge boundary holonomy on γ_Q is therefore the Q -fold weak endpoint holonomy:

$$\text{Hol}_{L_Y}(\gamma_Q) = \text{Hol}_{L_{T_3}}(\gamma_Q)^Q.$$

For the retained endpoint vector $\eta_Q = \eta_0^{\odot Q}$,

$$T_3 \eta_Q = -\frac{Q}{2} \eta_Q, \quad Y \eta_Q = \frac{Q}{2} \eta_Q.$$

With integer hypercharge $y = 6Y$, the hypercharge Wilson period is

$$q_Y = 3Q = 21.$$

Consequently

$$\Theta_N(21, 21) = \frac{1}{2}, \quad \delta\vartheta_Y = -\delta\vartheta_3.$$

For the photon-preserving massive-neutral branch,

$$v_Q = 0, \quad \delta\vartheta_Y = -\frac{q_Y}{q_3} \delta\vartheta_3.$$

Inside the identity chamber, both Wilson representatives obey

$$|\delta\vartheta_3| \leq \Theta_N(q_3, q_Y), \quad \Theta_N(q_3, q_Y) = \frac{1}{2} \min \left(1, \left| \frac{q_3}{q_Y} \right| \right).$$

Thus

$$\Xi_Q^{\text{harm}} \geq \frac{9Q^2}{8\pi^2(Q+1)\Theta_N(q_3, q_Y)^2},$$

with the $Q = 7$ specialization

$$\Xi_7^{\text{harm}} \geq \frac{441}{64\pi^2\Theta_N(q_3, q_Y)^2}.$$

For the massive-neutral amplitude

$$u_N = \frac{D_C}{(1 - R_{\text{dV}})C_Q},$$

the density condition becomes

$$\frac{Z_{\Gamma,0}}{L_Q} \geq \frac{9Q^2}{8\pi^2(Q+1)\Theta_N(q_3, q_Y)^2} \frac{D_C}{(1 - R_{\text{dV}})C_Q}.$$

At $Q = 7$,

$$\frac{Z_{\Gamma,0}}{L_Q} \geq \frac{441}{64\pi^2\Theta_N(q_3, q_Y)^2} \frac{D_C}{(1 - R_{\text{dV}})C_Q}.$$

With $q_3 = 21$, the same condition is

$$\frac{Z_{\Gamma,0}}{L_Q} \geq \frac{441}{16\pi^2} \frac{D_C}{(1 - R_{\text{dV}})C_Q} \quad (|q_Y| \leq 21),$$

and

$$\frac{Z_{\Gamma,0}}{L_Q} \geq \frac{q_Y^2}{16\pi^2} \frac{D_C}{(1 - R_{\text{dv}})C_Q} \quad (|q_Y| > 21).$$

For the endpoint-character value $q_Y = 21$, the density condition reduces to

$$\frac{Z_{\Gamma,0}}{L_Q} \geq \frac{441}{16\pi^2} \frac{D_C}{(1 - R_{\text{dv}})C_Q}.$$

Equivalently,

$$U_{\Gamma}^{(21)} = \frac{16\pi^2}{441} \frac{Z_{\Gamma,0}}{L_Q}$$

obeys

$$U_{\Gamma}^{(21)} \geq \frac{D_C}{(1 - R_{\text{dv}})C_Q}.$$

The corresponding shortest Wilson representatives satisfy

$$\delta\vartheta_3^2 = \delta\vartheta_Y^2 = \frac{1}{4U_{\Gamma}^{(21)}} \frac{D_C}{(1 - R_{\text{dv}})C_Q}.$$

The product DBI density factors the endpoint norm. Write

$$d\nu_{\Gamma,s} = d\mu_{\Gamma,s}^{\mathbb{CP}^2} ds_Q, \quad d\mu_{\Gamma,s}^{\mathbb{CP}^2} = \kappa_8 \omega_s e^{-\phi_s} \sqrt{\gamma_{\mathbb{CP}^2,s}} d^4 y.$$

For an S_Q^1 -invariant neutral endpoint section,

$$Z_{\Gamma,0} = L_Q \mathcal{Z}_{\Gamma,0}^{\mathbb{CP}^2},$$

where

$$\mathcal{Z}_{\Gamma,0}^{\mathbb{CP}^2} = \sum_s \int_{\mathbb{CP}_s^2} d\mu_{\Gamma,s}^{\mathbb{CP}^2} \text{tr}_{E_s} (\Pi_Q |\eta_Q|^2 \Pi_Q).$$

Hence

$$U_{\Gamma}^{(21)} = \frac{16\pi^2}{441} \mathcal{Z}_{\Gamma,0}^{\mathbb{CP}^2}.$$

The localized density condition becomes

$$\mathcal{Z}_{\Gamma,0}^{\mathbb{CP}^2} \geq \frac{441}{16\pi^2} \frac{D_C}{(1 - R_{\text{dv}})C_Q}.$$

Let $\mathfrak{z}_{\Gamma}^{\mathbb{CP}^2}$ denote the common CP2 density per finite endpoint state,

$$\mathfrak{z}_{\Gamma}^{\mathbb{CP}^2} = \sum_s \int_{\mathbb{CP}_s^2} d\mu_{\Gamma,s}^{\mathbb{CP}^2}.$$

The unit rank-one endpoint normalization gives

$$\text{tr}_{E_Q} (\Pi_Q |\eta_Q\rangle \langle \eta_Q| \Pi_Q) = 1,$$

and the full retained trace gives

$$\text{tr}_{E_Q} \Pi_Q = 2(Q + 1) = 16.$$

Thus

$$\mathcal{Z}_{\Gamma,0}^{\mathbb{CP}^2} = \mathfrak{z}_{\Gamma}^{\mathbb{CP}^2}, \quad \mathcal{Z}_{\Gamma,Q}^{\mathbb{CP}^2} = 16 \mathfrak{z}_{\Gamma}^{\mathbb{CP}^2}, \quad \tau_Q = 16.$$

The D8/O8 density datum is the scalar $\mathfrak{z}_\Gamma^{\mathbb{CP}^2}$. In a constant product profile, ω_s , ϕ_s , and $\gamma_{\mathbb{CP}^2,s}$ are constant on each retained sector, giving

$$\mathfrak{z}_\Gamma^{\mathbb{CP}^2} = \kappa_8 \sum_s \omega_s e^{-\phi_s} \text{Vol}(\mathbb{CP}_s^2, \gamma_{\mathbb{CP}^2,s}).$$

The density-dominated pass is therefore the effective volume inequality

$$\kappa_8 \sum_s \omega_s e^{-\phi_s} \text{Vol}(\mathbb{CP}_s^2, \gamma_{\mathbb{CP}^2,s}) \geq 9.724809426610 \dots$$

For a common CP2 metric and dilaton, this becomes

$$\kappa_8 e^{-\phi_0} \Omega_\Gamma \text{Vol}(\mathbb{CP}^2, \gamma_{\mathbb{CP}^2}) \geq 9.724809426610 \dots, \quad \Omega_\Gamma = \sum_s \omega_s.$$

Use the same WZ normalization as the CS source row,

$$\int_{\mathbb{CP}^2} h_Q^2 = 1.$$

Writing

$$J_{\Gamma,s} = r_{\Gamma,s} h_Q$$

gives

$$\text{Vol}(\mathbb{CP}_s^2, \gamma_{\mathbb{CP}^2,s}) = \frac{1}{2} \int_{\mathbb{CP}_s^2} J_{\Gamma,s}^2 = \frac{1}{2} r_{\Gamma,s}^2.$$

Thus the density condition is

$$\frac{\kappa_8}{2} \sum_s \omega_s e^{-\phi_s} r_{\Gamma,s}^2 \geq \frac{441}{16\pi^2} \frac{D_C}{(1 - R_{\text{dV}}) C_Q}.$$

The density-dominated metric pass is

$$\sum_s \omega_s e^{-\phi_s} r_{\Gamma,s}^2 \geq \frac{19.449618853220 \dots}{\kappa_8}.$$

For a common metric and dilaton,

$$r_\Gamma^2 \geq \frac{19.449618853220 \dots}{\kappa_8 e^{-\phi_0} \Omega_\Gamma}.$$

The density scalar fixes $U_\Gamma^{\mathbb{CP}^2}$. The point-row shape parameter comes from a separate normalized Hodge readout. Split the retained weak coframe into a unit endpoint component and a horizontal excess,

$$\theta_\Gamma^3 = \theta_{3,0} + \theta_{3,h}, \quad \langle \theta_{3,0}, \theta_{3,h} \rangle_\Gamma = 0.$$

Define

$$C_0^\Gamma = \sum_s \int_{\mathbb{CP}_s^2} d\mu_{\Gamma,s}^{\mathbb{CP}^2} \text{tr}_{E_s} (\Pi_Q |\theta_{3,0}|^2 \Pi_Q),$$

and

$$C_h^\Gamma = \sum_s \int_{\mathbb{CP}_s^2} d\mu_{\Gamma,s}^{\mathbb{CP}^2} \text{tr}_{E_s} (\Pi_Q |\theta_{3,h}|^2 \Pi_Q).$$

The point-evaluation row reads

$$\widehat{C}_{33}^\Gamma = \frac{C_0^\Gamma + C_h^\Gamma}{C_0^\Gamma} = 1 + \frac{Q^2 + 1}{(Q + 1)^2} q_\Gamma.$$

Hence the localized shape readout is

$$q_\Gamma = \frac{(Q+1)^2}{Q^2+1} \frac{C_h^\Gamma}{C_0^\Gamma}.$$

For $Q = 7$,

$$q_\Gamma = \frac{32}{25} \frac{C_h^\Gamma}{C_0^\Gamma}.$$

Define the dimensionless shape ratio

$$\varrho_\Gamma = \frac{C_h^\Gamma}{C_0^\Gamma}.$$

Then

$$q_\Gamma = \frac{32}{25} \varrho_\Gamma, \quad 0 < \varrho_\Gamma < \varrho_c, \quad \varrho_c = \frac{25}{32} q_c = 2.743448440896 \dots$$

The point-row amplitude as a function of the localized shape ratio is

$$u_\Gamma^{\text{sh}}(\varrho_\Gamma) = \frac{625 R_{\text{dV}}(1 + \varrho_\Gamma)^2 - 576 \varrho_\Gamma^2 - 900 \varrho_\Gamma}{625(1 - R_{\text{dV}})(1 + \varrho_\Gamma)}.$$

The charge chamber becomes the pair of localized inequalities

$$U_\Gamma^{\mathbb{CP}^2} \geq u_\Gamma^{\text{sh}}(\varrho_\Gamma), \quad 0 < \varrho_\Gamma < \varrho_c.$$

Equivalently, the localized density satisfies

$$\mathcal{Z}_{\Gamma,0}^{\mathbb{CP}^2} \geq \frac{441}{16\pi^2} u_\Gamma^{\text{sh}}(\varrho_\Gamma).$$

In the constant-profile metric row this becomes

$$\sum_s \omega_s e^{-\phi_s} r_{\Gamma,s}^2 \geq \frac{441}{8\pi^2 \kappa_8} u_\Gamma^{\text{sh}}(\varrho_\Gamma).$$

For a common metric and dilaton,

$$r_\Gamma^2 \geq \frac{441}{8\pi^2 \kappa_8 e^{-\phi_0} \Omega_\Gamma} u_\Gamma^{\text{sh}}(\varrho_\Gamma).$$

The point-evaluation row gives a concrete base matrix. Put

$$q = \lambda^2, \quad t_Q = -\frac{Q}{Q+1},$$

and

$$C_{33}^{\text{pt}} = 1 + \frac{Q^2 + 1}{(Q+1)^2} q.$$

Then

$$C_Q^{\text{pt}} = (Q+1)^2 C_{33}^{\text{pt}}, \quad \Delta_C^{\text{pt}} = (Q-1)^2 q(q+2),$$

so

$$u_N^{\text{pt}}(q) = \frac{R_{\text{dV}}(Q+1)^2 (C_{33}^{\text{pt}})^2 - (Q-1)^2 q(q+2)}{(1 - R_{\text{dV}})(Q+1)^2 C_{33}^{\text{pt}}}.$$

For $Q = 7$,

$$u_N^{\text{pt}}(q) = \frac{R_{\text{dV}}(25q+32)^2 - 576q(q+2)}{32(1 - R_{\text{dV}})(25q+32)}.$$

The zero of this amplitude is

$$q_c = 3.511614004347 \dots$$

The derivative is

$$\frac{du_N^{\text{pt}}}{dq} = -\frac{(14400 - 15625R_{\text{dV}})q^2 + (36864 - 40000R_{\text{dV}})q + (36864 - 25600R_{\text{dV}})}{32(1 - R_{\text{dV}})(25q + 32)^2}.$$

Since $0 < R_{\text{dV}} < 1$, the amplitude is strictly decreasing for $q > 0$, with

$$u_N^{\text{pt}}(0) = \frac{R_{\text{dV}}}{1 - R_{\text{dV}}} = 3.482268368857 \dots, \quad u_N^{\text{pt}}(q_c) = 0.$$

Together with the product DBI readout, this gives a density-dominated pass:

$$\mathcal{Z}_{\Gamma,0}^{\mathbb{CP}^2} \geq \frac{441}{16\pi^2} \frac{R_{\text{dV}}}{1 - R_{\text{dV}}} = 9.724809426610 \dots$$

In that regime the full interval $0 < q < q_c$ satisfies the endpoint-character chamber condition. For lower positive density, set

$$U_{\Gamma}^{\mathbb{CP}^2} = \frac{16\pi^2}{441} \mathcal{Z}_{\Gamma,0}^{\mathbb{CP}^2}.$$

Then the active interval is

$$q_{\text{sat}}(U_{\Gamma}^{\mathbb{CP}^2}) \leq q < q_c.$$

For $0 < q < q_c$, the photon-preserving harmonic lift has positive amplitude, and the $Q = 7$ density condition becomes

$$\frac{Z_{\Gamma,0}}{L_Q} \geq \frac{441 (R_{\text{dV}}(25q + 32)^2 - 576q(q + 2))}{2048\pi^2 \Theta_N(q_3, q_Y)^2 (1 - R_{\text{dV}})(25q + 32)}.$$

Define the normalized density

$$U_{\Gamma} = \frac{64\pi^2 \Theta_N(q_3, q_Y)^2}{441} \frac{Z_{\Gamma,0}}{L_Q}.$$

The chamber condition is $U_{\Gamma} \geq u_N^{\text{pt}}(q)$. On the saturation curve $U_{\Gamma} = u_N^{\text{pt}}(q)$, q solves

$$A_q q^2 + B_q(U_{\Gamma})q + C_q(U_{\Gamma}) = 0,$$

where

$$A_q = 576 - 625R_{\text{dV}},$$

$$B_q(U_{\Gamma}) = 1152 - 1600R_{\text{dV}} + 800(1 - R_{\text{dV}})U_{\Gamma},$$

and

$$C_q(U_{\Gamma}) = 1024((1 - R_{\text{dV}})U_{\Gamma} - R_{\text{dV}}).$$

For

$$0 \leq U_{\Gamma} \leq \frac{R_{\text{dV}}}{1 - R_{\text{dV}}},$$

the positive branch is

$$q_{\text{sat}}(U_{\Gamma}) = \frac{-B_q(U_{\Gamma}) + \sqrt{B_q(U_{\Gamma})^2 - 4A_q C_q(U_{\Gamma})}}{2A_q}.$$

The chamber pass is $q_{\text{sat}}(U_{\Gamma}) \leq q < q_c$. Given q and U_{Γ} , the Wilson representative is

$$\delta\vartheta_3^2 = \Theta_N(q_3, q_Y)^2 \frac{u_N^{\text{pt}}(q)}{U_{\Gamma}}.$$

A quadratic local expansion has the form

$$S_\Gamma^{(2)} = \int_{M_4} \left[-\frac{1}{4} \mathcal{K}_{AB}^\Gamma F_{\mu\nu}^A F^{B\mu\nu} + \frac{1}{2} (f_H^2)_{ab} \partial_\mu \theta_H^a \partial^\mu \theta_H^b \right. \\ \left. + \frac{1}{2} \langle r, \mathcal{M}_{\text{img}} r \rangle_\Gamma + L_2 \mathfrak{j}_A \langle r, \mathcal{J}_A \rangle_\Gamma + \frac{1}{2} \text{Str}_\chi \log \mathcal{B}_\Gamma^\chi(\theta_H) \right].$$

The current matrix is

$$\mathcal{K}_{AB}^\Gamma = \int_{\Gamma_{\text{int}}} d\nu_\Gamma \gamma_\Gamma^{-1}(\theta_\Gamma^A, \theta_\Gamma^B), \quad A, B = 3, Y, Q.$$

The image source functions are

$$\mathcal{J}_A = \gamma_\Gamma^{-1}(\theta_\Gamma^A, \Theta_{\text{img}}), \quad A = 3, Y.$$

Eliminating r gives

$$\Delta S_{\text{eff}} = -\frac{1}{2} L_2^2 \mathfrak{j}_A \mathcal{D}_{AB} \mathfrak{j}_B, \quad \mathcal{D}_{AB} = L_2^{-2} \langle \mathcal{J}_A, \mathcal{M}_{\text{img}}^{-1} \mathcal{J}_B \rangle_\Gamma.$$

Thus the same local expression yields

$$L_2^2 = \mathcal{K}_{33}^\Gamma, \quad L_Q^2 = (1, 1) \mathcal{K}_{(3,Y)}^\Gamma \begin{pmatrix} 1 \\ 1 \end{pmatrix},$$

$$\Delta_{\text{rank}} = \mathcal{D}_{33} \mathcal{D}_{YY} - \mathcal{D}_{3Y}^2, \quad \rho_D = \frac{\mathcal{D}_{3Y}}{\mathcal{D}_{33}},$$

and the Wilson kinetic entry

$$(f_H^2)_{ab} = \int_{\Gamma_{\text{int}}} d\nu_\Gamma g^{ss} \omega_{H,s} \omega_{H,s} \text{tr}_{\text{end}}(n_H^c T_c n_H^d T_d) \Pi_{ab}^{\text{Schur}}.$$

For the point-projector endpoint metric this row gives

$$Z_{\Gamma,H} = \frac{Q}{4} Z_{\Gamma,0}.$$

The charged determinant readout is

$$\mathcal{N}_\Gamma^\chi = \text{Str}_{\ker \mathcal{B}_\Gamma^\chi}(\tau_\alpha) = 2\sigma_\chi d_\chi F(z)$$

on the equal-threshold Borel–Weil charged pair. The single local coefficient test is therefore

$$\left(\mathcal{D}_{AB}, \frac{Z_{\Gamma,H}}{Z_{\Gamma,0}}, \mathcal{N}_\Gamma^\chi \right) = \left(\mathcal{D}_\sigma, \frac{Q}{4}, 2\sigma_\chi d_\chi F(z) \right).$$

7.3 Finite-Matrix Extraction

Choose a Γ -orthonormal basis $\{e_i\}_{i=1}^N$ for the sourced image subspace and define

$$M_{ij} = \langle e_i, \mathcal{M}_{\text{img}} e_j \rangle_\Gamma, \quad C_{Ai} = \langle e_i, \mathcal{J}_A \rangle_\Gamma, \quad A = 3, Y.$$

The image correction is

$$\mathcal{D}_{AB} = L_2^{-2} C_{Ai} (M^{-1})^{ij} C_{Bj}.$$

With

$$w_A^i = L_2^{-1} (M^{-1/2})^{ij} C_{Aj},$$

one has

$$\mathcal{D}_{AB} = w_A \cdot w_B.$$

The Schur branch condition is the finite-vector equation

$$w_Y = \rho_\sigma w_3, \quad \|w_3\|^2 = \delta_\sigma.$$

Thus any localized image spectrum produces the two Schur numbers by a direct matrix inversion. Sourced modes away from this line contribute to Δ_{rank} ; unsourced modes drop out of $\mathcal{D}_{AB}$.

For the charged endpoint determinant, let

$$(\mathcal{B}_{r,\Gamma}^\chi)^\dagger \mathcal{B}_{r,\Gamma}^\chi u_{\alpha r} = M_{\alpha r}^2 u_{\alpha r}, \quad r = 1, 2.$$

The equal-amplitude Borel–Weil row is the spectral equality

$$\{(\sigma_{\alpha 1}, d_{\alpha 1}, M_{\alpha 1})\}_\alpha = \{(\sigma_{\alpha 2}, d_{\alpha 2}, M_{\alpha 2})\}_\alpha.$$

It gives

$$\mathcal{N}_\Gamma^\chi = 2 \sum_\alpha \sigma_\alpha d_\alpha F(M_\alpha L_H).$$

The minimal endpoint row is

$$M_\alpha = M, \quad \mathcal{N}_\Gamma^\chi = 2\sigma_\chi d_\chi F(M L_H).$$

7.4 Minimal Finite Spectrum Package

The finite extraction has a minimal image realization on

$$\mathcal{H}_{\text{img},\sigma}^{\text{src}} = \text{span}\{e_\sigma\}, \quad \langle e_\sigma, e_\sigma \rangle_\Gamma = 1, \quad \mathcal{M}_{\text{img}} e_\sigma = m_\sigma e_\sigma.$$

The source couplings are

$$C_{3\sigma} = L_2 \sqrt{m_\sigma} \sqrt{\delta_\sigma}, \quad C_{Y\sigma} = L_2 \sqrt{m_\sigma} \sqrt{\delta_\sigma} \rho_\sigma.$$

Equivalently,

$$\frac{1}{L_2 \sqrt{m_\sigma}} \begin{pmatrix} C_{3\sigma} \\ C_{Y\sigma} \end{pmatrix} = g_\sigma.$$

For the two branches,

branch	$(g_\sigma)_3$	$(g_\sigma)_Y$
+	0.374811401563...	7.371454070999...
−	0.572116571654...	7.230005311453....

Substitution gives

$$\mathcal{D}_{AB} = \frac{C_{A\sigma} C_{B\sigma}}{m_\sigma L_2^2} = (g_\sigma)_A (g_\sigma)_B = (\mathcal{D}_\sigma)_{AB}.$$

In the constant-profile product row, the same density gives the normalized source mode

$$\psi_{\Gamma,0} = \left(L_Q \mathfrak{z}_\Gamma^{\mathbb{CP}^2} \right)^{-1/2} \eta_Q, \quad \int_{\Gamma_{\text{EW,int}}} d\nu_\Gamma \text{tr}_{E_Q} (\Pi_Q |\psi_{\Gamma,0}|^2 \Pi_Q) = 1.$$

Taking $e_\sigma = \psi_{\Gamma,0}$, the branch source current is

$$\mathcal{J}_A^{(\sigma)} = L_2 \sqrt{m_\sigma} (g_\sigma)_A e_\sigma^b, \quad A = 3, Y.$$

For the current Gram matrix

$$G_{AB}^{(0)} = \int_{\Gamma_{\text{EW,int}}} d\nu_{\Gamma} \psi_{\Gamma,0}^2 \gamma_{\Gamma}^{-1}(\theta_{\Gamma}^A, \theta_{\Gamma}^B),$$

the Gram-dual image one-form is

$$\Xi_{\sigma} = L_2 \psi_{\Gamma,0} (G^{(0)})^{-1 BC} (g_{\sigma})_C \theta_{\Gamma}^B.$$

It satisfies

$$L_2^{-1} \int d\nu_{\Gamma} \psi_{\Gamma,0} \gamma_{\Gamma}^{-1}(\theta_{\Gamma}^A, \Xi_{\sigma}) = (g_{\sigma})_A, \quad R_{\Xi,\sigma} = 0.$$

The constant-profile row therefore forms a single-pass data packet

$$\mathcal{P}_{\Gamma}^{\text{cp}} = \left(\kappa_8, e^{-\phi_0} \Omega_{\Gamma}, r_{\Gamma}, L_Q, \varrho_{\Gamma}, m_{\sigma}, z_{\chi}, \sigma_{\chi} d_{\chi} \right).$$

It returns

$$e^{-\phi_0} \Omega_{\Gamma} = \sum_{s \in \mathcal{S}_{\Gamma}} \omega_s e^{-\phi_s}, \quad \Omega_{\Gamma} = \sum_{s \in \mathcal{S}_{\Gamma}} \omega_s.$$

The Chan–Paton trace factor remains the finite endpoint multiplier $\tau_Q = 16$ in $Z_{\Gamma,Q}$. In the common-metric sector row,

$$e^{-\phi_0} = \frac{\sum_{s \in \mathcal{S}_{\Gamma}} \omega_s e^{-\phi_s}}{\sum_{s \in \mathcal{S}_{\Gamma}} \omega_s}.$$

The measure and CP2 density are

$$d\nu_{\Gamma} = \kappa_8 e^{-\phi_0} \Omega_{\Gamma} \sqrt{\gamma_{\mathbb{CP}^2}} d^4 y ds_Q, \quad \mathfrak{z}_{\Gamma}^{\mathbb{CP}^2} = \frac{\kappa_8}{2} e^{-\phi_0} \Omega_{\Gamma} r_{\Gamma}^2,$$

$$Z_{\Gamma,0} = L_Q \mathfrak{z}_{\Gamma}^{\mathbb{CP}^2}, \quad Z_{\Gamma,Q} = 16 L_Q \mathfrak{z}_{\Gamma}^{\mathbb{CP}^2}, \quad Z_{\Gamma,H} = \frac{Q}{4} Z_{\Gamma,0},$$

and

$$q_{\Gamma} = \frac{32}{25} \varrho_{\Gamma}, \quad U_{\Gamma}^{\mathbb{CP}^2} = \frac{16\pi^2}{441} \mathfrak{z}_{\Gamma}^{\mathbb{CP}^2}.$$

The shape parameter is the current-frame Rayleigh quotient

$$\varrho_{\Gamma} = \frac{C_{\Gamma}^{\Gamma}}{C_0^{\Gamma}}, \quad C_0^{\Gamma} = \langle P_0^{\Gamma} \theta_{\Gamma}^3, P_0^{\Gamma} \theta_{\Gamma}^3 \rangle_{\Gamma, \Pi_Q},$$

$$C_h^{\Gamma} = \langle P_h^{\Gamma} \theta_{\Gamma}^3, P_h^{\Gamma} \theta_{\Gamma}^3 \rangle_{\Gamma, \Pi_Q}, \quad C_0^{\Gamma} > 0.$$

Here P_0^{Γ} projects to the endpoint component and P_h^{Γ} projects to the horizontal $\mathbb{CP}^2$ component in the localized current frame. In sector form,

$$C_{0,s}^{\Gamma} = \omega_s e^{-\phi_s} \langle P_0^{\Gamma} \theta_{\Gamma,s}^3, P_0^{\Gamma} \theta_{\Gamma,s}^3 \rangle_{s, \Pi_Q}, \quad C_{h,s}^{\Gamma} = \omega_s e^{-\phi_s} \langle P_h^{\Gamma} \theta_{\Gamma,s}^3, P_h^{\Gamma} \theta_{\Gamma,s}^3 \rangle_{s, \Pi_Q}.$$

Therefore

$$C_0^{\Gamma} = \sum_s C_{0,s}^{\Gamma}, \quad C_h^{\Gamma} = \sum_s C_{h,s}^{\Gamma}, \quad \varrho_{\Gamma} = \frac{\sum_s C_{h,s}^{\Gamma}}{\sum_s C_{0,s}^{\Gamma}}. \quad (5.1)$$

On $\mathcal{S}_{\Gamma,0} = \{s : C_{0,s}^{\Gamma} > 0\}$, set

$$b_{\Gamma,s} = \frac{C_{0,s}^{\Gamma}}{C_0^{\Gamma}}, \quad \rho_{\Gamma,s} = \frac{C_{h,s}^{\Gamma}}{C_{0,s}^{\Gamma}}.$$

Then

$$b_{\Gamma,s} \geq 0, \quad \sum_{s \in \mathcal{S}_{\Gamma,0}} b_{\Gamma,s} = 1, \quad \varrho_{\Gamma} = \sum_{s \in \mathcal{S}_{\Gamma,0}} b_{\Gamma,s} \rho_{\Gamma,s}. \quad (5.2)$$

The density-shape pass is

$$0 < \varrho_\Gamma < 2.743448440896 \dots, \quad U_\Gamma^{\mathbb{CP}^2} \geq u_\Gamma^{\text{sh}}(\varrho_\Gamma).$$

The image and determinant entries are

$$\psi_\sigma = \psi_{\Gamma,0}, \quad \mathcal{M}_{\text{img}} \psi_\sigma = m_\sigma \psi_\sigma, \quad N_\Gamma^\chi = 2\sigma_\chi d_\chi F(z_\chi).$$

An endpoint-matrix realization makes the image mass a spectral readout:

$$\mathcal{M}_{\text{img}} = \mathcal{Q}_{\text{img}}^\dagger \mathcal{Q}_{\text{img}}, \quad \mathcal{Q}_{\text{img}} = \mathbf{1}_{\mathbb{CP}^2 \times S_Q^1} \otimes B_\sigma.$$

The product density cancels from the normalized image quotient:

$$m_\sigma = \frac{\langle \eta_Q, B_\sigma^\dagger B_\sigma \eta_Q \rangle_{E_Q}}{\langle \eta_Q, \eta_Q \rangle_{E_Q}}.$$

The one-line image sector is the eigenline condition

$$B_\sigma^\dagger B_\sigma \eta_Q = m_\sigma \eta_Q, \quad \Pi_\sigma^\perp B_\sigma^\dagger B_\sigma \eta_Q = 0, \quad m_\sigma > 0.$$

Let

$$\Pi_{\eta_Q} = \frac{|\eta_Q\rangle\langle\eta_Q|}{\langle\eta_Q, \eta_Q\rangle_{E_Q}}, \quad \Pi_{\eta_Q}^\perp = \mathbf{1}_{E_Q} - \Pi_{\eta_Q}.$$

The finite endpoint normal form is

$$B_\sigma^\dagger B_\sigma = m_\sigma \Pi_{\eta_Q} + \mathcal{B}_{\sigma,\perp}^\dagger \mathcal{B}_{\sigma,\perp}, \quad \mathcal{B}_{\sigma,\perp} = \Pi_{\eta_Q}^\perp \mathcal{B}_{\sigma,\perp} \Pi_{\eta_Q}^\perp.$$

The one-mode reduction has a separated complement whenever

$$\mathcal{B}_{\sigma,\perp}^\dagger \mathcal{B}_{\sigma,\perp} \geq M_{\sigma,\perp}^2 \Pi_{\eta_Q}^\perp, \quad M_{\sigma,\perp}^2 > m_\sigma.$$

Equivalently, define

$$g_{\sigma,\perp} = \frac{M_{\sigma,\perp}^2}{m_\sigma}.$$

The finite endpoint image row passes exactly for

$$g_{\sigma,\perp} > 1. \tag{5.3}$$

For a diagonal endpoint spectrum on $\Pi_{\eta_Q}^\perp E_Q$,

$$B_\sigma^\dagger B_\sigma e_{\sigma,\alpha} = \lambda_{\sigma,\alpha} e_{\sigma,\alpha}, \quad e_{\sigma,\alpha} \in \Pi_{\eta_Q}^\perp E_Q,$$

the ratio is

$$g_{\sigma,\perp} = \frac{\min_\alpha \lambda_{\sigma,\alpha}}{m_\sigma}. \tag{5.4}$$

The finite eigenline residual is

$$\varepsilon_{\sigma,Q}^2 = \frac{\left\| \Pi_{\eta_Q}^\perp B_\sigma^\dagger B_\sigma \eta_Q \right\|_{E_Q}^2}{m_\sigma^2 \|\eta_Q\|_{E_Q}^2}.$$

The one-line image condition is $\varepsilon_{\sigma,Q} = 0$. For the minimal positive endpoint block,

$$\mathcal{B}_\chi^\dagger \mathcal{B}_\chi = M_\chi^2 \mathbf{1}_{E_\chi}, \quad E_\chi = \mathbb{C}, \quad \sigma_\chi d_\chi = +1.$$

The determinant threshold coordinate is $z_\chi = M_\chi L_H$. Combining with the density target gives

$$F(z_\chi) = \Xi_\Gamma^{\text{cp}}, \quad \Xi_\Gamma^{\text{cp}} = \frac{21375.716460702053 \dots \lambda_F}{2w_H^8} \left(\frac{L_Q \mathfrak{z}_\Gamma^{\text{CP}^2}}{v_{\text{EW}}} \right)^4.$$

Since

$$F'(z) = -\frac{1}{3}e^{-z}z(1+z),$$

the minimal positive endpoint row passes exactly for

$$0 < \Xi_\Gamma^{\text{cp}} \leq 1, \quad z_\chi = F^{-1}(\Xi_\Gamma^{\text{cp}}).$$

The same determinant equation gives the neutral-density upper constraint

$$0 < Z_{\Gamma,0} = L_Q \mathfrak{z}_\Gamma^{\text{CP}^2} \leq Z_{\chi,\text{max}}.$$

Here

$$\begin{aligned} Z_{\chi,\text{max}} &= v_{\text{EW}} \left(\frac{2w_H^8}{21375.716460702053 \dots \lambda_F} \right)^{1/4} \\ &= 0.0983506713707568 \dots w_H^2 \lambda_F^{-1/4} v_{\text{EW}}. \end{aligned}$$

Together with the density-shape lower row, this gives the loop-length envelope

$$0 < L_Q \leq \frac{Z_{\chi,\text{max}}}{\mathfrak{z}_\Gamma^{\text{CP}^2}} \leq \frac{16\pi^2 Z_{\chi,\text{max}}}{441 u_\Gamma^{\text{sh}}(\varrho_\Gamma)}.$$

Equivalently, the CP2 radius must lie in the band

$$\frac{441 u_\Gamma^{\text{sh}}(\varrho_\Gamma)}{8\pi^2 \kappa_8 e^{-\phi_0} \Omega_\Gamma} \leq r_\Gamma^2 \leq \frac{2Z_{\chi,\text{max}}}{\kappa_8 e^{-\phi_0} \Omega_\Gamma L_Q}.$$

For the full sector-sum row define the weighted radius moment

$$\mathcal{R}_\Gamma^2 = \sum_{s \in \mathcal{S}_\Gamma} \omega_s e^{-\phi_s} r_{\Gamma,s}^2.$$

Equivalently, introduce the sector label space

$$\mathcal{H}_\mathcal{S} = \bigoplus_{s \in \mathcal{S}_\Gamma} \mathbb{C}|s\rangle$$

and diagonal sector operators

$$\mathbf{W}_\Gamma |s\rangle = \omega_s e^{-\phi_s} |s\rangle, \quad \mathbf{R}_\Gamma |s\rangle = r_{\Gamma,s}^2 |s\rangle.$$

Then

$$e^{-\phi_0} \Omega_\Gamma = \text{Tr}_\mathcal{S} \mathbf{W}_\Gamma, \quad \mathcal{R}_\Gamma^2 = \text{Tr}_\mathcal{S} (\mathbf{W}_\Gamma \mathbf{R}_\Gamma).$$

Then

$$\mathfrak{z}_\Gamma^{\text{CP}^2} = \frac{\kappa_8}{2} \mathcal{R}_\Gamma^2, \quad e^{-\phi_0} \Omega_\Gamma r_\Gamma^2 = \mathcal{R}_\Gamma^2 \quad \text{in the common-metric row.}$$

The sector-sum compactification band is

$$\frac{441 u_\Gamma^{\text{sh}}(\varrho_\Gamma)}{8\pi^2 \kappa_8} \leq \mathcal{R}_\Gamma^2 \leq \frac{2Z_{\chi,\text{max}}}{\kappa_8 L_Q}.$$

The determinant threshold depends on the weighted product

$$\mathcal{V}_{\Gamma,Q} = L_Q \mathcal{R}_{\Gamma}^2, \quad Z_{\Gamma,0} = \frac{\kappa_8}{2} \mathcal{V}_{\Gamma,Q}.$$

In trace form the local density outputs read

$$Z_{\Gamma,0} = \frac{\kappa_8 L_Q}{2} \text{Tr}_{\mathcal{S}}(\mathbf{W}_{\Gamma} \mathbf{R}_{\Gamma}), \quad Z_{\Gamma,Q} = 8\kappa_8 L_Q \text{Tr}_{\mathcal{S}}(\mathbf{W}_{\Gamma} \mathbf{R}_{\Gamma}),$$

$$Z_{\Gamma,H} = \frac{Q\kappa_8 L_Q}{8} \text{Tr}_{\mathcal{S}}(\mathbf{W}_{\Gamma} \mathbf{R}_{\Gamma}).$$

Set

$$\mathcal{T}_{\Gamma} = \text{Tr}_{\mathcal{S}}(\mathbf{W}_{\Gamma} \mathbf{R}_{\Gamma}).$$

The trace pass surface is

$$\mathcal{T}_{\Gamma} \geq \mathcal{T}_{\Gamma,\min}(\varrho_{\Gamma}) = \frac{441 u_{\Gamma}^{\text{sh}}(\varrho_{\Gamma})}{8\pi^2 \kappa_8}, \quad 0 < L_Q \leq \frac{2Z_{\chi,\max}}{\kappa_8 \mathcal{T}_{\Gamma}}.$$

Equivalently, with $0 < x_{\Gamma,Q} \leq 1$,

$$L_Q = \frac{2Z_{\chi,\max}}{\kappa_8 \mathcal{T}_{\Gamma}} x_{\Gamma,Q}, \quad \mathcal{T}_{\Gamma} \geq \mathcal{T}_{\Gamma,\min}(\varrho_{\Gamma}).$$

At lower-density saturation,

$$\mathcal{T}_{\Gamma} = \mathcal{T}_{\Gamma,\min}(\varrho_{\Gamma}), \quad L_Q = x_{\Gamma,Q} L_{Q,\max}(\varrho_{\Gamma}).$$

Define

$$\chi_{\Gamma} = \frac{\mathcal{T}_{\Gamma}}{\mathcal{T}_{\Gamma,\min}(\varrho_{\Gamma})} \geq 1.$$

Then

$$L_Q = \frac{x_{\Gamma,Q}}{\chi_{\Gamma}} L_{Q,\max}(\varrho_{\Gamma}), \quad x_{\Gamma,Q} = \chi_{\Gamma} \frac{L_Q}{L_{Q,\max}(\varrho_{\Gamma})}.$$

Therefore

$$0 < L_Q \leq \frac{L_{Q,\max}(\varrho_{\Gamma})}{\chi_{\Gamma}}.$$

In sector variables set

$$\zeta_{\Gamma,s} = \frac{\omega_s e^{-\phi_s} r_{\Gamma,s}^2}{\mathcal{T}_{\Gamma,\min}(\varrho_{\Gamma})} \geq 0.$$

Then

$$\chi_{\Gamma} = \sum_{s \in \mathcal{S}_{\Gamma}} \zeta_{\Gamma,s}, \quad L_Q = \frac{x_{\Gamma,Q}}{\sum_{s \in \mathcal{S}_{\Gamma}} \zeta_{\Gamma,s}} L_{Q,\max}(\varrho_{\Gamma}). \quad (6.1)$$

For a fixed loop length,

$$x_{\Gamma,Q} = \frac{L_Q}{L_{Q,\max}(\varrho_{\Gamma})} \sum_{s \in \mathcal{S}_{\Gamma}} \zeta_{\Gamma,s}, \quad 1 \leq \sum_{s \in \mathcal{S}_{\Gamma}} \zeta_{\Gamma,s} \leq \frac{L_{Q,\max}(\varrho_{\Gamma})}{L_Q}. \quad (6.2)$$

Define sector fractions

$$p_{\Gamma,s} = \frac{\zeta_{\Gamma,s}}{\chi_{\Gamma}}, \quad p_{\Gamma,s} \geq 0, \quad \sum_{s \in \mathcal{S}_{\Gamma}} p_{\Gamma,s} = 1.$$

Then each retained sector obeys

$$\omega_s e^{-\phi_s} r_{\Gamma,s}^2 = p_{\Gamma,s} \chi_{\Gamma} \mathcal{T}_{\Gamma,\min}(\varrho_{\Gamma}) = p_{\Gamma,s} \chi_{\Gamma} \frac{441 u_{\Gamma}^{\text{sh}}(\varrho_{\Gamma})}{8\pi^2 \kappa_8}. \quad (6.3)$$

Equivalently,

$$r_{\Gamma,s}^2 = \frac{441 p_{\Gamma,s} \chi_{\Gamma} u_{\Gamma}^{\text{sh}}(\varrho_{\Gamma})}{8\pi^2 \kappa_8 \omega_s e^{-\phi_s}}. \quad (6.4)$$

Collecting the local rows, define the finite D8/O8 data record

$$\mathcal{D}_{\Gamma} = (\mathcal{S}_{\Gamma}; \omega_s, \phi_s, p_{\Gamma,s}, b_{\Gamma,s}, \rho_{\Gamma,s}; \chi_{\Gamma}, x_{\Gamma,Q}, L_Q; m_{\sigma}, \varepsilon_{\sigma,Q}, \lambda_{\sigma,\alpha}).$$

It gives

$$\varrho_{\Gamma} = \sum_{s \in \mathcal{S}_{\Gamma,0}} b_{\Gamma,s} \rho_{\Gamma,s}, \quad \mathcal{T}_{\Gamma} = \chi_{\Gamma} \mathcal{T}_{\Gamma,\min}(\varrho_{\Gamma}), \quad L_Q = \frac{x_{\Gamma,Q}}{\chi_{\Gamma}} L_{Q,\max}(\varrho_{\Gamma}), \quad (6.5)$$

$$z_{\chi} = F^{-1}(x_{\Gamma,Q}^4), \quad \varepsilon_{\sigma,Q}^2 = \frac{\left\| \Pi_{\eta_Q}^{\perp} B_{\sigma}^{\dagger} B_{\sigma} \eta_Q \right\|_{E_Q}^2}{m_{\sigma}^2 \|\eta_Q\|_{E_Q}^2}, \quad g_{\sigma,\perp} = \frac{\min_{\alpha} \lambda_{\sigma,\alpha}}{m_{\sigma}}. \quad (6.6)$$

The admissible finite row satisfies

$$\chi_{\Gamma} \geq 1, \quad 0 < x_{\Gamma,Q} \leq 1, \quad 0 < \varrho_{\Gamma} < \varrho_c, \quad \varepsilon_{\sigma,Q} = 0, \quad g_{\sigma,\perp} > 1, \quad (6.7)$$

$$\sum_{s \in \mathcal{S}_{\Gamma}} p_{\Gamma,s} = 1, \quad \sum_{s \in \mathcal{S}_{\Gamma,0}} b_{\Gamma,s} = 1, \quad p_{\Gamma,s}, b_{\Gamma,s}, \rho_{\Gamma,s} \geq 0. \quad (6.8)$$

The corresponding localized-action map starts from the quadratic data

$$\mathcal{Q}_{\Gamma}^{(2)} = \left(\mathcal{S}_{\Gamma}; \omega_s, \phi_s, \gamma_{\Gamma,s}^{\mathbb{CP}^2}; \theta_{\Gamma,s}^3, P_0^{\Gamma}, P_h^{\Gamma}, \Pi_Q; B_{\sigma}, \eta_Q, L_Q \right).$$

Let $\widehat{\gamma}_s$ be the unit $\mathbb{CP}^2$ reference metric in sector s , and define

$$\text{Av}_{s,\Pi_Q}(f) = \frac{\int_{\mathbb{CP}_s^2} \sqrt{\widehat{\gamma}_s} \text{tr}_{E_s}(\Pi_Q f \Pi_Q)}{\int_{\mathbb{CP}_s^2} \sqrt{\widehat{\gamma}_s} \text{tr}_{E_s}(\Pi_Q)}. \quad (6.9)$$

For each retained sector, set

$$r_{\Gamma,s}^2 = \text{Av}_{s,\Pi_Q} \left(\frac{1}{4} \text{tr}_{\widehat{\gamma}_s} \gamma_{\Gamma,s}^{\mathbb{CP}^2} \right), \quad \mathcal{T}_{\Gamma} = \sum_s \omega_s e^{-\phi_s} r_{\Gamma,s}^2. \quad (6.10)$$

Then

$$\varrho_{\Gamma} = \frac{\sum_s C_{h,s}^{\Gamma}}{\sum_s C_{0,s}^{\Gamma}}, \quad \chi_{\Gamma} = \frac{\mathcal{T}_{\Gamma}}{\mathcal{T}_{\Gamma,\min}(\varrho_{\Gamma})}, \quad x_{\Gamma,Q} = \frac{\kappa_8 L_Q \mathcal{T}_{\Gamma}}{2Z_{\chi,\max}}. \quad (6.11)$$

The endpoint spectrum supplies

$$m_{\sigma} = \frac{\langle \eta_Q, B_{\sigma}^{\dagger} B_{\sigma} \eta_Q \rangle_{E_Q}}{\langle \eta_Q, \eta_Q \rangle_{E_Q}}, \quad \{\lambda_{\sigma,\alpha}\} = \text{Spec} \left(\Pi_{\eta_Q}^{\perp} B_{\sigma}^{\dagger} B_{\sigma} \Pi_{\eta_Q}^{\perp} \right). \quad (6.12)$$

The same endpoint operator gives

$$\varepsilon_{\sigma,Q}^2 = \frac{\left\| \Pi_{\eta_Q}^{\perp} B_{\sigma}^{\dagger} B_{\sigma} \eta_Q \right\|_{E_Q}^2}{m_{\sigma}^2 \|\eta_Q\|_{E_Q}^2}.$$

Together with

$$p_{\Gamma,s} = \frac{\omega_s e^{-\phi_s} r_{\Gamma,s}^2}{\mathcal{T}_{\Gamma}}, \quad b_{\Gamma,s} = \frac{C_{0,s}^{\Gamma}}{\sum_t C_{0,t}^{\Gamma}}, \quad \rho_{\Gamma,s} = \frac{C_{h,s}^{\Gamma}}{C_{0,s}^{\Gamma}}, \quad (6.13)$$

these equations define

$$\mathfrak{E}_\Gamma(\mathcal{Q}_\Gamma^{(2)}) = \mathcal{D}_\Gamma. \quad (6.14)$$

The algebraic inverse local packet is explicit on the support

$$\mathcal{S}_{\Gamma,+} = \{s : p_{\Gamma,s} > 0\}.$$

Choose $\mathcal{C}_\Gamma > 0$, endpoint-normalized one-forms $\widehat{e}_{0,s}$ and $\widehat{e}_{h,s}$, and set

$$\gamma_{\Gamma,s}^{\mathbb{CP}^2} = r_{\Gamma,s}^2 \widehat{\gamma}_s, \quad r_{\Gamma,s}^2 = \frac{p_{\Gamma,s} \chi_\Gamma \mathcal{T}_{\Gamma,\min}(\varrho_\Gamma)}{\omega_s e^{-\phi_s}}. \quad (6.15)$$

The coframe pieces may be chosen as

$$P_0^\Gamma \theta_{\Gamma,s}^3 = \left(\frac{b_{\Gamma,s} \mathcal{C}_\Gamma}{\omega_s e^{-\phi_s}} \right)^{1/2} \widehat{e}_{0,s}, \quad P_h^\Gamma \theta_{\Gamma,s}^3 = \left(\frac{b_{\Gamma,s} \rho_{\Gamma,s} \mathcal{C}_\Gamma}{\omega_s e^{-\phi_s}} \right)^{1/2} \widehat{e}_{h,s}. \quad (6.16)$$

With $e_0 = \eta_Q / \|\eta_Q\|_{E_Q}$ and an orthonormal complement $\{e_\alpha\} \subset \Pi_{\eta_Q}^\perp E_Q$, take

$$B_\sigma^\dagger B_\sigma = m_\sigma |e_0\rangle \langle e_0| + \sum_\alpha \lambda_{\sigma,\alpha} |e_\alpha\rangle \langle e_\alpha|. \quad (6.17)$$

Then

$$\mathfrak{I}_\Gamma(\mathcal{D}_\Gamma; \mathcal{C}_\Gamma, \widehat{e}) = \mathcal{Q}_\Gamma^{(2)}, \quad \mathfrak{E}_\Gamma(\mathfrak{I}_\Gamma(\mathcal{D}_\Gamma; \mathcal{C}_\Gamma, \widehat{e})) = \mathcal{D}_\Gamma. \quad (6.18)$$

The charged determinant extension uses the same threshold coordinate. Let

$$z_\chi = F^{-1}(x_{\Gamma,Q}^4), \quad M_\chi = \frac{z_\chi}{L_H}, \quad E_\chi = \mathbb{C}, \quad \sigma_\chi d_\chi = +1. \quad (6.19)$$

Set

$$P_1^\chi = \text{span}\{u, v\}, \quad P_2^\chi = \text{span}\{u^2, v^2\}, \quad (6.20)$$

and

$$B_\chi^\dagger B_\chi = M_\chi^2 \mathbf{1}_{E_\chi}, \quad \mathcal{B}_{r,\Gamma}^\chi = \mathbf{1}_{P_r^\chi} \otimes B_\chi, \quad r = 1, 2. \quad (6.21)$$

Then

$$\mathcal{N}_\Gamma^\chi = 2F(M_\chi L_H) = 2x_{\Gamma,Q}^4. \quad (6.22)$$

The extended local packet is

$$\widetilde{\mathcal{Q}}_\Gamma^{(2)} = \left(\mathcal{Q}_\Gamma^{(2)}; P_1^\chi, P_2^\chi, E_\chi, B_\chi, L_H, \sigma_\chi d_\chi \right). \quad (6.23)$$

The finite package carries the density, coframe, image, and determinant rows. For concrete localized data

$$\mathcal{A}_\Gamma = (P_s[G + B], F_s, \phi_s, \omega_s, \Pi_s; \iota_\Gamma^* C_1; B_\sigma, B_\chi, L_Q, L_H)_{s \in \mathcal{S}_\Gamma},$$

the action-matching residual is the finite vector

$$\mathcal{R}_\Gamma(\mathcal{A}_\Gamma; \mathcal{D}_\Gamma) = (R_T, R_x, R_\varrho, R_\epsilon, R_g, R_\chi), \quad (6.24)$$

where

$$R_T = \frac{\mathcal{T}_\Gamma(\mathcal{A}_\Gamma)}{\mathcal{T}_{\Gamma,\min}(\varrho_\Gamma)} - \chi_\Gamma, \quad R_x = \frac{\kappa_8 L_Q \mathcal{T}_\Gamma(\mathcal{A}_\Gamma)}{2Z_{\chi,\max}} - x_{\Gamma,Q}, \quad (6.25)$$

$$R_\varrho = \frac{C_h^\Gamma(\mathcal{A}_\Gamma)}{C_0^\Gamma(\mathcal{A}_\Gamma)} - \varrho_\Gamma, \quad R_\epsilon = \varepsilon_{\sigma,Q}, \quad (6.26)$$

$$R_g = \max\left\{0, 1 - \frac{\min_\alpha \lambda_{\sigma,\alpha}}{m_\sigma}\right\}, \quad R_\chi = \frac{M_\chi L_H - z_\chi}{1 + z_\chi}. \quad (6.27)$$

The localized action produces the finite packet exactly when

$$\mathcal{R}_\Gamma(\mathcal{A}_\Gamma; \mathcal{D}_\Gamma) = 0. \quad (6.28)$$

A localized finite normal form realizing an admissible row is explicit. Write

$$W_s = \omega_s e^{-\phi_s}, \quad \mathcal{S}_{\Gamma,+} = \{s : p_{\Gamma,s} > 0\}.$$

On $\mathcal{S}_{\Gamma,+}$, take endpoint-normalized one-forms $\widehat{e}_{0,s}, \widehat{e}_{h,s}$ and set

$$P_s[G + B]_{\mathbb{CP}^2} = r_{\Gamma,s}^2 \widehat{\gamma}_s, \quad r_{\Gamma,s}^2 = \frac{p_{\Gamma,s} \chi_\Gamma \mathcal{T}_{\Gamma,\min}(\varrho_\Gamma)}{W_s}.$$

The retained current pullback is

$$\Pi_{\text{EW}} \iota_\Gamma^* C_1|_s = q_\Gamma \theta_{\Gamma,s}^3, \quad q_\Gamma > 0,$$

with

$$P_0^\Gamma \theta_{\Gamma,s}^3 = \left(\frac{b_{\Gamma,s} \mathcal{C}_\Gamma}{W_s}\right)^{1/2} \widehat{e}_{0,s}, \quad P_h^\Gamma \theta_{\Gamma,s}^3 = \left(\frac{b_{\Gamma,s} \rho_{\Gamma,s} \mathcal{C}_\Gamma}{W_s}\right)^{1/2} \widehat{e}_{h,s}.$$

The endpoint image and determinant blocks are

$$B_\sigma^\dagger B_\sigma = m_\sigma |e_0\rangle \langle e_0| + \sum_\alpha \lambda_{\sigma,\alpha} |e_\alpha\rangle \langle e_\alpha|, \quad \frac{\min_\alpha \lambda_{\sigma,\alpha}}{m_\sigma} > 1,$$

$$B_\chi^\dagger B_\chi = \left(\frac{z_\chi}{L_H}\right)^2 \mathbf{1}_{E_\chi}, \quad L_Q = \frac{x_{\Gamma,Q}}{\chi_\Gamma} L_{Q,\max}(\varrho_\Gamma).$$

Then

$$\mathcal{T}_\Gamma(\mathcal{A}_\Gamma^{\text{nf}}) = \chi_\Gamma \mathcal{T}_{\Gamma,\min}(\varrho_\Gamma), \quad \frac{C_h^\Gamma(\mathcal{A}_\Gamma^{\text{nf}})}{C_0^\Gamma(\mathcal{A}_\Gamma^{\text{nf}})} = \varrho_\Gamma,$$

$$\varepsilon_{\sigma,Q} = 0, \quad M_\chi L_H = z_\chi, \quad \mathcal{R}_\Gamma(\mathcal{A}_\Gamma^{\text{nf}}; \mathcal{D}_\Gamma) = 0.$$

A global Type IIA source datum can be written as

$$\mathfrak{G}_\Gamma = (X_{10}, \mathcal{I}_{O8}, \iota_{\Gamma,s}, E_s, F_s, \phi_s, G, B, C_1, F_0, \Omega)_{s \in \mathcal{S}_\Gamma},$$

with localized DBI/CS/Payen readout

$$\mathfrak{L}_\Gamma(\mathfrak{G}_\Gamma) = \mathcal{A}_\Gamma.$$

The finite normal form is the local image of this global datum when

$$\mathfrak{L}_\Gamma(\mathfrak{G}_\Gamma) = \mathcal{A}_\Gamma^{\text{nf}}.$$

Component form:

$$P_s[G + B]_{\mathbb{CP}^2} = r_{\Gamma,s}^2 \widehat{\gamma}_s, \quad \Pi_{\text{EW}} \iota_{\Gamma,s}^* C_1 = q_\Gamma \theta_{\Gamma,s}^3, \\ \Pi_{\text{EW}} \iota_{\Gamma,s}^* F_2 = q_\Gamma d_\Gamma \theta_{\Gamma,s}^3, \quad \text{End}_\Omega(E_s, F_s) = (\Pi_Q, e_0, E_\chi, B_\sigma, B_\chi).$$

The source and lower-charge rows are

$$dF_0 = \sum_s N_{8,s} \delta_{\Gamma_s} + \sum_a Q_{O8,a} \delta_{O8,a}, \quad \sum_s N_{8,s} + \sum_a Q_{O8,a} = 0,$$

$$Q_{\text{lower}}^{\text{tot}} = \left[\sum_s \text{ch}(E_s) e^{F_s/2\pi} \sqrt{\frac{\widehat{A}(T\Gamma_s)}{\widehat{A}(N\Gamma_s)}} + \sum_a Q_{O8,a}^{\text{curv}} \right]_{<8} = 0.$$

Equivalently, define

$$\mathcal{R}_{\Gamma}^{\text{glob}}(\mathfrak{G}_{\Gamma}) = \left(\mathfrak{L}_{\Gamma}(\mathfrak{G}_{\Gamma}) - \mathcal{A}_{\Gamma}^{\text{nf}}, dF_0 - \sum_s N_{8,s} \delta_{\Gamma_s} - \sum_a Q_{O8,a} \delta_{O8,a}, Q_{\text{lower}}^{\text{tot}} \right).$$

Then

$$\mathcal{R}_{\Gamma}^{\text{glob}}(\mathfrak{G}_{\Gamma}) = 0 \implies \mathcal{R}_{\Gamma}(\mathcal{A}_{\Gamma}; \mathcal{D}_{\Gamma}) = 0.$$

The minimal endpoint-real type-I' datum has

$$Q_{O8,L} = Q_{O8,R} = -16, \quad N_{8,L} = N_{8,R} = 16.$$

Thus

$$\sum_{\epsilon=L,R} (N_{8,\epsilon} + Q_{O8,\epsilon}) = 0, \quad k_{\epsilon} = \frac{N_{8,\epsilon}}{2} - 8 = 0.$$

At $Q = 7$,

$$\begin{aligned} W_7 &= H^0(\mathbb{CP}^1, \mathcal{O}(7)), & \dim W_7 &= 8, \\ E_7 &= (W_7 \oplus W_7^{\vee})^{\Omega}, & \text{rk}_{\mathbb{C}}(E_7^{\mathbb{C}}) &= 16. \end{aligned}$$

The minimal spin^c $\mathbb{CP}^2$ source has $m = 0$, $f_0 = 13/2$, and

$$\mathfrak{q}_{C_1} = 16 \left(\frac{f_0^2}{2} - \frac{1}{16} \right) = 337.$$

For the two endpoints,

$$Q_{D8/O8}^{<8} = 208h_Q + 673h_Q^2.$$

A bulk lower-source completion

$$Q_{\text{bulk}}^{<8} = -208h_Q - 673h_Q^2$$

gives

$$Q_{\text{lower}}^{\text{tot}} = 0.$$

Consequently, a datum $\mathfrak{G}_{\Gamma}^{(7)}$ with

$$\mathfrak{L}_{\Gamma}(\mathfrak{G}_{\Gamma}^{(7)}) = \mathcal{A}_{\Gamma}^{\text{nf}}$$

satisfies

$$\mathcal{R}_{\Gamma}^{\text{glob}}(\mathfrak{G}_{\Gamma}^{(7)}) = 0.$$

The localized type-I' readout realizing the finite normal form is explicit. Set

$$W_s = \omega_s e^{-\phi_s}, \quad P_s B_{\mathbb{CP}^2} = 0,$$

and choose

$$P_s G_{\mathbb{CP}^2} = r_{\Gamma,s}^2 \widehat{\gamma}_s, \quad W_s r_{\Gamma,s}^2 = p_{\Gamma,s} \chi_{\Gamma} \mathcal{T}_{\Gamma, \min}(\varrho_{\Gamma}).$$

Then

$$\mathcal{T}_{\Gamma}^{\text{rd}} = \sum_s W_s r_{\Gamma,s}^2 = \chi_{\Gamma} \mathcal{T}_{\Gamma, \min}(\varrho_{\Gamma}).$$

The RR/current row is

$$\Pi_{\text{EW}} \iota_{\Gamma,s}^* C_1 = q_{\Gamma} \theta_{\Gamma,s}^3, \quad \Pi_{\text{EW}} \iota_{\Gamma,s}^* F_2 = q_{\Gamma} d_{\Gamma} \theta_{\Gamma,s}^3.$$

Choose endpoint-normalized coframes by

$$P_0^\Gamma \theta_{\Gamma,s}^3 = \left(\frac{b_{\Gamma,s} \mathcal{C}_\Gamma}{W_s} \right)^{1/2} \widehat{e}_{0,s}, \quad P_h^\Gamma \theta_{\Gamma,s}^3 = \left(\frac{b_{\Gamma,s} \rho_{\Gamma,s} \mathcal{C}_\Gamma}{W_s} \right)^{1/2} \widehat{e}_{h,s}.$$

The readout gives

$$C_{0,s}^{\Gamma,\text{rd}} = b_{\Gamma,s} \mathcal{C}_\Gamma, \quad C_{h,s}^{\Gamma,\text{rd}} = b_{\Gamma,s} \rho_{\Gamma,s} \mathcal{C}_\Gamma,$$

and hence

$$\frac{\sum_s C_{h,s}^{\Gamma,\text{rd}}}{\sum_s C_{0,s}^{\Gamma,\text{rd}}} = \sum_s b_{\Gamma,s} \rho_{\Gamma,s} = \varrho_\Gamma.$$

The endpoint operators are

$$B_\sigma^\dagger B_\sigma = m_\sigma |e_0\rangle \langle e_0| + \sum_\alpha \lambda_{\sigma,\alpha} |e_\alpha\rangle \langle e_\alpha|, \quad \min_\alpha \lambda_{\sigma,\alpha} > m_\sigma,$$

$$B_\chi^\dagger B_\chi = \left(\frac{z_\chi}{L_H} \right)^2 \mathbf{1}_{E_\chi}, \quad L_Q = \frac{x_{\Gamma,Q}}{\chi_\Gamma} L_{Q,\max}(\varrho_\Gamma).$$

The residual coordinates satisfy

$$R_T = R_\varrho = R_\epsilon = R_g = R_\chi = 0,$$

and

$$R_x = \frac{\kappa_8 L_Q \chi_\Gamma \mathcal{T}_{\Gamma,\min}(\varrho_\Gamma)}{2Z_{\chi,\max}} - x_{\Gamma,Q} = 0.$$

Therefore

$$\mathfrak{L}_\Gamma(\mathfrak{G}_{\Gamma,\text{loc}}^{(7)}) = \mathcal{A}_\Gamma^{\text{nf}}.$$

The extension from the local readout to a global type-I' representative separates into a period condition and an endpoint spectral condition. Let ω_Q be the integral generator on S_Q^1 , normalized by $\int_{S_Q^1} \omega_Q = 1$, and set

$$\ell_{\Gamma,s} = \int_{S_Q^1} \theta_{\Gamma,s}^3.$$

The RR one-form periods obey

$$R_{\text{per},s} = \frac{q_\Gamma \ell_{\Gamma,s}}{2\pi} - n_{\Gamma,s} - \nu_{\Gamma,s}, \quad n_{\Gamma,s} \in \mathbb{Z}, \quad \nu_{\Gamma,s} \in \mathbb{Q}/\mathbb{Z}.$$

A common q_Γ lies on the rational-period locus

$$q_\Gamma \ell_{\Gamma,s} = 2\pi(n_{\Gamma,s} + \nu_{\Gamma,s}) \quad \text{for every retained } s.$$

The metric, dilaton, and B -field rows extend through disjoint endpoint collars by partition of unity with preservation of the local readout values. For the endpoint operator, let $D_{\partial,\Gamma}^{(7)}$ be the real boundary Dirac–Dolbeault operator on

$$E_7 \oplus E_\sigma \oplus (P_1^\chi \oplus P_2^\chi) \otimes E_\chi.$$

The endpoint spectral condition is

$$\begin{aligned} R_{\text{DD}} &= \left\| \Pi_{\eta_Q} D_{\partial,\Gamma}^{(7)2} \Pi_{\eta_Q} - m_\sigma \Pi_{\eta_Q} \right\| + \left\| \Pi_{\eta_Q}^\perp D_{\partial,\Gamma}^{(7)2} \Pi_{\eta_Q} \right\| \\ &+ \left\| \Pi_\chi D_{\partial,\Gamma}^{(7)2} \Pi_\chi - \left(\frac{z_\chi}{L_H} \right)^2 \Pi_\chi \right\|, \quad \Pi_\chi = P_1^\chi \oplus P_2^\chi. \end{aligned}$$

The global representative satisfies

$$R_{\text{per},s} = 0 \quad (\forall s), \quad R_{\text{DD}} = 0, \quad Q_{\text{lower}}^{\text{tot}} = 0.$$

Thus

$$\Xi_{\Gamma}^{\text{cp}} = \left(\frac{Z_{\Gamma,0}}{Z_{\chi,\text{max}}} \right)^4 = \left(\frac{\kappa_8 \mathcal{V}_{\Gamma,Q}}{2Z_{\chi,\text{max}}} \right)^4.$$

The determinant pass is equivalently

$$0 < \mathcal{V}_{\Gamma,Q} \leq \frac{2Z_{\chi,\text{max}}}{\kappa_8}, \quad z_{\chi} = F^{-1} \left[\left(\frac{\kappa_8 \mathcal{V}_{\Gamma,Q}}{2Z_{\chi,\text{max}}} \right)^4 \right].$$

The normalized endpoint coordinate is

$$x_{\Gamma,Q} = \frac{Z_{\Gamma,0}}{Z_{\chi,\text{max}}} = \frac{\kappa_8 \mathcal{V}_{\Gamma,Q}}{2Z_{\chi,\text{max}}} = \frac{\kappa_8 L_Q \mathcal{T}_{\Gamma}}{2Z_{\chi,\text{max}}}.$$

Then

$$\Xi_{\Gamma}^{\text{cp}} = x_{\Gamma,Q}^4, \quad z_{\chi} = F^{-1}(x_{\Gamma,Q}^4), \quad 0 < x_{\Gamma,Q} \leq 1.$$

Combining the sector-sum lower row with the determinant upper row gives

$$\frac{L_Q}{L_{Q,\text{max}}(\varrho_{\Gamma})} \leq x_{\Gamma,Q} \leq 1.$$

Define

$$r_{\min}^2(\varrho_{\Gamma}) = \frac{441 u_{\Gamma}^{\text{sh}}(\varrho_{\Gamma})}{8\pi^2 \kappa_8 e^{-\phi_0} \Omega_{\Gamma}}, \quad r_{\max}^2(L_Q) = \frac{2Z_{\chi,\text{max}}}{\kappa_8 e^{-\phi_0} \Omega_{\Gamma} L_Q}.$$

The compactification interval

$$\mathcal{I}_{r^2}(\varrho_{\Gamma}, L_Q) = [r_{\min}^2(\varrho_{\Gamma}), r_{\max}^2(L_Q)]$$

is populated exactly when

$$0 < L_Q \leq L_{Q,\text{max}}(\varrho_{\Gamma}), \quad L_{Q,\text{max}}(\varrho_{\Gamma}) = \frac{16\pi^2 Z_{\chi,\text{max}}}{441 u_{\Gamma}^{\text{sh}}(\varrho_{\Gamma})}.$$

The matching endpoint determinant spectrum is a paired charged oscillator:

$$\mathcal{E}_1^{\chi} = \text{span}\{u, v\} \otimes E_{\chi}, \quad \mathcal{E}_2^{\chi} = \text{span}\{u^2, v^2\} \otimes E_{\chi},$$

with

$$(\mathcal{B}_{r,\Gamma}^{\chi})^{\dagger} \mathcal{B}_{r,\Gamma}^{\chi} = M_{\chi}^2 \mathbf{1}_{2d_{\chi}}, \quad r = 1, 2.$$

For positive determinant sign this gives

$$\mathcal{N}_{\Gamma}^{\chi} = 2d_{\chi} F(M_{\chi} L_H).$$

The minimal row has $d_{\chi} = 1$, hence

$$\mathcal{N}_{\Gamma}^{\chi} = 2F(z_{\chi}), \quad z_{\chi} = M_{\chi} L_H.$$

The local spectrum package identifies e_{σ} and the paired oscillator block E_{χ} inside the same D8/O8 boundary spectrum.

7.5 Single-Block Boundary Complex

A local complex that generates the spectrum package has Hilbert space

$$\mathcal{H}_\Gamma^{\text{loc}} = \text{span}\{e_\sigma\} \oplus (P_1^\chi \oplus P_2^\chi) \otimes E_\chi \oplus \mathcal{H}_\perp,$$

where

$$P_1^\chi = \text{span}\{u, v\}, \quad P_2^\chi = \text{span}\{u^2, v^2\}.$$

Let

$$\mathcal{Q}_{\Gamma, \sigma} = \sqrt{m_\sigma} \Pi_{e_\sigma} \oplus (\mathbf{1}_{P_1^\chi} \oplus \mathbf{1}_{P_2^\chi}) \otimes B_\chi \oplus Q_\perp.$$

Then

$$\mathcal{M}_{\text{img}} = \mathcal{Q}_{\Gamma, \sigma}^\dagger \mathcal{Q}_{\Gamma, \sigma} |_{\text{span}\{e_\sigma\}}, \quad \mathcal{B}_{r, \Gamma}^\chi = \mathbf{1}_{P_r^\chi} \otimes B_\chi.$$

The current-source row is

$$\mathcal{J}_A^{\text{src}} = C_{A\sigma} e_\sigma^b, \quad A = 3, Y,$$

with

$$C_{3\sigma} = L_2 \sqrt{m_\sigma \delta_\sigma}, \quad C_{Y\sigma} = \rho_\sigma C_{3\sigma}.$$

The charged block obeys

$$B_\chi^\dagger B_\chi = M_\chi^2 \mathbf{1}_{E_\chi}, \quad \text{Str}_{E_\chi} \mathbf{1} = d_\chi, \quad \sigma_\chi = +1.$$

It follows that

$$F(M_\chi L_H) = \frac{\mathcal{N}_\Gamma^\chi}{2d_\chi}.$$

Using the density target,

$$F(M_\chi L_H) = \frac{21375.716460702053 \dots \lambda_F}{2d_\chi w_H^8} \left(\frac{Z_{\Gamma,0}}{v_{\text{EW}}} \right)^4.$$

For $d_\chi = 1$, the right hand side is Ξ_Γ^{cp} . Values in $(0, 1]$ determine

$$M_\chi = \frac{z_\chi}{L_H}, \quad F(z_\chi) = \Xi_\Gamma^{\text{cp}}.$$

The image mass m_σ enters the source coupling $C_{A\sigma}$ and cancels from $\mathcal{D}_{AB}$. The local action supplies m_σ and the profile e_σ .

7.6 Profile-To-Matrix Closure

Identify the image eigenline with

$$e_\sigma = \psi_\sigma, \quad \int_{\Gamma_{\text{int}}} d\nu_\Gamma |\psi_\sigma|^2 = 1.$$

For the weighted current Gram matrix

$$\mathcal{G}_{AB}^{(\psi)} = \int_{\Gamma_{\text{int}}} d\nu_\Gamma \psi_\sigma^2 \gamma_\Gamma^{-1}(\theta_\Gamma^A, \theta_\Gamma^B), \quad A, B = 3, Y,$$

define

$$c_\sigma^B = \left(\mathcal{G}^{(\psi)-1} \right)^{BA} (g_\sigma)_A.$$

The minimal source profile is

$$\Xi_{\sigma, \text{min}} = L_2 \psi_\sigma c_\sigma^B \theta_{\Gamma B}, \quad \Theta_{\text{img}, \sigma} = \sqrt{m_\sigma} \Xi_{\sigma, \text{min}}.$$

Then

$$\begin{aligned} C_{A\sigma} &= \int_{\Gamma_{\text{int}}} d\nu_{\Gamma} \psi_{\sigma} \gamma_{\Gamma}^{-1}(\theta_{\Gamma}^A, \Theta_{\text{img},\sigma}) \\ &= L_2 \sqrt{m_{\sigma}} \mathcal{G}_{AB}^{(\psi)} c_{\sigma}^B = L_2 \sqrt{m_{\sigma}} (g_{\sigma})_A. \end{aligned}$$

Thus

$$M_{11} = m_{\sigma}, \quad \mathcal{D}_{AB} = \frac{C_{A\sigma} C_{B\sigma}}{m_{\sigma} L_2^2} = (g_{\sigma})_A (g_{\sigma})_B.$$

The source-profile norm is

$$\frac{\|\Xi_{\sigma,\text{min}}\|_{\Gamma}^2}{L_2^2} = g_{\sigma}^T \mathcal{G}^{(\psi)-1} g_{\sigma}.$$

In an orthonormal weighted current frame this gives

branch	$\ \Xi_{\sigma,\text{min}}\ _{\Gamma}^2 / L_2^2$
+	54.478818707582...
-	52.600294175197....

The image row is therefore reduced to the local production of ψ_{σ} , m_{σ} , and $\mathcal{G}^{(\psi)}$.

If a Schur slope came from an effective weak period,

$$\rho_{\sigma} = 2Q_{\text{eff}}^{(\sigma)} + 1.$$

The two branches give

$$Q_{\text{eff}}^{(+)} = 9.333551007611\dots, \quad Q_{\text{eff}}^{(-)} = 5.818646994052\dots$$

Both values lie outside the integral weak-period lattice. A period relabel is therefore excluded; the viable realization rows are Green-source coframe deformation or projection inside the localized image sector.

7.7 Boundary Operator Package

Collect the local realization data as

$$\mathcal{B}_{\Gamma} = (d\nu_{\Gamma}, \theta_{\Gamma}^a, \theta_{\Gamma}^Y, \eta_0, \Delta_{\Gamma,j}, \mu_{\Gamma}^2).$$

The branch-operator realization is then governed by the boundary system

$$\begin{aligned} \text{current :} & \quad \int_{\Gamma_{\text{EW,int}}} d\nu_{\Gamma} \gamma_{\Gamma}^{-1}(\theta_{\Gamma}^a, \theta_{\Gamma}^b) = L_{2,\Gamma}^2 \delta^{ab}, \\ \text{neutral :} & \quad L_Q^2 = (1, 1) \mathcal{L}_{\Gamma} \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad Q\eta_0 = 0, \\ \text{endpoint :} & \quad Z_{\mathcal{E},1} = \int_{\Gamma_{\text{EW,int}}} d\nu_{\Gamma} |\eta_0|^2 \|\hat{\eta}_0 \odot \hat{\eta}_0\|^2 = Z_0 I_0, \\ \text{complement :} & \quad \ker \Delta_{\Gamma,j} = \text{Im } A_j, \quad M_{\perp,j}^2 > 0, \\ \text{image :} & \quad \mathcal{D}_{AB} = L_2^{-2} \langle \mathcal{J}_A, \mathcal{M}_{\text{img}}^{-1} \mathcal{J}_B \rangle_{\Gamma}, \quad A, B = 3, Y, \\ & \quad \Delta_{\text{rank}} = 0, \quad (\mathcal{D}_{33}, \mathcal{D}_{3Y} / \mathcal{D}_{33}) \in \{(\delta_+, \rho_+), (\delta_-, \rho_-)\}, \\ \text{auxiliary :} & \quad S_{\text{aux}}^{\Gamma} \rightsquigarrow \mu_{\Gamma}^2 \begin{pmatrix} 0 & A_j^{\dagger} \\ A_j & -A_j A_j^{\dagger} \end{pmatrix}. \end{aligned}$$

A D10 realization that hosts the branch operator supplies the displayed boundary system. The absolute electromagnetic normalization, the neutral mass, and the Schur coefficient are outputs once $\mathcal{B}_{\Gamma}$ is computed.

Proposition 4 (Boundary-current closure constraint). *A D10 IIA realization hosting the branch operator with unit current metric supplies a localized weak-current coframe satisfying*

$$\int_{\Gamma_{\text{EW, int}}} d\nu_{\Gamma} \gamma_{\Gamma}^{-1}(\theta_{\Gamma}^a, \theta_{\Gamma}^b) = L_{2,\Gamma}^2 \delta^{ab}.$$

When this closure equation holds, define

$$e_{\Gamma}^a = \frac{\theta_{\Gamma}^a}{L_{2,\Gamma}}.$$

Then

$$\langle e_{\Gamma}^a, e_{\Gamma}^b \rangle_{\text{cur}} = \delta^{ab}, \quad \kappa_{\text{cur}} = 1,$$

and

$$A_j = \sum_a T_a^{(j)} \otimes e_{\Gamma}^a, \quad A_j^{\dagger} A_j = j(j+1) \mathbf{1}_{V_j}.$$

Proof. The displayed norm equation makes e_{Γ}^a orthonormal in the retained current Hilbert space. Substitution into the current operator leaves the finite $\text{SU}(2)$ Casimir $\sum_a T_a^{(j)} T_a^{(j)} = j(j+1) \mathbf{1}$. $\square$

8 Product-To-Flag Interface

The source-backed D10 base is $\mathbb{CP}^2 \times \mathbb{CP}^1$. The active refinement is the flag

$$F_{1,2}(\mathbb{C}^3) = \text{SU}(3)/T^2 = P(Q) \longrightarrow \mathbb{CP}^2,$$

where Q is the tautological quotient bundle. With

$$h = c_1(\mathcal{O}_{\mathbb{CP}^2}(1)), \quad \eta = c_1(\mathcal{O}_{P(Q)}(1)),$$

the projective-bundle relation is

$$\eta^2 - h\eta + h^2 = 0.$$

For a retained primitive class

$$\zeta = Ph + Q\eta,$$

the cubic is

$$\zeta^3 = 3PQ(P+Q)h^2\eta.$$

This gives a D10 flag-ring normalization candidate for the boundary/current sector.

The KLT flag metric uses

$$J = -ae^{12} + be^{34} - ce^{56}, \quad a, b, c > 0,$$

with

$$d^2 = abc.$$

The $\mathbb{CP}^2$ -base-symmetric line is

$$a = b = A, \quad c = \lambda A.$$

On this line,

$$W_2^- = -\frac{2iA^2(1-\lambda)}{3d} (e^{12} - e^{34} - 2\lambda e^{56}), \quad d^2 = \lambda A^3.$$

The KLT admissibility expression reduces to

$$\frac{-5\lambda^2 + 12\lambda - 4}{8A\lambda}, \quad \frac{2}{5} \leq \lambda \leq 2.$$

The D10 stationary value λ_* must lie in this interval.

9 Finite Direction, Branched Base, And S^3 Fiber

The finite-direction reading places the neutral $O(1)$ field as a two-sheet connection. For a finite field Φ_0 ,

$$\ell_H = \frac{1}{\|\Phi_0\|} = \frac{\sqrt{2}}{v_{EW}}.$$

This supplies a language for the scalar Hessian and the electroweak scale. The same Q -neutral $O(1)$ component protects the photon and provides the branch $j = \frac{1}{2}$ sector.

The branched-projective clue supplies the four-dimensional base replacement

$$\mathbb{CP}^2/\text{complex conjugation} = S^4.$$

The quotient map $\mathbb{CP}^2 \rightarrow S^4$ is a two-fold branched covering with branch locus $\mathbb{RP}^2$. Away from $\mathbb{RP}^2$ it has two local sheets; along $\mathbb{RP}^2$ the sheets meet. This gives a geometric setting for double-cover memory, branch/interface data, and spin or twisted sectors.

The corresponding seven-dimensional Hopf model is

$$S^3 \longrightarrow S^7 \longrightarrow S^4, \quad 3 + 4 = 7.$$

Replacing the spherical base by the projective base asks for an S^3 -type parent over $\mathbb{CP}^2$:

$$S^3\text{-type fiber} \longrightarrow K_7^{\text{br}} \longrightarrow \mathbb{CP}^2.$$

A precise geometric test object is the pullback

$$K_7^{\text{br}} = \mathbb{CP}^2 \times_{S^4} S^7.$$

Over the smooth locus of the branched map, K_7^{br} gives an S^3 -fibered seven-dimensional space over $\mathbb{CP}^2$; over $\mathbb{RP}^2$ it carries the special branch/interface data.

The D10 compactification keeps the six-dimensional projective object

$$K_6 = \mathbb{CP}^2 \times \mathbb{CP}^1$$

with retained C_1 , F_2 , current, boundary, and branch data. Local $SU(2)$ double-cover memory can be carried by current, flat, spin, branch, or boundary sectors inside this D10 object. The flag target

$$\mathbb{CP}^1 \longrightarrow F_{1,2}(\mathbb{C}^3) = SU(3)/T^2 \longrightarrow \mathbb{CP}^2$$

ties the weak projective fiber to the projective base.

The D9 projection keeps the $\mathbb{CP}^2$ base and the electromagnetic stabilizer:

$$K_{5,\text{eff}} = \mathbb{CP}^2 \times S_Q^1.$$

The weak projective direction is projected by the neutral $O(1)$ section,

$$\eta_0 \in H^0(\mathbb{CP}^1, \mathcal{O}(1)), \quad Q\eta_0 = 0.$$

The ladder is therefore

$$\begin{array}{ccccc} S^3 & \longrightarrow & S^7 & \longrightarrow & S^4 \\ S^3\text{-type} & \longrightarrow & K_7^{\text{br}} & \longrightarrow & \mathbb{CP}^2 \\ \mathbb{CP}^1 & \longrightarrow & K_6 & \longrightarrow & \mathbb{CP}^2 \\ & & \downarrow Q & & \\ S_Q^1 & \longrightarrow & K_5 & & \mathbb{CP}^2. \end{array}$$

The branched $\mathbb{CP}^2 \rightarrow S^4$ map is the comparison between the spherical Hopf base and the projective base; the S^3 -type fiber supplies the seven-dimensional parent.

The pure Hopf branch gives a useful control endpoint. Keeping S^7 as the parent and projecting to the electromagnetic stabilizer leads to

$$S^4 \times S_Q^1, \quad \text{Spin}(5) \times \text{U}(1)_Q \simeq \text{Sp}(2) \times \text{U}(1)_Q$$

at the double-cover level, or $\text{SO}(5) \times \text{U}(1)_Q$ at the adjoint level. The projective lift

$$\mathbb{CP}^2 \simeq \text{SU}(3)/S(\text{U}(2) \times \text{U}(1))$$

supplies the eight-generator $\text{SU}(3)$ action and the $\mathbb{Z}_3$ center entering

$$(\text{SU}(3)_c \times \text{U}(1)_Q)/\mathbb{Z}_3.$$

Thus the S^7/S^4 chain is the quaternionic control branch. The $\mathbb{CP}^2$ lift is the color branch.

10 Numerical Consequences

At $c = 1$,

$$X_+(3/4) = \frac{1}{8}(-3 + \sqrt{57}) = 0.568729\dots, \quad X_+(2) = -1 + \sqrt{3} = 0.732051\dots$$

Hence

$$\sin^2 \theta_W = 1 - \frac{X_+(3/4)}{X_+(2)} = 0.223101\dots,$$

and

$$\tan^2 \theta_{\text{pole}} = 0.2871691363429836.$$

The same definitions give the custodial tree-level identity. Since

$$\cos^2 \theta_W = \frac{X_+(3/4)}{X_+(2)}, \quad \frac{M_W^2}{M_Z^2} = \frac{X_+(3/4)}{X_+(2)},$$

one has

$$\rho_{\text{cust}} = \frac{M_W^2}{M_Z^2 \cos^2 \theta_W} = 1.$$

The absolute electromagnetic normalization is assigned to the D10 boundary length matrix. With

$$L_Q^2 = (1, 1) \mathcal{L}_\Gamma \begin{pmatrix} 1 \\ 1 \end{pmatrix}, \quad \alpha_\Gamma = \frac{C^2}{4\pi L_Q^2},$$

alpha is obtained from the same localized current calculation that gives M_Z and the photon null direction.

11 Conclusion

The D10 boundary-current route has a compact operator core. The Borel–Weil weak sectors supply $j = 0, \frac{1}{2}, 1$. The finite current zero modes give $A_j^\dagger A_j = J$. A single Stueckelberg boundary Schur completion gives the block $\mathcal{A}(J)$. The Q -boundary sector protects the photon and descends the global form to $(\text{SU}(3)_c \times \text{U}(1)_Q)/\mathbb{Z}_3$. The D8-like boundary Γ_{EW} also supplies a natural location for the Romans jump, the RR/current vertical length, and the auxiliary field producing the negative diagonal.

The finite electroweak Connes layer supplies the scalar-vacuum dictionary: $\ell_H = \sqrt{2}/v_{\text{EW}}$, $Q\eta_0 = 0$, the standard $W/Z/\gamma$ Hessian, and the local (h, σ) extension. This fixes the finite sheet scale. A stationary D10 realization is specified by the common compactification potential

$$V_{\text{eff}}(\rho, \lambda, v_F, \theta_H, \sigma, \dots)$$

with a stable or metastable D10 threshold window above the D9 endpoint. The KO6 layer supplies the finite-triple tests; fermion data are assigned to the full Standard-Model lift.

The resulting conditional manuscript is concentrated on one boundary/current sector. Its realization data are the localized vertical current, the unit current metric, the Schur auxiliary coefficient and complement gap, and the finite/KK scalar Hessian on the D10 interface.

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